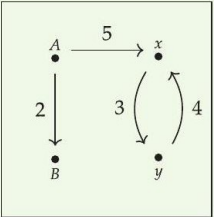


fong-spivak-invitation — formula report

4061 inline formulas (1 shown: the ones that did not render), 262 display equations, 68 tables, 505 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E09292 [conf 0.001]	291	■ 0.001 ● C +79	G_{1}=\begin{array}{ c } \hline \text{李} \\ \hline \end{array} \quad G_{2}=\begin{array}{ c } \hline f \\ \hline \end{array} \quad G_{3}=\begin{array}{ ccc } \bullet & \xrightarrow{f} & \bullet \\ \bullet & \xrightarrow{i} & \bullet \\ \hline f \circ h = g \circ i \end{array} \quad G_{4}=\begin{array}{ ccc } \bullet & \xrightarrow{f} & \bullet \\ \bullet & \xrightarrow{g} & \bullet \\ \hline \text{no equations} \end{array}	(not rendered)	
fong-spivak-invitation_E0962 [conf 0.001]	113	■ 0.001 ● C +8	I=\left\lvert\begin{array}{ccc} 1 & a & 2 \\ \downarrow & \triangleright & 3 \\ & & 4 \\ & & b \end{array}\right.		
fong-spivak-invitation_E09189 [conf 0.013]	273	■ 0.013 ● C +7	G=\left \begin{array}{ccc} 1 & & 2 \\ & a^{\downarrow}e^{\downarrow} & \\ & & v_3 \end{array}\right		
fong-spivak-invitation_E09135 [conf 0.019]	215	■ 0.019 ● C +8	\begin{aligned} \mu_X &:= \left(X + X \xrightarrow{\mathrm{id}_X} X \xleftarrow{\mathrm{id}_X} X \right) \\ \eta_X &:= \left(\emptyset \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X \right) \\ \delta_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{\mathrm{id}_X} X + X \right) \\ \epsilon_X &:= \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \emptyset \right) \end{aligned}		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_ E00096 [conf 0.022]	146	■ 0.022 ●	$\operatorname{Col}(\Phi)=\begin{array}{ c} \stackrel{5}{\rightarrow} \\ 2\downarrow \quad 3\left(\bigcup_y^x 4\right. \\ \bullet \\ \dot{a} \\ a \\ \dot{y} \end{array}$		$\operatorname{Col}(\Phi) =$ 
fong-spivak-invitation_ E00199 [conf 0.051]	291	■ 0.051 ● S +0	$A, \quad A \text{ : } f, \quad A \text{ : } g, \quad A \text{ : } f \text{ : } h, \quad A \text{ : } g \text{ : } i, \quad B, \quad B \text{ : } h, \quad C, \quad C \text{ : } i, \quad D$	$A, \quad A \text{ : } f, \quad A \text{ : } g, \quad A \text{ : } f \text{ : } h, \quad A \text{ : } g \text{ : } i, \quad B, \quad B \text{ : } h, \quad C, \quad C \text{ : } i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ f \circ h, \quad A \circ g \circ i, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-invitation_ E00257 (A.2) [conf 0.064]	323	■ 0.064 ● S -1	$\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longleftarrow} X + X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longrightarrow} X \stackrel{\mathrm{id}}{\longleftarrow} X$	$\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longleftarrow} X + X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longrightarrow} X \stackrel{\mathrm{id}}{\longleftarrow} X$	$\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longleftarrow} X + X \stackrel{[\mathrm{id}, \mathrm{id}]}{\longrightarrow} X \stackrel{\mathrm{id}}{\longleftarrow} X$
fong-spivak-invitation_ E00187 [conf 0.087]	287	■ 0.087 ● C -5	$\begin{aligned} & (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3)) \\ &= \mathcal{X}(x_1, x_2) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_2, y_3) \\ &= \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{Y}(y_2, y_3) \\ &\leq \mathcal{X}(x_1, x_3) \otimes \mathcal{Y}(y_1, y_3) \\ &= (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)). \end{aligned}$	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3))$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_2, y_3)$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{Y}(y_2, y_3)$ $\leq \mathcal{X}(x_1, x_3) \otimes \mathcal{Y}(y_1, y_3)$ $= (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)).$	$(\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_2, y_2)) \otimes (\mathcal{X} \times \mathcal{Y})((x_2, y_2), (x_3, y_3))$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_2, y_3)$ $= \mathcal{X}(x_1, x_2) \otimes \mathcal{X}(x_2, x_3) \otimes \mathcal{Y}(y_1, y_2) \otimes \mathcal{Y}(y_2, y_3)$ $\leq \mathcal{X}(x_1, x_3) \otimes \mathcal{Y}(y_1, y_3)$ $= (\mathcal{X} \times \mathcal{Y})((x_1, y_1), (x_3, y_3)).$
fong-spivak-invitation_ E00095	146	■ 0.151 ● C +37	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X}, \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y}, \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X}, \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y}. \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X} \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y} \end{cases}$	$\operatorname{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X}, \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y}, \\ \emptyset & \text{if } a \in \mathcal{Y}, b \in \mathcal{X}, \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y}. \end{cases}$
fong-spivak-invitation_ E00085	138	■ 0.156 ● K +0	$\text{Chassis: } (\text{load} \times \text{velocity}) \rightarrow (\text{torque} \times \text{speed} \times \$).$	$\text{Chassis: } (\text{load} \times \text{velocity}) \rightarrow (\text{torque} \times \text{speed} \times \$).$	$\text{Chassis: } (\text{load} \times \text{velocity}) \rightarrow (\text{torque} \times \text{speed} \times \$).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E08281	291	■0.169 ●W +1	$\mathcal{D}:=\left\lvert\!\!\!\begin{array}{c} s \\ \bullet \\ z \end{array}\right\rvert \\ sqs; s; s = s; s \\ \end{array}\right.\right.$	$\mathcal{D} := \left\lvert\!\!\!\begin{array}{c} s \\ \bullet \\ z \end{array}\right\rvert \\ sqs; s; s = s; s$	$\mathcal{D} := \left\lvert\!\!\!\begin{array}{c} s \\ \bullet \\ z \end{array}\right\rvert \\ s \mathbin{\mathcal{S}} s \mathbin{\mathcal{S}} s \mathbin{\mathcal{S}} s = s \mathbin{\mathcal{S}} s$
fong-spivak-invitation_E08020 (2.3)	056	■0.184 ●	$\begin{array}{l} \begin{array}{l} h_1 := \{2 \oslash, 3 \oslash, 10 \oslash, J \bullet, Q \bullet\}, \\ \& , J \bullet, Q \bullet\}, \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \bullet \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 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\\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \& \& i_1 := \{4 \bullet, 4 \bullet, 6 \oslash, 6 \oslash, 10 \oslash\}, \\ \$		

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E00134	213	0.235 C +91	$\begin{array}{rrrr} \operatorname{in}(\mu) & := 2, & \operatorname{in}(\eta) & := 0, \\ \operatorname{in}(\delta) & := 1, & \operatorname{in}(\epsilon) & := 1, \\ \operatorname{out}(\mu) & := 1, & \operatorname{out}(\eta) & := 1, \\ \operatorname{out}(\delta) & := 2, & \operatorname{out}(\epsilon) & := 0. \end{array}$	$\operatorname{in}(\mu) := 2, \quad \operatorname{in}(\eta) := 0, \quad \operatorname{in}(\delta) := 1, \quad \operatorname{in}(\epsilon) := 1, ,$	$\begin{array}{cccc} \operatorname{in}(\mu) := 2, & \operatorname{in}(\eta) := 0, & \operatorname{in}(\delta) := 1, & \operatorname{in}(\epsilon) := 1, \\ \operatorname{out}(\mu) := 1, & \operatorname{out}(\eta) := 1, & \operatorname{out}(\delta) := 2, & \operatorname{out}(\epsilon) := 0. \end{array}$
fong-spivak-invitation_E00116	180	0.238 C -3	$\text{and } X \mapsto \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \text{ and } -a- \mapsto (a).$	$- \mapsto (1) \quad \text{and} \quad X \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad -a- \mapsto (a).$	$\longrightarrow \mapsto \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \times \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \text{---} \overline{a} \text{---} \mapsto \begin{pmatrix} a \end{pmatrix}.$
fong-spivak-invitation_E00023	058	0.254	$\overline{\frac{3.14}{-1}} = \frac{3.14}{-1} \quad \overline{0} = \text{nothing}$	$\overline{\frac{3.14}{-1}} = \frac{3.14}{-1} \quad \overline{0} = \text{nothing}$	$\overline{\frac{3.14}{-1}} = \frac{3.14}{-1} \quad \overline{0} = \text{nothing}$
fong-spivak-invitation_E00227	306	0.259 C +43	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \\ & \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b, c, d), (1, e)) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \\ & \otimes (\epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y)) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x, a) \otimes \mathcal{X}(a, b) \otimes \mathcal{X}(c, d) \otimes \mathcal{X}(b, c) \otimes \mathcal{X}(d, e) \otimes \mathcal{X}(e, y) \\ & = \mathcal{X}(x, y), \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a, 1), (b, c, d)) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \\ & \otimes ((b, c, d), (1, e)) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} x(x, a) \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ & = x(x, y), \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a, 1), (b, c, d)) \\ & \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b, c, d), (1, e)) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x, (a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d) \\ & \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ & = \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x, a) \otimes \mathcal{X}(a, b) \otimes \mathcal{X}(c, d) \otimes \mathcal{X}(b, c) \otimes \mathcal{X}(d, e) \otimes \mathcal{X}(e, y) \\ & = \mathcal{X}(x, y), \end{aligned}$
fong-spivak-invitation_E00040	078	0.267 W +1	$\begin{aligned} & \mathcal{C}_f(c, d) \otimes_W \mathcal{C}_f(d, e) = f(\mathcal{C}(c, d)) \otimes_W f(\mathcal{C}(d, e)) \\ & \leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \quad (f \text{ is monoidal monotone}) \\ & \leq f(\mathcal{C}(c, e)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ & = \mathcal{C}_f(c, e) \quad (\text{definition of } \mathcal{C}_f). \end{aligned}$	$\begin{aligned} \mathcal{C}_f(c, d) \otimes_W \mathcal{C}_f(d, e) &= f(\mathcal{C}(c, d)) \otimes_W f(\mathcal{C}(d, e)) && (\text{definition of } \mathcal{C}_f) \\ &\leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) && (f \text{ is monoidal monotone}) \\ &\leq f(\mathcal{C}(c, e)) && (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ &= \mathcal{C}_f(c, e) && (\text{definition of } \mathcal{C}_f). \end{aligned}$	$\mathcal{C}_f(c, d) \otimes_W \mathcal{C}_f(d, e) = f(\mathcal{C}(c, d)) \otimes_W f(\mathcal{C}(d, e)) \quad (\text{definition of } \mathcal{C}_f)$ $\leq f(\mathcal{C}(c, d) \otimes_V \mathcal{C}(d, e)) \quad (f \text{ is monoidal monotone})$ $\leq f(\mathcal{C}(c, e)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category})$ $= \mathcal{C}_f(c, e) \quad (\text{definition of } \mathcal{C}_f).$
fong-spivak-invitation_E00256	320	0.281 C +4	$\left[\operatorname{id}_A, !_A \right] \circ \iota_A = \left[\operatorname{id}_A \circ \iota_A, !_A \circ \iota_A \right] = \left[\iota_A, !_A \circ \iota_A \right] = \left[\iota_A, \iota_{\emptyset} \right] = \operatorname{id}_{A+\emptyset}.$	$[\operatorname{id}_A, !_A] \circ \iota_A = [\operatorname{id}_A \circ \iota_A, !_A \circ \iota_A] = [\iota_A, !_A \circ \iota_A] = [\iota_A, \iota_{\emptyset}] = \operatorname{id}_{A+\emptyset}.$	$[\operatorname{id}_A, !_A] \circ \iota_A = [\operatorname{id}_A \circ \iota_A, !_A \circ \iota_A] = [\iota_A, !_A \circ \iota_A] = [\iota_A, \iota_{\emptyset}] = \operatorname{id}_{A+\emptyset}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0043 (2.22)	080	0.305 N +0	$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right).$		$(\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_2, y_2) \right) \otimes (\mathcal{X} \times \mathcal{Y}) \left((x_2, y_2), (x_3, y_3) \right) \leq (\mathcal{X} \times \mathcal{Y}) \left((x_1, y_1), (x_3, y_3) \right).$
fong-spivak-invitation_EQ0112 (5.6)	177	0.315 C +5	$G_R := \{\bigcirc-, \circ-, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \{\bigcirc-, \circ-, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \left\{ \supset-, \circ-, \subset-, \rightarrow \right\} \cup \left\{ \neg a- \mid a \in R \right\},$
fong-spivak-invitation_EQ0059 (3.8)	108	0.323 N +0	$\mathcal{C} := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ \bullet \\ z \\ \hline s; s = s \end{array}$	$\mathcal{C} := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ \bullet \\ z \\ \hline s; s = s \end{array}$	$\mathcal{C} := \begin{array}{ c } \hline s \\ \mathcal{Q} \\ \bullet \\ z \\ \hline s \circ s = s \end{array}$
fong-spivak-invitation_EQ0118	184	0.327	$G_R := \{\supset-, \circ-, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \{\supset-, \circ-, \subset, \cdots\} \cup \{-a- \mid a \in R\},$	$G_R := \left\{ \supset-, \circ-, \subset-, \rightarrow \right\} \cup \left\{ \neg a- \mid a \in R \right\},$
fong-spivak-invitation_EQ0037	074	0.341 N +0	$X := \text{Ob}(\mathcal{X}), \quad d(x, y) := \mathcal{X}(x, y).$	$X := \text{Ob}(\mathcal{X}), \quad d(x, y) := \mathcal{X}(x, y).$	$X := \text{Ob}(\mathcal{X}), \quad d(x, y) := \mathcal{X}(x, y).$
fong-spivak-invitation_EQ0260	332	0.372 K +0	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$	$\bigcup_{i \in I} A_i = \bigcup_{i \in I} (B_i \cap Y) = \left(\bigcup_{i \in I} B_i \right) \cap Y \in \mathbf{Op}_{? \cap Y}.$
fong-spivak-invitation_EQ0075 (3.15)	121	0.380 C +40	$\mathcal{G} := \text{mid} \stackrel{\text{Email}}{\underset{\text{received_by}}{\longrightarrow}} \text{Address}$	<i>(not rendered)</i>	$\mathcal{G} := \begin{array}{ c } \hline \text{Email} \xrightarrow{\text{sent_by}} \text{Address} \\ \hline \text{received_by} \\ \hline \text{no equations} \end{array}$
fong-spivak-invitation_EQ0141	221	0.382 W +1	$\begin{aligned} & \operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V') \\ & (V, A, s, t, \ell) \longmapsto (V', A, (s \div f), (t; f), \ell). \end{aligned}$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V')$ $(V, A, s, t, \ell) \longmapsto (V', A, (s \div f), (t; f), \ell).$	$\operatorname{Circ}(f) : \operatorname{Circ}(V) \longrightarrow \operatorname{Circ}(V')$ $(V, A, s, t, \ell) \longmapsto (V', A, (s \div f), (t; f), \ell).$
fong-spivak-invitation_EQ0214	299	0.401 W -2	$\begin{aligned} & \Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book}) \\ & \quad = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true} \\ & \quad \& = \Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}. \end{aligned}$	$\Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book})$ $\quad = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true}$ $\Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}.$	$\Lambda(\text{monoid}, \text{nothing}) = \Lambda(\text{monoid}, \text{this book})$ $\quad = \Lambda(\text{preorder}, \text{this book}) = \Lambda(\text{category}, \text{this book}) = \text{true}$ $\Lambda(\text{preorder}, \text{nothing}) = \Lambda(\text{category}, \text{nothing}) = \text{false}.$
fong-spivak-invitation_EQ0024	059	0.473 U +2	$\left. \left. \frac{x}{y} \right _z = \frac{x \otimes y}{z} \right\} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right\} \right.$	$\left. \left. \frac{x}{y} \right _z = \frac{x \otimes y}{z} \right\} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right\} \right.$	$\left. \left. \frac{x}{y} \right _z = \frac{x \otimes y}{z} \right\} = \left\{ \frac{x}{y \otimes z} = \left\{ \frac{y}{z} \right\} \right.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged;
Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

• not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0255	320	■ 0.480 ● N +1	$\mathrm{id}_{\{A\}}+\mathrm{id}_{\{B\}}=\left[\mathrm{id}_{\{A\}}^{\circ}\iota_A,\mathrm{id}_{\{B\}}^{\circ}\iota_B\right]=\mathrm{id}_{\{A+B\}}.$	$\mathrm{id}_A+\mathrm{id}_B=\left[\mathrm{id}_A^{\circ}\iota_A,\mathrm{id}_B^{\circ}\iota_B\right]=\left[\iota_A,\iota_B\right]=\mathrm{id}_{A+B}.$	$\mathrm{id}_A+\mathrm{id}_B=\left[\mathrm{id}_A^{\circ}\iota_A,\mathrm{id}_B^{\circ}\iota_B\right]=\left[\iota_A,\iota_B\right]=\mathrm{id}_{A+B}.$
fong-spivak-invitation_EQ0210	296	■ 0.483 ● W -2	$H:=\stackrel{4}{\underbrace{\stackrel{d}{\rightarrow}}_{\rightarrow}}\stackrel{5}{\rightarrow}e$	$H:=\stackrel{4}{\underbrace{\stackrel{d}{\rightarrow}}_{\rightarrow}}\stackrel{5}{\rightarrow}e$	$H:=\stackrel{4}{\underbrace{\stackrel{d}{\rightarrow}}_{\rightarrow}}\stackrel{5}{\rightarrow}e$
fong-spivak-invitation_EQ0186	284	■ 0.486 ● C +6	$\begin{gathered} g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}), \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}), \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}), \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}). \end{gathered}$	$g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}), \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}), \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}), \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}).$	$g(\text{false})+g(\text{false})=\infty+\infty\geq\infty=g(\text{false}\wedge\text{false}), \\ g(\text{false})+g(\text{true})=\infty+0\geq\infty=g(\text{false}\wedge\text{true}), \\ g(\text{true})+g(\text{false})=0+\infty\geq\infty=g(\text{true}\wedge\text{false}), \\ g(\text{true})+g(\text{true})=0+0\geq0=g(\text{true}\wedge\text{true}).$
fong-spivak-invitation_EQ0092	143	■ 0.504 ●	$(\Phi;\Psi);\Upsilon=\Phi;(\Psi;\Upsilon).$	$(\Phi;\Psi);\Upsilon=\Phi;(\Psi;\Upsilon).$	$(\Phi;\Psi);\Upsilon=\Phi;(\Psi;\Upsilon).$
fong-spivak-invitation_EQ0077	122	■ 0.507 ● C -7	$\left.\frac{\mathbf{1}}{\mathrm{Em}_6}\right \,$	<i>(not rendered)</i>	$\frac{\mathbf{1}}{\mathrm{Em}_6}\,$
fong-spivak-invitation_EQ0071	118	■ 0.519 ● W +2	$\text{Set}\stackrel{-\times B}{\rightleftarrows}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}(A,C^B).$	$\text{Set}\stackrel{-\times B}{\rightleftarrows}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}(A,C^B).$	$\text{Set}\stackrel{-\times B}{\rightleftarrows}\text{Set}\quad\text{Set}(A\times B,C)\cong\text{Set}(A,C^B).$
fong-spivak-invitation_EQ0206	293	■ 0.529 ● N +0	$F(a):=a^{\prime},\quad G(a):=a^{\prime},\quad F(b):=b^{\prime},\quad G(b):=b^{\prime},\quad F(f):=f_1,\quad G(f):=f_2.$	$F(a):=a',\quad G(a):=a',\quad F(b):=b',\quad G(b):=b',\quad F(f):=f_1,\quad G(f):=f_2.$	$F(a):=a',\quad G(a):=a',\quad F(b):=b',\quad G(b):=b',\quad F(f):=f_1,\quad G(f):=f_2.$
fong-spivak-invitation_EQ0074	121	■ 0.559 ● N +0	$\Sigma_1:\mathcal{C}-\text{Inst}\rightarrow\text{Set}\quad\text{and}\quad\Pi_1:\mathcal{C}-\text{Inst}\rightarrow\text{Set}.$	$\Sigma_1:\mathcal{C}-\text{Inst}\rightarrow\text{Set}\quad\text{and}\quad\Pi_1:\mathcal{C}-\text{Inst}\rightarrow\text{Set}.$	$\Sigma_1:\mathcal{C}\text{-Inst}\rightarrow\text{Set}\quad\text{and}\quad\Pi_1:\mathcal{C}\text{-Inst}\rightarrow\text{Set}.$
fong-spivak-invitation_EQ0239	313	■ 0.561 ● U -4	$c_n:=-c:(1+{-c}):(1+1+{-c}): \ldots:(1+\cdots+1+<c):1\rightarrow n.$	$c_n:=-c:(1+{-c}):(1+1+{-c}): \ldots:(1+\cdots+1+<c):1\rightarrow n.$	$c_n:=-c:(1+{-c}):(1+1+{-c}): \ldots:(1+\cdots+1+<c):1\rightarrow n.$
fong-spivak-invitation_EQ0073 (3.14)	121	■ 0.576 ● K +0	$!:C\rightarrow\mathbf{1}.$	$!:C\rightarrow\mathbf{1}.$	$!:C\rightarrow\mathbf{1}.$
fong-spivak-invitation_EQ0025	059	■ 0.583 ● W +0	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$	$\frac{\frac{x}{y}}{z}=\frac{\frac{x}{y}}{\frac{z}{z}}$
fong-spivak-invitation_EQ0132	206	■ 0.584 ● K +0	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$	$\{(v,d)\mid d\in D(v),\text{ where }v=v_1\text{ or }v=v_2\}.$
fong-spivak-invitation_EQ0057	098	■ 0.592 ● N +0	$\{A,B,C,D,f,g,h,i,f;h\}.$	$\{A,B,C,D,f,g,h,i,f;h\}.$	$\{A,B,C,D,f,g,h,i,f;h\}.$
fong-spivak-invitation_EQ0123 (5.14)	189	■ 0.602 ● K +0	$\mathbf{SFG}_R^+:=\text{Free}\left(G_R\sqcup G_R^{\text{op}}\right).$	$\mathbf{SFG}_R^+:=\text{Free}\left(G_R\sqcup G_R^{\text{op}}\right).$	$\mathbf{SFG}_R^+:=\text{Free}\left(G_R\sqcup G_R^{\text{op}}\right).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation-EQ0221	301	0.622	$\begin{aligned} \Phi: \Psi &= \left(M_X^3 * M_\Phi * M_Y^2 \right) * \left(M_Y^2 * M_\Psi * M_Z^3 \right) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= \left(M_X^3 * M_\Phi * M_Y^2 \right) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix}. \end{aligned}$	$\begin{aligned} \Phi \circ \Psi &= (M_X^3 * M_\Phi * M_Y^2) * (M_Y^2 * M_\Psi * M_Z^3) \\ &= M_X^3 * M_\Phi * M_Y^4 * M_\Psi * M_Z^3 \\ &= (M_X^3 * M_\Phi * M_Y^2) * M_\Psi * M_Z^3 \\ &= \Phi * M_\Psi * M_Z^3 \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & 0 & \infty \\ 4 & \infty & \infty & 4 \end{pmatrix} \begin{pmatrix} 0 & 2 & 4 & 5 \\ 4 & 0 & 2 & 3 \\ 2 & 4 & 0 & 1 \\ 1 & 3 & 5 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix} \begin{pmatrix} \infty & \infty & \infty & \infty \\ 2 & 4 & 0 & 1 \\ 4 & 6 & 8 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 22 & 24 & 20 & 21 \\ 16 & 18 & 14 & 15 \\ 19 & 21 & 17 & 18 \\ 11 & 13 & 9 & 10 \end{pmatrix}. \end{aligned}$	
fong-spivak-invitation-EQ0010 (1.7)	044	0.623 C +21	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-invitation-EQ0155 (7.7)	242	0.623 N +2	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$	$X = \{(\text{true}, \text{true}), (\text{true}, \text{false}), (\text{false}, \text{true})\} \subseteq \mathbb{B} \times \mathbb{B}.$
fong-spivak-invitation-EQ0156	247	0.628 N +1	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$	$B(p; \epsilon) := \{p' \in \mathbb{R}^2 \mid d(p, p') < \epsilon\}.$
fong-spivak-invitation-EQ0124	190	0.641 N +2	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$\mathrm{B}(g \circ (h^{\mathrm{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-invitation-EQ0080 (4.2)	134	0.641 C +11	$\Phi: X^{\mathrm{op}} \times y \rightarrow \mathbf{Bool}.$	$\Phi: X^{\mathrm{op}} \times y \rightarrow \mathbf{Bool}.$	$\Phi: \mathcal{X}^{\mathrm{op}} \times \mathcal{Y} \rightarrow \mathbf{Bool}.$
fong-spivak-invitation-EQ0097	150	0.644 C +3	$\begin{aligned} S \times (T \times U) &:= \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \\ &=? (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}. \end{aligned}$	$\begin{aligned} S \times (T \times U) &:= \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \\ &=? (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}. \end{aligned}$	$\begin{aligned} S \times (T \times U) &:= \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \\ &=? (S \times T) \times U := \{((s, t), u) \mid s \in S, t \in T, u \in U\}. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0224	305	0.649 N -1	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$	$\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) \leq \bigvee_{y \in \mathcal{X}} \mathcal{X}(x, z) = \mathcal{X}(x, z).$
fong-spivak-invitation_EQ0055 (3.5)	096	0.649	$\mathbf{2} := \text{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \text{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$	$\mathbf{2} := \text{Free} \left(\begin{array}{ ccc } \hline v_1 & f_1 & v_2 \\ \bullet & & \bullet \\ \hline \end{array} \right)$
fong-spivak-invitation_EQ0107 (5.4)	167	0.670 N +0	$G + G' := ((V \sqcup V'), [\text{in}, \text{in'}], [\text{out}, \text{out'}], (\iota \sqcup \iota')).$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in'}], [\text{out}, \text{out'}], (\iota \sqcup \iota')).$	$G + G' := ((V \sqcup V'), [\text{in}, \text{in'}], [\text{out}, \text{out'}], (\iota \sqcup \iota')).$
fong-spivak-invitation_EQ0151 (7.3)	239	0.683 N +0	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C).$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C).$	$\mathcal{C}(A \times C, D) \cong \mathcal{C}(A, D^C).$
fong-spivak-invitation_EQ0191	288	0.683 N +0	$\emptyset \otimes v \cong v \otimes \emptyset \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$	$0 \otimes v \cong v \otimes 0 \cong \left(v \otimes \bigvee_{a \in \emptyset} a \right) = \bigvee_{a \in \emptyset} (v \otimes a) = 0.$
fong-spivak-invitation_EQ0200	291	0.687 S +0	$A, \quad A: f, \quad A: g, \quad A: j, \quad B, \quad B: h, \quad C, \quad C: i, \quad D$	$A, \quad A: f, \quad A: g, \quad A: j, \quad B, \quad B: h, \quad C, \quad C: i, \quad D$	$A, \quad A \circ f, \quad A \circ g, \quad A \circ j, \quad B, \quad B \circ h, \quad C, \quad C \circ i, \quad D$
fong-spivak-invitation_EQ0089	142	0.688 C +7	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}.$	$\mathrm{U}_{\mathcal{P}}; \Phi = \Phi = \Phi; \mathrm{U}_{\mathcal{Q}}.$	$\mathrm{U}_{\mathcal{P}} \circ \Phi = \Phi = \Phi \circ \mathrm{U}_{\mathcal{Q}}.$
fong-spivak-invitation_EQ0170 (7.18)	263	0.706 K +0	$\forall (t : T). \exists (s : S). f(s) = t$	$\forall (t : T). \exists (s : S). f(s) = t$	$\forall (t : T). \exists (s : S). f(s) = t$
fong-spivak-invitation_EQ0061 (3.10)	113	0.708 C +13	$\mathbf{Gr} := \mid \stackrel{\text{Arrow}}{\text{source}} \stackrel{\text{Vertex}}{\text{target}} \bullet$	$\mathbf{Gr} := \mid \stackrel{\text{Arrow}}{\text{source}} \stackrel{\text{Vertex}}{\text{target}} \bullet$	$\mathbf{Gr} := \begin{array}{c} \text{Arrow} \xrightarrow[\text{target}]{\text{source}} \text{Vertex} \\ \text{no equations} \end{array}$
fong-spivak-invitation_EQ0018	049	0.726 N -1	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}.$	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}.$	$\text{fix}_j := \{p \in P \mid j(p) \cong p\}.$
fong-spivak-invitation_EQ0149 (7.2)	234	0.727 N +0	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0).$	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0).$	$((A = \text{go}) \Rightarrow \ddot{S} > 1) \wedge ((A = \text{stop}) \Rightarrow \ddot{S} = 0).$
fong-spivak-invitation_EQ0082	135	0.730 C +8	$\Phi: x^{\text{op}} \times y \rightarrow \mathcal{V}.$	$\Phi: x^{\text{op}} \times y \rightarrow \mathcal{V}.$	$\Phi: \mathcal{X}^{\text{op}} \times \mathcal{Y} \rightarrow \mathcal{V}.$
fong-spivak-invitation_EQ0248	316	0.734 N +0	$B(g_9^\circ(h^{\text{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$B(g_9^\circ(h^{\text{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$	$B(g \circ (h^{\text{op}})) = \{(x, y) \mid S(g)(x) = S(h)(y)\}.$
fong-spivak-invitation_EQ0114	178	0.740 N +2	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$	$\left(\begin{array}{cc} A & 0 \\ 0 & B \end{array} \right)$
fong-spivak-invitation_EQ0138	220	0.748 N +1	$C: -\{\text{light, switch, battery}\} \sqcup \{x\Omega \mid x \in \mathbb{R}^+\}.$	$C: -\{\text{light, switch, battery}\} \sqcup \{x\Omega \mid x \in \mathbb{R}^+\}.$	$C := \{\text{light, switch, battery}\} \sqcup \{x\Omega \mid x \in \mathbb{R}^+\}.$
fong-spivak-invitation_EQ0143 (6.21)	226	0.754 C +20	$\circ_i: \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$	$\circ_i: \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$	$\circ_i: \mathcal{O}(s_1, \dots, s_m; t_i) \times \mathcal{O}(t_1, \dots, t_n; t) \rightarrow \mathcal{O}(t_1, \dots, t_{i-1}, s_1, \dots, s_m, t_{i+1}, \dots, t_n; t);$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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fong-spivak-invitation-EQ0198	290	■ 0.762 ● N +0	$\stackrel{\bullet}{v_1} \xrightarrow{\bullet} \stackrel{\bullet}{v_2} \xrightarrow{\bullet} \dots \xrightarrow{\bullet} \stackrel{\bullet}{v_n}$	$v_1 \xrightarrow{f_1} v_2 \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} v_n$	$v_1 \xrightarrow{f_1} v_2 \xrightarrow{f_2} \dots \xrightarrow{f_{n-1}} v_n$
fong-spivak-invitation-EQ0209	296	■ 0.772 ● C +3	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b}$	$G := \stackrel{1}{\bullet} \xrightarrow{a} \stackrel{2}{\bullet} \xrightarrow{b}$	$G := \left \begin{array}{c} 1 \xrightarrow{a} 2 \xrightarrow{b} 3 \end{array} \right $
fong-spivak-invitation-EQ0160	253	■ 0.775 ● N +2	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{g: U \rightarrow X \mid g \text{ is continuous and } (g \circ f)(u) = u \text{ for all } u \in U\}.$
fong-spivak-invitation-EQ0217	300	■ 0.776 ●	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes y(y, y') \leq \Phi(x', y').$
fong-spivak-invitation-EQ0022 (nothing)	057	■ 0.778 ● N +0	$\bar{I}$	$\bar{I}$	$\overline{I}$
fong-spivak-invitation-EQ0128	198	■ 0.782 ● N +2	$\mathcal{C} := \stackrel{A}{\bullet} \xrightarrow{f} \stackrel{B}{\bullet}$	$\mathcal{C} := \stackrel{A}{\bullet} \xrightarrow{f} \stackrel{B}{\bullet}$	$\mathcal{C} := \left \begin{array}{c} A \xrightarrow{f} B \end{array} \right $
fong-spivak-invitation-EQ0148	231	■ 0.787 ● N +2	$\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$	$\operatorname{Circ}(\varphi): \operatorname{Circ}(2) \times \operatorname{Circ}(2) \times \operatorname{Circ}(2) \rightarrow \operatorname{Circ}(0).$
fong-spivak-invitation-EQ0003	026	■ 0.790 ● C +5	$S := \begin{array}{ccc} 11 & 12 & 13 \\ \bullet & \bullet & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \end{array}$	$S := \begin{array}{ccc} 11 & 12 & 13 \\ \bullet & \bullet & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \end{array}$	$S := \left \begin{array}{ccc} 11 & 12 & 13 \\ \bullet & \bullet & \\ 21 & 22 & 23 \\ \bullet & \bullet & \bullet \end{array} \right $
fong-spivak-invitation-EQ0066	114	■ 0.798 ● W -1	$\begin{array}{c} G := \\ \begin{array}{c cc} \text{Arrow} & \text{source} & \text{target} \\ \hline a & 1 & 2 \\ b & 2 & 3 \end{array} \\ H := \\ \begin{array}{c cc} \text{Arrow} & \text{source} & \text{target} \\ \hline c & 4 & 5 \\ d & 4 & 5 \\ e & 5 & 5 \end{array} \\ \text{Vertex} \\ \begin{array}{c} 4 \\ 5 \end{array} \end{array}$	$\begin{array}{c} G := \\ \begin{array}{c cc} \text{Arrow} & \text{source} & \text{target} \\ \hline a & 1 & 2 \\ b & 2 & 3 \end{array} \\ H := \\ \begin{array}{c cc} \text{Arrow} & \text{source} & \text{target} \\ \hline c & 4 & 5 \\ d & 4 & 5 \\ e & 5 & 5 \end{array} \\ \text{Vertex} \\ \begin{array}{c} 4 \\ 5 \end{array} \end{array}$	$\begin{array}{c} G := \\ \begin{array}{c cc} \text{Arrow} & \text{source} & \text{target} \\ \hline a & 1 & 2 \\ b & 2 & 3 \end{array} \\ H := \\ \begin{array}{c cc} \text{Arrow} & \text{source} & \text{target} \\ \hline c & 4 & 5 \\ d & 4 & 5 \\ e & 5 & 5 \end{array} \\ \text{Vertex} \\ \begin{array}{c} 4 \\ 5 \end{array} \end{array}$
fong-spivak-invitation-EQ0159 (7.10)	251	■ 0.806 ● N +0	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}.$	$\operatorname{Sec}_f(U) := \{s: U \rightarrow X \mid (s \circ f)(u) = u \text{ for all } u \in U\}.$
fong-spivak-invitation-EQ0139	221	■ 0.813 ● N +0	$\operatorname{Circ}: (\operatorname{FinSet}, +) \longrightarrow (\operatorname{Set}, \times)$	$\operatorname{Circ}: (\operatorname{FinSet}, +) \longrightarrow (\operatorname{Set}, \times)$	$\operatorname{Circ}: (\mathbf{FinSet}, +) \longrightarrow (\mathbf{Set}, \times)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_E00225	305	0.820 C -4	$\left(\alpha^{-1} ; \alpha\right)(x, z)=\bigvee_{\left(y, 1\right) \in \mathcal{X} \times \mathbf{1}} \alpha(x,(y, 1)) \otimes \alpha^{-1}((y, 1), z)=\bigvee_{y \in \mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z)=\mathcal{X}(x, z) .$	$(\alpha^{-1};\alpha)(x,z)=\bigvee_{(y,1)\in\mathcal{X}\times\mathbf{1}}\alpha(x,(y,1))\otimes\alpha^{-1}((y,1),z)=\bigvee_{y\in\mathcal{X}}\mathcal{X}(x,y)\otimes\mathcal{X}(y,z)=\mathcal{X}(x,z).$	$(\alpha^{-1} \circ \alpha)(x, z) = \bigvee_{(y,1)\in\mathcal{X}\times\mathbf{1}} \alpha(x, (y, 1)) \otimes \alpha^{-1}((y, 1), z) = \bigvee_{y\in\mathcal{X}} \mathcal{X}(x, y) \otimes \mathcal{X}(y, z) = \mathcal{X}(x, z).$
fong-spivak-invitation_E00100	157	0.831 N +0	$\alpha: \beta:=\iota_{A \sqcup C}^{*}\left(\iota_{A \sqcup B}\right)!(\alpha) \vee\left(\iota_{B \sqcup C}\right)!(\beta)) .$	$\alpha: \beta:=\iota_{A \sqcup C}^{*}\left(\iota_{A \sqcup B}\right)!(\alpha) \vee\left(\iota_{B \sqcup C}\right)!(\beta)) .$	$\alpha \circ \beta:=\iota_{A \sqcup C}^{*}\left(\iota_{A \sqcup B}\right)!(\alpha) \vee\left(\iota_{B \sqcup C}\right)!(\beta)) .$
fong-spivak-invitation_E00083	135	0.833 N +0	$\mathcal{X}\left(x^{\prime}, x\right) \otimes \Phi(x, y) \otimes y\left(y, y^{\prime}\right) \leq \Phi\left(x^{\prime}, y^{\prime}\right) .$	$\mathcal{X}\left(x^{\prime}, x\right) \otimes \Phi(x, y) \otimes y\left(y, y^{\prime}\right) \leq \Phi\left(x^{\prime}, y^{\prime}\right) .$	$\mathcal{X}(x', x) \otimes \Phi(x, y) \otimes \mathcal{Y}(y, y') \leq \Phi(x', y').$
fong-spivak-invitation_E00021	057	0.839 W -2	$\frac{\bar{x}}{\frac{y}{y}} \quad \overline{x \otimes y}$	$\frac{\bar{x}}{\frac{y}{y}} \quad \overline{x \otimes y}$	$\frac{x}{y} \quad \overline{x \otimes y}$
fong-spivak-invitation_E00150	236	0.840 C +8	$\{p \in P \mid \text { likes_cats }(p)\} .$	<i>(not rendered)</i>	$\{p \in P \mid \text { likes_cats }(p)\} .$
fong-spivak-invitation_E00047	083	0.847	$2 \mathrm{H}_{2} \mathrm{~} \rightarrow \left(2 \mathrm{Na} \multimap\left(2 \mathrm{NaOH}+\mathrm{H}_{2}\right)\right) .$	$2\mathrm{H}_2\mathrm{O}\rightarrow(2\mathrm{Na}\multimap(2\mathrm{NaOH}+\mathrm{H}_2)).$	$2\mathrm{H}_2\mathrm{O}\rightarrow(2\mathrm{Na}\multimap(2\mathrm{NaOH}+\mathrm{H}_2)).$
fong-spivak-invitation_E00088 (4.8)	142	0.851 N +0	$\mathrm{U}_{\mathcal{X}}\left(\mathrm{U}_{\mathcal{X}}\right)(x, y):=\mathfrak{X}(x, y) .$	$\mathrm{U}_{\mathcal{X}}(x, y):=\mathfrak{X}(x, y).$	$\mathrm{U}_{\mathcal{X}}(x, y):=\mathfrak{X}(x, y).$
fong-spivak-invitation_E00090 (4.9)	142	0.868 W +1	$\begin{aligned} \Phi(p, q) \&=I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \\ \leq \bigvee_{p_1 \in P} &(\mathcal{P}(p, p_1) \otimes \Phi(p_1, q))=(\mathrm{U}_{\mathcal{P}} ; \Phi)(p, q) . \end{aligned}$	$\Phi(p, q)=I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \\ \leq \bigvee_{p_1 \in P} &(\mathcal{P}(p, p_1) \otimes \Phi(p_1, q))=(\mathrm{U}_{\mathcal{P}} ; \Phi)(p, q) .$	$\Phi(p, q)=I \otimes \Phi(p, q) \leq \mathcal{P}(p, p) \otimes \Phi(p, q) \\ \leq \bigvee_{p_1 \in P} &(\mathcal{P}(p, p_1) \otimes \Phi(p_1, q))=(\mathrm{U}_{\mathcal{P}} \circ \Phi)(p, q) .$
fong-spivak-invitation_E00099	152	0.880 N +0	$(f \times g) ; \left(f^{\prime} \times g^{\prime}\right)=(f ; g) \times\left(f^{\prime} ; g^{\prime}\right) .$	$(f \times g) ;\left(f^{\prime} \times g^{\prime}\right)=(f ; g) \times\left(f^{\prime} ; g^{\prime}\right) .$	$(f \times g) \circ\left(f^{\prime} \times g^{\prime}\right)=(f \circ g) \times\left(f^{\prime} \circ g^{\prime}\right) .$
fong-spivak-invitation_E00167	260	0.881 K +0	$s: S \mid p(s) \vdash q(s) \quad \text { or } \quad p(s) \vdash_{s: S} q(s) .$	$s: S \mid p(s) \vdash q(s) \quad \text { or } \quad p(s) \vdash_{s: S} q(s) .$	$s: S \mid p(s) \vdash q(s) \quad \text { or } \quad p(s) \vdash_{s: S} q(s) .$
fong-spivak-invitation_E00129 (6.2)	200	0.882 S +0	$[f, g](x)=\begin{cases} f(x) & \text { if } x=\iota_A(a) \text { for some } a \in A, \\ g(x) & \text { if } x=\iota_B(b) \text { for some } b \in B . \end{cases}$	$[f, g](x)=\begin{cases} f(x) & \text { if } x=\iota_A(a) \text { for some } a \in A, \\ g(x) & \text { if } x=\iota_B(b) \text { for some } b \in B. \end{cases}$	$[f, g](x)=\begin{cases} f(x) & \text { if } x=\iota_A(a) \text { for some } a \in A, \\ g(x) & \text { if } x=\iota_B(b) \text { for some } b \in B. \end{cases}$
fong-spivak-invitation_E00144	228	0.883 K +0	$\begin{aligned} (g \circ i f)\left(x_1, \ldots, x_{i-1}, w_1, \ldots, w_m, x_{i+1}, \ldots, x_n\right) \\ =g\left(x_1, \ldots, x_{i-1}, f\left(w_1, \ldots, w_m\right), x_{i+1}, \ldots, x_n\right) \end{aligned}$	$(g \circ i f)\left(x_1, \ldots, x_{i-1}, w_1, \ldots, w_m, x_{i+1}, \ldots, x_n\right) \\ =g\left(x_1, \ldots, x_{i-1}, f\left(w_1, \ldots, w_m\right), x_{i+1}, \ldots, x_n\right)$	$(g \circ_i f)\left(x_1, \ldots, x_{i-1}, w_1, \ldots, w_m, x_{i+1}, \ldots, x_n\right) \\ =g\left(x_1, \ldots, x_{i-1}, f\left(w_1, \ldots, w_m\right), x_{i+1}, \ldots, x_n\right)$
fong-spivak-invitation_E00119	188	0.887 U -1	$\mathrm{B}(-\mathrm{C})=\{(x,(x, x)) \mid x \in R\} .$	$\mathrm{B}(-\mathrm{C})=\{(x,(x, x)) \mid x \in R\} .$	$\mathrm{B}(\neg\!\!\!\lhd)=\{(x,(x, x)) \mid x \in R\} .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0007	040	0.967 C +6	$f(a) \vee f(b) \nsubseteq f(a \vee b)$.	$f(a) \vee f(b) \not\subseteq f(a \vee b)$.	$f(a) \vee f(b) \not\cong f(a \vee b)$.
fong-spivak-invitation_EQ0101	158	0.969 N -1	$(\mathcal{X} \times \mathcal{Y}) \left((x, y), \left(x^{\prime}, y^{\prime} \right) \right) := \mathcal{X} \left(x, x^{\prime} \right) \otimes \mathcal{Y} \left(y, y^{\prime} \right)$.	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$.	$(\mathcal{X} \times \mathcal{Y})((x, y), (x', y')) := \mathcal{X}(x, x') \otimes \mathcal{Y}(y, y')$.
fong-spivak-invitation_EQ0094 (4.11)	145	0.970 N +0	$\mathcal{P}(p, G(q)) \cong Q(F(p), q)$.	$\mathcal{P}(p, G(q)) \cong Q(F(p), q)$.	$\mathcal{P}(p, G(q)) \cong \mathcal{Q}(F(p), q)$.
fong-spivak-invitation_EQ0213	299	0.971 N -1	$\lim_{\mathcal{J}} \mathcal{D} := \left\{ \left(d_1, d_2 \right) \mid d_1 \in X \text{ and } d_2 \in Y \right\}$,	$\lim_{\mathcal{J}} D := \{ (d_1, d_2) \mid d_1 \in X \text{ and } d_2 \in Y \}$,	$\lim_{\mathcal{J}} D := \{ (d_1, d_2) \mid d_1 \in X \text{ and } d_2 \in Y \}$,
fong-spivak-invitation_EQ0259	328	0.972 S +0	$p \ ; \ h_3=r \ ; \ g \ ; \ h_3=r \ ; \ h_2 \ ; \ g^{\prime}=q \ ; \ f^{\prime} \ ; \ g^{\prime}$,	$p;h_3=r;g;h_3=r;h_2;g'=q;f';g'$,	$p \circ h_3 = r \circ g \circ h_3 = r \circ h_2 \circ g' = q \circ f' \circ g'$,
fong-spivak-invitation_EQ0152 (7.5)	241	0.972 K +0	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true} , \text{false} \}$.	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true} , \text{false} \}$.	$\Omega_{\text{Set}} := \mathbb{B} = \{ \text{true}, \text{false} \}$.
fong-spivak-invitation_EQ0060 (3.13)	117	0.979 K +0	$f(p) \leq q \text{ iff } p \leq g(q)$.	$f(p) \leq q \text{ iff } p \leq g(q)$.	$f(p) \leq q \text{ iff } p \leq g(q)$.
fong-spivak-invitation_EQ0111	176	0.979	$\begin{array}{l} D \ x + 3 \ D^2 \ y - 2 \ z = 0 \\ D^2 \ y + 5 \ D \ z = 0 \end{array}$	$\begin{array}{l} D x + 3 D^2 y - 2 z = 0 \\ D^2 y + 5 D z = 0 \end{array}$	$\begin{array}{l} D x + 3 D^2 y - 2 z = 0, \\ D^2 y + 5 D z = 0, \end{array}$
fong-spivak-invitation_EQ0104 (5.2)	163	0.979 N +0	$i \longmapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m, \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m, \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$	$i \mapsto \begin{cases} f(i) & \text{if } 1 \leq i \leq m, \\ m' + g(i) & \text{if } m + 1 \leq i \leq m + n. \end{cases}$
fong-spivak-invitation_EQ0016	047	0.980 N -1	$f_{\{!\}} \left(A^{\{ \prime \}} \right) := \left\{ b \in B \mid \text{there exists } a \in A^{\{ \prime \}} \text{ such that } f(a) = b \right\}$.	$f_{!}(A') := \{ b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b \}$.	$f_{!}(A') := \{ b \in B \mid \text{there exists } a \in A' \text{ such that } f(a) = b \}$.
fong-spivak-invitation_EQ0076	122	0.982 W +0	$\begin{array}{l} 1 \\ \hline \text{Bob-Grace-Pat-Emmy} \\ \text{Sue-Doug} \end{array} \quad \& \quad \begin{array}{l} \text{Sue-Doug} \end{array}$	$\frac{1}{\text{Bob-Grace-Pat-Emmy} \quad \text{Sue-Doug}} \quad \bigg \quad \begin{array}{l} \text{Sue-Doug} \end{array}$	$\frac{\mathbf{1}}{\text{Bob-Grace-Pat-Emmy} \quad \text{Sue-Doug}} \quad \bigg \quad \begin{array}{l} \text{Sue-Doug} \end{array}$
fong-spivak-invitation_EQ0034	067	0.983 K +0	$\text{Cost} := ([0, \infty], \geq, 0, +)$,	$\text{Cost} := ([0, \infty], \geq, 0, +)$,	$\text{Cost} := ([0, \infty], \geq, 0, +)$,
fong-spivak-invitation_EQ0131	206	0.985 C +28	$\operatorname{colim}_{\mathcal{J}} \mathcal{D} := \{ (v, d) \mid v \in V \text{ and } d \in \mathcal{D}(v) \} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{ (v, d) \mid v \in V \text{ and } d \in D(v) \} / \sim$	$\operatorname{colim}_{\mathcal{J}} D := \{ (v, d) \mid v \in V \text{ and } d \in D(v) \} / \sim$
fong-spivak-invitation_EQ0247	316	0.986 N -2	$\left\{ (x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y) \right\}$.	$\{ (x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y) \}$.	$\{ (x, y) \mid \text{there exists } z \in R^n \text{ such that } S(g)(x) = z \text{ and } z = S(h)(y) \}$.
fong-spivak-invitation_EQ0017	047	0.988 K +0	$f_{\{*\}} \left(A^{\{ \prime \}} \right) := \left\{ b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A^{\{ \prime \}} \right\}$.	$f_*(A') := \{ b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A' \}$.	$f_*(A') := \{ b \in B \mid \text{for all } a \text{ such that } f(a) = b, \text{ we have } a \in A' \}$.
fong-spivak-invitation_EQ0069	118	0.988 C +8	$\alpha_{c,d} \colon \mathcal{C}(c, R(d)) \stackrel{\cong}{\rightarrow} \mathcal{D}(L(c), d)$	$\alpha_{c,d} \colon \mathcal{C}(c, R(d)) \stackrel{\cong}{\rightarrow} \mathcal{D}(L(c), d)$	$\alpha_{c,d} \colon \mathcal{C}(c, R(d)) \stackrel{\cong}{\rightarrow} \mathcal{D}(L(c), d)$
fong-spivak-invitation_EQ0228	307	0.992 K +0	$\sigma_{m,n}(i) := \begin{cases} i + n & \text{if } i \leq m, \\ i - m & \text{if } m + 1 \leq i. \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i + n & \text{if } i \leq m, \\ i - m & \text{if } m + 1 \leq i. \end{cases}$	$\sigma_{m,n}(i) := \begin{cases} i + n & \text{if } i \leq m, \\ i - m & \text{if } m + 1 \leq i. \end{cases}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_ E08126 (and)	191	■0.992 ●K +0	\{(\theta,(x,x))\mid x\in R\}\subseteq R^{\{0\}}\times R^{\{2\}}\quad \text{and }\quad \{((x,x),0)\mid x\in R\}\subseteq R^2\times R^0.	$\{(0,(x,x))\mid x\in R\}\subseteq R^0\times R^2\quad\text{and}\quad\{((x,x),0)\mid x\in R\}\subseteq R^2\times R^0.$	$\{(0,(x,x))\mid x\in R\}\subseteq R^0\times R^2\quad\text{and}\quad\{((x,x),0)\mid x\in R\}\subseteq R^2\times R^0.$
fong-spivak-invitation_ E08161 (7.13)	255	■0.992 ●K +0	\Omega(U):=\left\{U^{\prime}\in\mathbf{Op}\mid U^{\prime}\subseteq U\right\}.	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}.$	$\Omega(U) := \{U' \in \mathbf{Op} \mid U' \subseteq U\}.$
fong-spivak-invitation_ E08042	079	■0.993 ●K +0	\mathrmcal{X}\left(x_{\scriptscriptstyle 1},\,x_{\scriptscriptstyle 2}\right)\leq y\left(F\left(x_{\scriptscriptstyle 1}\right),\,F\left(x_{\scriptscriptstyle 2}\right)\right),	$\mathcal{X}(x_1,x_2)\leq y(F(x_1),F(x_2)),$	$\mathcal{X}(x_1,x_2)\leq \mathcal{Y}(F(x_1),F(x_2)),$
fong-spivak-invitation_ E08164 (7.15)	257	■0.994 ●K +0	(U\wedge V):=U\cap V\quad\text{and }\quad (U\vee V):=U\cup V.	$(U\wedge V) := U\cap V\quad\text{and}\quad (U\vee V) := U\cup V.$	$(U\wedge V) := U\cap V\quad\text{and}\quad (U\vee V) := U\cup V.$
fong-spivak-invitation_ E08171	265	■0.994 ●N +0	\mathbb{I}\,\mathbb{R}:=\{[d,u]\subseteq\mathbb{R}\mid d\leq u\}.	$\mathbb{IR} := \{[d,u] \subseteq \mathbb{R} \mid d \leq u\}.$	$\mathbb{IR} := \{[d,u] \subseteq \mathbb{R} \mid d \leq u\}.$
fong-spivak-invitation_ E08192	289	■0.994 ●	\begin{aligned} I_X * M(x,y) &= \bigvee_{x'\in X} I_X(x,x')\otimes M(x',y) \\ &= (I_X(x,x)\otimes M(x,y))\vee\left(\bigvee_{x'\in X,x'\neq x} I_X(x,x')\otimes M(x',y)\right) \\ &= (I\otimes M(x,y))\vee\left(\bigvee_{x'\in X,x'\neq x} 0\otimes M(x',y)\right) \\ &= M(x,y)\vee 0 = M(x,y) \end{aligned}	$I_X * M(x,y) = \bigvee_{x'\in X} I_X(x,x')\otimes M(x',y)$ $= (I_X(x,x)\otimes M(x,y))\vee\left(\bigvee_{x'\in X,x'\neq x} I_X(x,x')\otimes M(x',y)\right)$ $= (I\otimes M(x,y))\vee\left(\bigvee_{x'\in X,x'\neq x} 0\otimes M(x',y)\right)$ $= M(x,y)\vee 0 = M(x,y)$	$I_X * M(x,y) = \bigvee_{x'\in X} I_X(x,x')\otimes M(x',y)$ $= (I_X(x,x)\otimes M(x,y))\vee\left(\bigvee_{x'\in X,x'\neq x} I_X(x,x')\otimes M(x',y)\right)$ $= (I\otimes M(x,y))\vee\left(\bigvee_{x'\in X,x'\neq x} 0\otimes M(x',y)\right)$ $= M(x,y)\vee 0 = M(x,y).$
fong-spivak-invitation_ E08113 (5.7)	178	■0.995 ●N +0	M ; N(a,c):=\sum_b\limits_{b\in\underline{n}}M(a,b)\times N(b,c).	$M;N(a,c) := \sum_{b\in\underline{n}} M(a,b)\times N(b,c).$	$M\circ N(a,c) := \sum_{b\in\underline{n}} M(a,b)\times N(b,c).$
fong-spivak-invitation_ E08056 (3.6)	096	■0.995 ●N +2	\mathbf{Free}\stackrel{v_1}{\bullet}\stackrel{f_1}{\longrightarrow}\stackrel{v_2}{\bullet}\stackrel{f_2}{\longrightarrow}\stackrel{v_3}{\bullet}	$\mathbf{3} := \mathbf{Free} \mid \overset{v_1}{\bullet} \overset{f_1}{\rightarrow} \overset{v_2}{\bullet} \overset{f_2}{\rightarrow} \overset{v_3}{\bullet}$	$\mathbf{3} := \mathbf{Free} \Bigg \mid \overset{v_1}{\bullet} \overset{f_1}{\rightarrow} \overset{v_2}{\bullet} \overset{f_2}{\rightarrow} \overset{v_3}{\bullet}$
fong-spivak-invitation_ E08195	290	■0.996 ●K +0	t(p):=\begin{cases} v&\text{ if }n=0,\\ t(a_n)&\text{ if }n\geq 1.\end{cases}	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$	$t(p) := \begin{cases} v & \text{if } n = 0, \\ t(a_n) & \text{if } n \geq 1. \end{cases}$
fong-spivak-invitation_ E08082 (1.5)	023	■0.996 ●C +3	A=\bigcup_{p\in P}A_p\quad\text{and }\quad \text{if } p\neq q \text{ then } A_p\cap A_q=\varnothing.	$A = \bigcup_{p\in P} A_p\quad\text{and}\quad \text{if } p\neq q \text{ then } A_p\cap A_q = \emptyset.$	$A = \bigcup_{p\in P} A_p\quad\text{and}\quad \text{if } p\neq q \text{ then } A_p\cap A_q = \emptyset.$
fong-spivak-invitation_ E08189	288	■0.996 ●K +0	A\cap B\subseteq (\overline{B}\cup C)\cap B=(\overline{B}\cap B)\cup(C\cap B)=C\cap B\subseteq C.	$A\cap B\subseteq (\bar{B}\cup C)\cap B=(\bar{B}\cap B)\cup(C\cap B)=C\cap B\subseteq C.$	$A\cap B\subseteq (\bar{B}\cup C)\cap B=(\bar{B}\cap B)\cup(C\cap B)=C\cap B\subseteq C.$
fong-spivak-invitation_ E08136	217	■0.996 ●C +15	\varphi_{c_1,c_2}: F(c_1)\otimes_{\mathcal{D}} F(c_2)\rightarrow F(c_1\otimes_{\mathcal{C}} c_2),	$\varphi_{c_1,c_2}: F(c_1)\otimes_{\mathcal{D}} F(c_2)\rightarrow F(c_1\otimes_{\mathcal{C}} c_2),$	$\varphi_{c_1,c_2}: F(c_1)\otimes_{\mathcal{D}} F(c_2)\rightarrow F(c_1\otimes_{\mathcal{C}} c_2),$
fong-spivak-invitation_ E08089	042	■0.997 ●N +0	\lceil x / 3\rceil\leq y\text{ if and only if } x\leq 3y.	$\lceil x/3\rceil\leq y \text{ if and only if } x\leq 3y.$	$\lceil x/3\rceil\leq y \text{ if and only if } x\leq 3y.$
Conf. MathPix confidence: ■≥0.9 ▣0.5–0.9 ■<0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation-EQ0053	087	0.997 C +7	$\left(\begin{array}{cc}\text{ false } & \text{ false } \\ \text{ false } & \text{ true } \end{array}\right) * \left(\begin{array}{ccc}\text{ true } & \text{ true } & \text{ false } \\ \text{ true } & \text{ false } & \text{ true } \end{array}\right) = \left(\begin{array}{ccc}\text{ false } & \text{ false } & \text{ false } \\ \text{ true } & \text{ false } & \text{ true } \\ \text{ true } & \text{ true } & \text{ true } \end{array}\right)$	$\begin{pmatrix} \text{false} & \text{false} \\ \text{false} & \text{true} \\ \text{true} & \text{true} \end{pmatrix} * \begin{pmatrix} \text{true} & \text{true} & \text{false} \\ \text{true} & \text{false} & \text{true} \end{pmatrix} = \begin{pmatrix} \text{false} & \text{false} & \text{false} \\ \text{true} & \text{false} & \text{true} \\ \text{true} & \text{true} & \text{true} \end{pmatrix}$	
fong-spivak-invitation-EQ0231	310	0.997 N +0	$[], \quad \text{quad}[a], \quad \text{quad}[b], \quad \text{quad}[a, a], \quad \text{quad}[a, b], \quad \text{quad} \ldots, \quad \text{quad}[b, a, b, b, a, b, a, a, a, a], \quad \text{quad} \ldots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$	$[], \quad [a], \quad [b], \quad [a, a], \quad [a, b], \quad \dots, \quad [b, a, b, b, a, b, a, a, a, a], \quad \dots$
fong-spivak-invitation-EQ0049	086	0.997 N +1	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad \text{quad} X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v)$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v)$	$d(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v), \quad X(U, V) := \bigwedge_{u \in U} \bigvee_{v \in V} X(u, v).$
fong-spivak-invitation-EQ0215	300	0.997	$\left(\mathcal{X}^{\mathrm{op}} \times y\right)((x, y), (x', y')) \leq \mathcal{V}(\Phi(x, y), \Phi(x', y')).$	$(\mathcal{X}^{\mathrm{op}} \times y)((x, y), (x', y')) \leq \mathcal{V}(\Phi(x, y), \Phi(x', y')).$	$(\mathcal{X}^{\mathrm{op}} \times \mathcal{Y})((x, y), (x', y')) \leq \mathcal{V}(\Phi(x, y), \Phi(x', y')).$
fong-spivak-invitation-EQ0108	169	0.998 N +0	$a \subset \bullet \supset b$	$a \subset \bullet \supset b$	$a \curvearrowright \bullet \curvearrowleft b$
fong-spivak-invitation-EQ0166 (7.17)	260	0.998 K +0	$\left(\left \Omega^S\right ,\leq^S\right).$	$\left(\left \Omega^S\right ,\leq^S\right).$	$\left(\left \Omega^S\right ,\leq^S\right).$
fong-spivak-invitation-EQ0011	045	0.998 C +14	$g(\bigwedge A) \cong \bigwedge g(A).$	$g(\bigwedge A) \cong \bigwedge g(A).$	$g\left(\bigwedge A\right) \cong \bigwedge g(A).$
fong-spivak-invitation-EQ0106	166	0.998 N +0	$\iota^{\mathrm{\prime\prime}}(x) = \begin{cases} \iota(x) & \text{if } \iota(x) \in I, \\ \iota'(x) & \text{if } \iota(x) \in \underline{n}, \\ \iota'(x) & \text{if } x \in O'. \end{cases}$	$\iota''(x) = \begin{cases} \iota(x) & \text{if } \iota(x) \in I, \\ \iota'(\iota(x)) & \text{if } \iota(x) \in \underline{n}, \\ \iota'(x) & \text{if } x \in O'. \end{cases}$	$\iota''(x) = \begin{cases} \iota(x) & \text{if } \iota(x) \in I, \\ \iota'(\iota(x)) & \text{if } \iota(x) \in \underline{n}, \\ \iota'(x) & \text{if } x \in O'. \end{cases}$
fong-spivak-invitation-EQ0008 (1.6)	041	0.999 C +3	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$	$f(p) \leq q \quad \text{if and only if} \quad p \leq g(q).$
fong-spivak-invitation-EQ0212	298	0.999 W +0	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4).$	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4).$	$(c_1, d_1) \xrightarrow{(f_1, g_1)} (c_2, d_2) \xrightarrow{(f_2, g_2)} (c_3, d_3) \xrightarrow{(f_3, g_3)} (c_4, d_4).$
fong-spivak-invitation-EQ0232	310	0.999 N +0	$G := \{\rho_{m, n} \colon m \rightarrow n \mid m, n \in \mathbb{N}\}, \quad \text{quad } s \left(\rho_{m, n} \right) := m, \quad \text{quad } t \left(\rho_{m, n} \right) := n ;$	$G := \{\rho_{m, n} \colon m \rightarrow n \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m, n}) := m, \quad t(\rho_{m, n}) := n;$	$G := \{\rho_{m, n} \colon m \rightarrow n \mid m, n \in \mathbb{N}\}, \quad s(\rho_{m, n}) := m, \quad t(\rho_{m, n}) := n;$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0253	318	0.999 W -2	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \dots + 1_R}_{m \text{ summands}}) * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) \\ &= 1_R * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) + \dots + 1_R * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) \\ &= \underbrace{1_R + \dots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \dots + 1_R}_{m \text{ summands}}) * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) \\ &= 1_R * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) + \dots + 1_R * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) \\ &= \underbrace{1_R + \dots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$	$\begin{aligned} f(m) *_R f(n) &= (\underbrace{1_R + \dots + 1_R}_{m \text{ summands}}) * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) \\ &= 1_R * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) + \dots + 1_R * (\underbrace{1_R + \dots + 1_R}_{n \text{ summands}}) \\ &= \underbrace{1_R + \dots + 1_R}_{mn \text{ summands}} = f(m * n). \end{aligned}$
fong-spivak-invitation_EQ0146	230	0.999 C +17	$(f \circ \mathrm{id}) := (\mathrm{id} \otimes, \dots, \otimes \mathrm{id} \otimes g \otimes \mathrm{id} \otimes, \dots, \otimes \mathrm{id}); f,$	$(f \circ ig) := (\mathrm{id} \otimes, \dots, \otimes \mathrm{id} \otimes g \otimes \mathrm{id} \otimes, \dots, \otimes \mathrm{id}); f,$	$(f \circ_i g) := (\mathrm{id} \otimes, \dots, \otimes \mathrm{id} \otimes g \otimes \mathrm{id} \otimes, \dots, \otimes \mathrm{id}) \circ f,$
fong-spivak-invitation_EQ0015	047	0.999 K +0	$f(\text{left}(g(\text{left}(q_{\{0\}})\text{right}))\text{right}) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$	$f(g(q_0)) = \bigwedge \{q \in Q \mid g(q_0) \leq g(q)\} \leq \bigwedge \{q_0\} = q_0,$
fong-spivak-invitation_EQ0081 (4.3)	134	0.999 K +0	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$	$b \wedge c \leq d \quad \text{iff} \quad b \leq (c \Rightarrow d).$
fong-spivak-invitation_EQ0091 (4.10)	142	0.999	$\begin{aligned} \mathcal{P}(\text{left}(p, p_{\{1\}})\text{right}) \otimes \Phi(\text{left}(p_{\{1\}}, q)\text{right}) &= \mathcal{P}(\text{left}(p, p_{\{1\}})\text{right}) \otimes \Phi(\text{left}(p_{\{1\}}, q)\text{right}) \otimes I \\ &\leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes Q(q, q) \leq \Phi(p, q). \end{aligned}$	$\begin{aligned} \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) &= \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes I \\ &\leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes Q(q, q) \leq \Phi(p, q). \end{aligned}$	$\begin{aligned} \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) &= \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes I \\ &\leq \mathcal{P}(p, p_1) \otimes \Phi(p_1, q) \otimes Q(q, q) \leq \Phi(p, q). \end{aligned}$
fong-spivak-invitation_EQ0157	248	0.999 N +1	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}.$	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}.$	$\mathbf{Op}_1 := \{\emptyset, \{1\}, X\} \quad \text{and} \quad \mathbf{Op}_2 := \{\emptyset, \{2\}, X\}.$
fong-spivak-invitation_EQ0183	276	0.999 W +0	$f^*(P) = \{\{x \in \mathbb{Z} \mid x \leq 0\}, \{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text{and} \quad f^*(Q) = \{\{0\}, \{x \in \mathbb{Z} \mid x \neq 0\}\}.$	$f^*(P) = \{\{x \in \mathbb{Z} \mid x \leq 0\}, \{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text{and} \quad f^*(Q) = \{\{0\}, \{x \in \mathbb{Z} \mid x \neq 0\}\}.$	$f^*(P) = \{\{x \in \mathbb{Z} \mid x \leq 0\}, \{x \in \mathbb{Z} \mid x \geq 1\}\} \quad \text{and} \quad f^*(Q) = \{\{0\}, \{x \in \mathbb{Z} \mid x \neq 0\}\}.$
fong-spivak-invitation_EQ0245	316	0.999 N -1	$B + C := \{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid (w, x) \in B \text{ and } (y, z) \in C\}.$	$B + C := \{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid (w, x) \in B \text{ and } (y, z) \in C\}.$	$B + C := \{(w, y, x, z) \in R^{m+p} \times R^{n+q} \mid (w, x) \in B \text{ and } (y, z) \in C\}.$
fong-spivak-invitation_EQ0039	078	0.999 W -2	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f); \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f); \end{array}$	$\begin{array}{l} I_W \leq f(I_V) \quad (f \text{ is monoidal monotone}) \\ \leq f(\mathcal{C}(c, c)) \quad (\mathcal{C} \text{ is } \mathcal{V}\text{-category}) \\ = \mathcal{C}_f(c, c) \quad (\text{definition of } \mathcal{C}_f); \end{array}$

Conf. MathPix confidence:

0.999

0.5-0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged;
Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0142 (6.16)	221	■ 0.999 ● N +0	$\begin{aligned} \operatorname{Circ}(V) \times \operatorname{Circ}(V') &\longrightarrow \operatorname{Circ}(V + V') \\ ((V, A, s, t, \ell), (V', A', s', t', \ell')) &\longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']) \end{aligned}$	$\psi_{V,V'} : \operatorname{Circ}(V) \times \operatorname{Circ}(V') \longrightarrow \operatorname{Circ}(V + V') ;$ $((V, A, s, t, \ell), (V', A', s', t', \ell')) \longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']) .$	$\psi_{V,V'} : \operatorname{Circ}(V) \times \operatorname{Circ}(V') \longrightarrow \operatorname{Circ}(V + V') ;$ $((V, A, s, t, \ell), (V', A', s', t', \ell')) \longmapsto (V + V', A + A', s + s', t + t', [\ell, \ell']) .$
fong-spivak-invitation_EQ0052 (2.26)	087	■ 1.000 ● N -1	$(M * N)(x, z) := \bigvee_{y \in Y} M(x, y) \otimes N(y, z)$	$(M * N)(x, z) := \bigvee_{y \in Y} M(x, y) \otimes N(y, z).$	$(M * N)(x, z) := \bigvee_{y \in Y} M(x, y) \otimes N(y, z).$
fong-spivak-invitation_EQ0054	087	■ 1.000 ● W +2	$I_X(x, y) := \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$	$I_X(x, y) := \begin{cases} I & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$	$I_X(x, y) := \begin{cases} I & \text{if } x = y, \\ 0 & \text{if } x \neq y. \end{cases}$
fong-spivak-invitation_EQ0063	113	■ 1.000 ● C -3	$\alpha_{\text{Vertex}} : G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{Arrow}) \rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}} : G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{Arrow}) \rightarrow H(\text{Arrow}),$	$\alpha_{\text{Vertex}} : G(\text{Vertex}) \rightarrow H(\text{Vertex}) \quad \text{and} \quad \alpha_{\text{Arrow}} : G(\text{Arrow}) \rightarrow H(\text{Arrow}),$
fong-spivak-invitation_EQ0067 (3.12)	116	■ 1.000 ● N +0	$\text{Gr} \xrightarrow{F} \text{DDS} \xrightarrow{I} \text{Set}.$	$\text{Gr} \xrightarrow{F} \text{DDS} \xrightarrow{I} \text{Set}.$	$\text{Gr} \xrightarrow{F} \text{DDS} \xrightarrow{I} \text{Set}.$
fong-spivak-invitation_EQ0181	273	■ 1.000 ● K +0	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4.$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4.$	$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3, \quad 4 \leq 4.$
fong-spivak-invitation_EQ0220	301	■ 1.000 ● N +1	$\begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix}$	$\begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix}.$	$= \begin{pmatrix} 17 & 20 & 20 \\ 11 & 14 & 14 \\ 14 & 17 & 17 \\ 12 & 9 & 15 \end{pmatrix}.$
fong-spivak-invitation_EQ0258	324	■ 1.000 ●	$\begin{aligned} \varphi_{S',T'}((\operatorname{im} f \times \operatorname{im} g)(A \times B)) &= \varphi_{S',T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\}) \\ &= \{(\varphi_{S,T}(f(a)), \varphi_{S,T}(g(b))) \mid a \in A, b \in B\} \\ &= \varphi_{S,T}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\}) \\ &= \varphi_{S,T}(\varphi_{S,T}(A \times B)) \end{aligned}$	$\varphi_{S',T'}((\operatorname{im} f \times \operatorname{im} g)(A \times B)) = \varphi_{S',T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \{(\varphi_{S,T}(f(a)), \varphi_{S,T}(g(b))) \mid a \in A, b \in B\}$ $= \varphi_{S,T}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \varphi_{S,T}(\varphi_{S,T}(A \times B)).$	$\varphi_{S',T'}((\operatorname{im} f \times \operatorname{im} g)(A \times B)) = \varphi_{S',T'}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \{(\varphi_{S,T}(f(a)), \varphi_{S,T}(g(b))) \mid a \in A, b \in B\}$ $= \varphi_{S,T}(\{f(a) \mid a \in A\} \times \{g(b) \mid b \in B\})$ $= \varphi_{S,T}(\varphi_{S,T}(A \times B)).$
fong-spivak-invitation_EQ0001 (1.4)	020	■ 1.000 ● K +0	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$	$\Phi(A) \vee \Phi(B) \leq \Phi(A \vee B).$
fong-spivak-invitation_EQ0004	026	■ 1.000 ● K +0	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d.$	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d.$	$f(11)=a, \quad f(12)=a, \quad f(13)=b, \quad f(21)=c, \quad f(22)=d, \quad f(23)=d.$
fong-spivak-invitation_EQ0006	037	■ 1.000 ● N +0	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$	$\left\{ \frac{1}{n+1} \mid n \in \mathbb{N} \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \subseteq \mathbb{R}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0012	045	■ 1.000 ● C +7	$f(\bigvee A) \cong \bigvee f(A)$	$f(\bigvee A) \cong \bigvee f(A)$	$f\left(\bigvee A\right) \cong \bigvee f(A).$
fong-spivak-invitation_EQ0013 (1.8)	046	■ 1.000 ● K +0	$f(p) := \bigwedge \{q \in Q \mid p \leq g(q)\} \ ;$	$f(p) := \bigwedge \{q \in Q \mid p \leq g(q)\};$	$f(p) := \bigwedge \{q \in Q \mid p \leq g(q)\};$
fong-spivak-invitation_EQ0014	047	■ 1.000 ● N +0	$p_{\{0\}} \leq \bigwedge \left\{ g(q) \in P \mid p_{\{0\}} \leq g(q) \right\} \cong g \left(\bigwedge \left\{ q \in Q \mid p_{\{0\}} \leq g(q) \right\} \right) = g \left(f \left(p_{\{0\}} \right) \right),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g \left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\} \right) = g \left(f \left(p_0 \right) \right),$	$p_0 \leq \bigwedge \{g(q) \in P \mid p_0 \leq g(q)\} \cong g \left(\bigwedge \{q \in Q \mid p_0 \leq g(q)\} \right) = g \left(f \left(p_0 \right) \right),$
fong-spivak-invitation_EQ0019 (2.2)	055	■ 1.000 ● N +0	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$	$m * e = m, \quad e * m = m, \quad (m * n) * p = m * (n * p)$
fong-spivak-invitation_EQ0026 (2.8)	060	■ 1.000 ● N +0	$t \leq v + w, \quad w + u \leq x + z, \quad v + x \leq y,$	$t \leq v + w, \quad w + u \leq x + z, \quad v + x \leq y,$	$t \leq v + w, \quad w + u \leq x + z, \quad v + x \leq y,$
fong-spivak-invitation_EQ0027 (2.9)	060	■ 1.000 ● N +0	$t + u \leq y + z \ .$	$t + u \leq y + z.$	$t + u \leq y + z.$
fong-spivak-invitation_EQ0028 (2.10)	060	■ 1.000 ● N +0	$t + u \leq v + w + u \leq v + x + z \leq y + z \ .$	$t + u \leq v + w + u \leq v + x + z \leq y + z.$	$t + u \leq v + w + u \leq v + x + z \leq y + z.$
fong-spivak-invitation_EQ0029	061	■ 1.000 ● K +0	$\mathrm{H}_2 \mathrm{O}, \quad \mathrm{NaCl}, \quad 2 \mathrm{NaOH}, \quad \mathrm{CH}_4 + 3 \mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$	$\mathrm{H}_2\mathrm{O}, \quad \mathrm{NaCl}, \quad 2\mathrm{NaOH}, \quad \mathrm{CH}_4 + 3\mathrm{O}_2$
fong-spivak-invitation_EQ0030	061	■ 1.000 ● N +0	$2 \mathrm{H}_2 \mathrm{O} + 2 \mathrm{Na} \rightarrow 2 \mathrm{NaOH} + \mathrm{H}_2.$	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2.$	$2\mathrm{H}_2\mathrm{O} + 2\mathrm{Na} \rightarrow 2\mathrm{NaOH} + \mathrm{H}_2.$
fong-spivak-invitation_EQ0031 (2.11)	062	■ 1.000 ● N +0	$y + k \rightarrow y^{\prime} + k^{\prime}, \quad x + y^{\prime} \rightarrow z^{\prime}, \quad z^{\prime} + k^{\prime} \rightarrow z + k.$	$y + k \rightarrow y' + k', \quad x + y' \rightarrow z', \quad z' + k' \rightarrow z + k.$	$y + k \rightarrow y' + k', \quad x + y' \rightarrow z', \quad z' + k' \rightarrow z + k.$
fong-spivak-invitation_EQ0032 (2.12)	062	■ 1.000 ● K +0	$x + y + k \rightarrow x + y^{\prime} + k^{\prime} \rightarrow z^{\prime} + k^{\prime} \rightarrow z + k.$	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$	$x + y + k \rightarrow x + y' + k' \rightarrow z' + k' \rightarrow z + k.$
fong-spivak-invitation_EQ0033 (2.14)	063	■ 1.000 ● K +0	$\mathrm{H}_2 \mathrm{O} + \mathrm{NaCl} \rightarrow \mathrm{H}_2 \mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow \mathrm{H}_2\mathrm{O}$	$\mathrm{H}_2\mathrm{O} + \mathrm{NaCl} \rightarrow \mathrm{H}_2\mathrm{O}$
fong-spivak-invitation_EQ0035	069	■ 1.000 ● W +1	$d(x) := \left[\begin{array}{l} \text{false} \\ \text{true} \end{array} \right] \text{ if } x > 0, \quad u(x) := \left[\begin{array}{l} \text{false} \\ \text{true} \end{array} \right] \text{ if } x = \infty,$	$d(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x > 0, \\ \text{true} & \text{if } x = 0, \end{array} \right. \quad u(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x = \infty, \\ \text{true} & \text{if } x < \infty. \end{array} \right.$	$d(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x > 0, \\ \text{true} & \text{if } x = 0, \end{array} \right. \quad u(x) := \left\{ \begin{array}{ll} \text{false} & \text{if } x = \infty, \\ \text{true} & \text{if } x < \infty. \end{array} \right.$
fong-spivak-invitation_EQ0036	073	■ 1.000 ●	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$	$d_L(U, V) := \sup_{u \in U} \inf_{v \in V} d(u, v),$
fong-spivak-invitation_EQ0038 (2.20)	076	■ 1.000 ● N +0	$M_Y := \begin{bmatrix} \nearrow & x & y & z \\ x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{bmatrix}$	$M_Y := \begin{array}{c ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$	$M_Y := \begin{array}{c ccc} \nearrow & x & y & z \\ \hline x & 0 & 4 & 3 \\ y & 3 & 0 & \infty \\ z & \infty & 4 & 0 \end{array}$
fong-spivak-invitation_EQ0041 (2.21)	078	■ 1.000 ● N +0	$f(x) := \begin{cases} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{cases}$	$f(x) := \left\{ \begin{array}{ll} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{array} \right.$	$f(x) := \left\{ \begin{array}{ll} \text{true} & \text{if } x = 0, \\ \text{false} & \text{if } x > 0. \end{array} \right.$
fong-spivak-invitation_EQ0044 (2.23)	082	■ 1.000 ● C +8	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$	$(a \otimes v) \leq w \quad \text{iff} \quad a \leq (v \multimap w)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0045	083	■ 1.000 ● N +1	$\text{a}+\text{x} \geq \text{y} \quad \text{iff} \quad \text{a} \geq \text{y}-\text{x} \quad \text{iff} \quad \text{max}(0,\text{a}) \geq \text{max}(0,\text{y}-\text{x}) \quad \text{iff} \quad \text{a} \geq (\text{x} \multimap \text{y})$	$a+x \geq y \quad \text{iff} \quad a \geq y-x \quad \text{iff} \quad \max(0,a) \geq \max(0,y-x) \quad \text{iff} \quad a \geq (x \multimap y)$	$a+x \geq y \quad \text{iff} \quad a \geq y-x \quad \text{iff} \quad \max(0,a) \geq \max(0,y-x) \quad \text{iff} \quad a \geq (x \multimap y)$
fong-spivak-invitation_EQ0050	086	■ 1.000 ● N +0	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a$	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a$	$v \multimap w := \bigvee_{\{a \in V \mid a \otimes v \leq w\}} a$
fong-spivak-invitation_EQ0051 (2.25)	087	■ 1.000 ● N +0	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k)$	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k)$	$(M * N)(i, k) = \sum_j M(i, j) * N(j, k)$
fong-spivak-invitation_EQ0058	107	■ 1.000 ● S +0	$f^{\prime} ; h^{\prime} = F(f) ; F(h) = F(f ; h) = F(g ; i) = F(g) ; F(i) = g^{\prime} ; i^{\prime}$	$f';h' = F(f);F(h) = F(f;h) = F(g;i) = F(g);F(i) = g';i'$	$f' \circ h' = F(f) \circ F(h) = F(f \circ h) = F(g \circ i) = F(g) \circ F(i) = g' \circ i'$
fong-spivak-invitation_EQ0060 (3.9)	110	■ 1.000 ● C +8	$F(f) ; \alpha_d = \alpha_c ; G(f)$	$F(f); \alpha_d = \alpha_c; G(f)$	$F(f) \circ \alpha_d = \alpha_c \circ G(f)$
fong-spivak-invitation_EQ0078	124	■ 1.000 ● N +2	$X \times Y := \{(x, y) \mid x \in X, y \in Y\}$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\}$	$X \times Y := \{(x, y) \mid x \in X, y \in Y\}$
fong-spivak-invitation_EQ0084 (4.5)	137	■ 1.000 ● N +2	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17$	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17$	$\Phi(B, x) = 11, \quad \Phi(A, z) = 20, \quad \Phi(C, y) = 17$
fong-spivak-invitation_EQ0086 (4.7)	140	■ 1.000 ● N +1	$(\Phi ; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r)$	$(\Phi; \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r)$	$(\Phi \circ \Psi)(p, r) := \bigvee_{q \in Q} \Phi(p, q) \wedge \Psi(q, r)$
fong-spivak-invitation_EQ0087	141	■ 1.000 ● C +13	$(\Phi ; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r))$	$(\Phi; \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r))$	$(\Phi \circ \Psi)(p, r) = \bigvee_{q \in Q} (\Phi(p, q) \otimes \Psi(q, r))$
fong-spivak-invitation_EQ0098	150	■ 1.000 ● N +1	$\alpha_{s,t,u} : \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\}$	$\alpha_{s,t,u} : \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\}$	$\alpha_{s,t,u} : \{(s, (t, u)) \mid s \in S, t \in T, u \in U\} \xrightarrow{\cong} \{((s, t), u) \mid s \in S, t \in T, u \in U\}$
fong-spivak-invitation_EQ0102	158	■ 1.000 ● N -1	$(\Phi \times \Psi) (\left(\left(x_1, y_1 \right), \left(x_2, y_2 \right) \right) := \Phi \left(x_1, x_2 \right) \otimes \Psi \left(y_1, y_2 \right)$	$(\Phi \times \Psi) ((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2)$	$(\Phi \times \Psi)((x_1, y_1), (x_2, y_2)) := \Phi(x_1, x_2) \otimes \Psi(y_1, y_2)$
fong-spivak-invitation_EQ0105	164	■ 1.000 ● N -1	$R ; S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists (j \in \underline{n}). (i, j) \in R \text{ and } (j, k) \in S\}$	$R; S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists (j \in \underline{n}). (i, j) \in R \text{ and } (j, k) \in S\}$	$R \circ S := \{(i, k) \in \underline{m} \times \underline{p} \mid \exists (j \in \underline{n}). (i, j) \in R \text{ and } (j, k) \in S\}$
fong-spivak-invitation_EQ0109 (5.5)	170	■ 1.000 ● N +0	$G := \left\{ \rho_{m,n} \mid m, n \in \mathbb{N} \right\}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$	$G := \{ \rho_{m,n} \mid m, n \in \mathbb{N} \}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$	$G := \{ \rho_{m,n} \mid m, n \in \mathbb{N} \}, \quad s(\rho_{m,n}) := m, \quad t(\rho_{m,n}) := n,$
fong-spivak-invitation_EQ0115	178	■ 1.000 ● K +0	$A = \left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right), \quad B = \left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array} \right)$	$A = \left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right), \quad B = \left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array} \right)$	$A = \left(\begin{array}{lll} 3 & 3 & 1 \\ 2 & 0 & 4 \end{array} \right), \quad B = \left(\begin{array}{llll} 2 & 5 & 6 & 1 \end{array} \right)$
fong-spivak-invitation_EQ0117	181	■ 1.000 ●	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k)$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k)$	$\sum_{j \in \underline{n}} \alpha(i, j) * \beta(j, k)$
fong-spivak-invitation_EQ0120 (5.11)	188	■ 1.000 ● K +0	$\mathrm{B}(g) = \{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n$	$B(g) = \{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n$	$B(g) = \{(x, S(g)(x)) \mid x \in R^m\} \subseteq R^m \times R^n$
fong-spivak-invitation_EQ0121 (5.12)	188	■ 1.000 ● K +0	$\mathrm{B} \left(g^{\mathrm{op}} \right) := \{ (S(g)(x), x) \mid x \in R^m \} \subseteq R^n \times R^m$	$B(g^{\mathrm{op}}) := \{(S(g)(x), x) \mid x \in R^m\} \subseteq R^n \times R^m$	$B(g^{\mathrm{op}}) := \{(S(g)(x), x) \mid x \in R^m\} \subseteq R^n \times R^m$
fong-spivak-invitation_EQ0125	190	■ 1.000 ● N +0	$\mathrm{B} \left((g^{\mathrm{op}}) ; h \right) = \{ (S(g)(x), S(h)(x)) \mid x \in R^m \}$	$B((g^{\mathrm{op}}); h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\}$	$B((g^{\mathrm{op}}) \circ h) = \{(S(g)(x), S(h)(x)) \mid x \in R^m\}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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fong-spivak-invitation_ EQ0188	288	1.000 K +0	$A=(A\cap\bar{B})\cup(A\cap B)\subseteq\bar{B}\cup C$.	$A=(A\cap\bar{B})\cup(A\cap B)\subseteq\bar{B}\cup C.$	$A=(A\cap\bar{B})\cup(A\cap B)\subseteq\bar{B}\cup C.$
fong-spivak-invitation_ EQ0190	288	1.000 W +0	$\left(\begin{array}{ll}1&0\\0&1\end{array}\right),\quad\left(\begin{array}{cc}\text{true}&\text{false}\\\text{false}&\text{true}\end{array}\right),\quad\text{and}\quad\left(\begin{array}{cc}0&\infty\\\infty&0\end{array}\right).$	$\left(\begin{array}{cc}1&0\\0&1\end{array}\right),\quad\left(\begin{array}{cc}\text{true}&\text{false}\\\text{false}&\text{true}\end{array}\right),\quad\text{and}\quad\left(\begin{array}{cc}0&\infty\\\infty&0\end{array}\right).$	$\left(\begin{array}{cc}1&0\\0&1\end{array}\right),\quad\left(\begin{array}{cc}\text{true}&\text{false}\\\text{false}&\text{true}\end{array}\right),\quad\text{and}\quad\left(\begin{array}{cc}0&\infty\\\infty&0\end{array}\right).$
fong-spivak-invitation_ EQ0194	289	1.000 N +1	$\begin{gathered}M_X=\left(\begin{array}{cccc}0&\infty&3&\infty\\2&0&\infty&5\\0&\infty&\infty&6\\0&\infty&\infty&6\end{array}\right),\\M_X^2=\left(\begin{array}{cccc}0&6&3&\infty\\2&0&5&5\\5&3&0&8\\11&9&6&0\end{array}\right),\\M_X^3=M_X^4=\left(\begin{array}{cccc}0&6&3&11\\2&0&5&5\\5&3&0&8\\11&9&6&0\end{array}\right).\end{gathered}$	$M_X=\left(\begin{array}{cccc}0&\infty&3&\infty\\2&0&\infty&5\\0&\infty&\infty&6\\0&\infty&\infty&6\end{array}\right),\quad M_X^2=\left(\begin{array}{cccc}0&6&3&\infty\\2&0&5&5\\5&3&0&8\\11&9&6&0\end{array}\right),\\M_X^3=M_X^4=\left(\begin{array}{cccc}0&6&3&11\\2&0&5&5\\5&3&0&8\\11&9&6&0\end{array}\right).$	$M_X=\left(\begin{array}{cccc}0&\infty&3&\infty\\2&0&\infty&5\\0&\infty&\infty&6\\0&\infty&\infty&6\end{array}\right),\quad M_X^2=\left(\begin{array}{cccc}0&6&3&\infty\\2&0&5&5\\5&3&0&8\\11&9&6&0\end{array}\right),\\M_X^3=M_X^4=\left(\begin{array}{cccc}0&6&3&11\\2&0&5&5\\5&3&0&8\\11&9&6&0\end{array}\right).$
fong-spivak-invitation_ EQ0196	290	1.000 N +0	$(p;q):=\left(v,a_1,\ldots,a_m,b_1,\ldots,b_n\right)$.	$(p;q):=(v,a_1,\ldots,a_m,b_1,\ldots,b_n).$	$(p\circledcirc q):=(v,a_1,\ldots,a_m,b_1,\ldots,b_n).$
fong-spivak-invitation_ EQ0197	290	1.000 K +0	$\left(v,a_1,\ldots,a_m,b_1,\ldots,b_n,c_1,\ldots,c_o\right)$.	$(v,a_1,\ldots,a_m,b_1,\ldots,b_n,c_1,\ldots,c_o).$	$(v,a_1,\ldots,a_m,b_1,\ldots,b_n,c_1,\ldots,c_o).$
fong-spivak-invitation_ EQ0203	292	1.000 W +0	$\begin{array}{lll}(1,1)&(1,2)&(1,3)\\(2,1)&(2,2)&(2,3)\\(3,1)&(3,2)&(3,3)\end{array}$	$\begin{array}{ccc}(1,1)&(1,2)&(1,3)\\(2,1)&(2,2)&(2,3)\\(3,1)&(3,2)&(3,3)\end{array}$	$\begin{array}{ccc}(1,1)&(1,2)&(1,3)\\(2,1)&(2,2)&(2,3)\\(3,1)&(3,2)&(3,3)\end{array}$
fong-spivak-invitation_ EQ0204	292	1.000 N +0	$f^{-1}(1)=b,\quad f^{-1}(2)=a,\quad f^{-1}(3)=c$.	$f^{-1}(1)=b,\quad f^{-1}(2)=a,\quad f^{-1}(3)=c.$	$f^{-1}(1)=b,\quad f^{-1}(2)=a,\quad f^{-1}(3)=c.$
fong-spivak-invitation_ EQ0205	293	1.000 W +0	$\begin{gathered}A^{\prime}\mapsto A,\quad f^{\prime}\mapsto f,\quad g^{\prime}\mapsto g,\quad h^{\prime}\mapsto h,\quad i^{\prime}\mapsto i,\\B^{\prime}\mapsto B,\quad h^{\prime}\mapsto h,\quad C^{\prime}\mapsto C,\quad i^{\prime}\mapsto i,\quad D^{\prime}\mapsto D.\end{gathered}$	$A'\mapsto A,\quad f'\mapsto f,\quad g'\mapsto g,\quad f';h'\mapsto f;h,\quad g';i'\mapsto f;h,\\B'\mapsto B,\quad h'\mapsto h,\quad C'\mapsto C,\quad i'\mapsto i,\quad D'\mapsto D.$	$A'\mapsto A,\quad f'\mapsto f,\quad g'\mapsto g,\quad f'\circledcirc h'\mapsto f\circledcirc h,\quad g'\circledcirc i'\mapsto f\circledcirc h,\\B'\mapsto B,\quad h'\mapsto h,\quad C'\mapsto C,\quad i'\mapsto i,\quad D'\mapsto D.$
fong-spivak-invitation_ EQ0208	293	1.000 S +0	$((F;G);H)(c)=H((F;G)(c))=H(G(F(c)))=(G;H)(F(c))=(F;(G;H))(c).$	$((F;G);H)(c)=H((F;G)(c))=H(G(F(c)))=(G;H)(F(c))=(F;(G;H))(c).$	$((F\circledcirc G)\circledcirc H)(c)=H((F\circledcirc G)(c))=H(G(F(c)))=(G\circledcirc H)(F(c))=(F\circledcirc (G\circledcirc H))(c).$
fong-spivak-invitation_ EQ0211	297	1.000	$\left(f_1;f_2\right)^B(g)=g;\left(f_1;f_2\right)=\left(g;f_1\right);f_2=\left(f_1^B;f_2^B\right)(g)$	$\left(f_1;f_2\right)^B(g)=g;\left(f_1;f_2\right)=\left(g;f_1\right);f_2=\left(f_1^B;f_2^B\right)(g)$	$\left(f_1\circledcirc f_2\right)^B(g)=g\circledcirc\left(f_1\circledcirc f_2\right)=\left(g\circledcirc f_1\right)\circledcirc f_2=\left(f_1^B\circledcirc f_2^B\right)(g)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation-EQ0246	316	■1.000 ●	$\begin{aligned} \mathrm{B}(g) &= \{(x, z) \in R^m \times R^n \mid S(g)(x) = z\}, \\ \mathrm{B}(h^{\mathrm{op}}) &= \{(z, y) \in R^n \times R^\ell \mid z = S(h)(y)\}, \end{aligned}$	$\begin{aligned} \mathrm{B}(g) &= \{(x, z) \in R^m \times R^n \mid S(g)(x) = z\}, \\ \mathrm{B}(h^{\mathrm{op}}) &= \{(z, y) \in R^n \times R^\ell \mid z = S(h)(y)\}, \end{aligned}$	$\begin{aligned} \mathrm{B}(g) &= \{(x, z) \in R^m \times R^n \mid S(g)(x) = z\}, \\ \mathrm{B}(h^{\mathrm{op}}) &= \{(z, y) \in R^n \times R^\ell \mid z = S(h)(y)\}, \end{aligned}$
fong-spivak-invitation-EQ0249	316	■1.000 ●W -1	$\begin{aligned} \mathrm{B}\left(g^{\mathrm{op}}\right) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}, \\ \mathrm{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}, \end{aligned}$	$\begin{aligned} \mathrm{B}\left(g^{\mathrm{op}}\right) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}, \\ \mathrm{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}, \end{aligned}$	$\begin{aligned} \mathrm{B}(g^{\mathrm{op}}) &= \{(y, x) \in R^n \times R^m \mid y = S(g)(x)\}, \\ \mathrm{B}(h) &= \{(x, z) \in R^m \times R^p \mid S(h)(x) = z\}, \end{aligned}$
fong-spivak-invitation-EQ0251	316	■1.000 ●N +1	$\mathrm{B}\left(\left(\left(g^{\mathrm{op}}\right); h\right) ; h\right)=\left(\left(S(g)(x), S(h)(x)\right) \mid x \in R^m\right) \text { . }$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right); h\right)=\{(S(g)(x), S(h)(x)) \mid x \in R^m\}.$	$\mathrm{B}\left(\left(g^{\mathrm{op}}\right) \circ h\right)=\{(S(g)(x), S(h)(x)) \mid x \in R^m\}.$
fong-spivak-invitation-EQ0252	318	■1.000 ●C -12	$f(m)=f(\underbrace{1+1+\cdots+1}_m\text{ \textit{ summands }})=\underbrace{f(1)+f(1)+\cdots+f(1)}_m\text{ \textit{ summands }}=\underbrace{1_{\mathrm{R}}+1_{\mathrm{R}}+\cdots+1_{\mathrm{R}}}_m\text{ \textit{ summands }} \text{ . }$	$f(m)=f(\underbrace{1+1+\cdots+1}_m\text{ \textit{ summands }})=\underbrace{f(1)+f(1)+\cdots+f(1)}_m\text{ \textit{ summands }}=\underbrace{1_R+1_R+\cdots+1_R}_m\text{ \textit{ summands }}.$	$f(m)=f(\underbrace{1+1+\cdots+1}_m\text{ \textit{ summands }})=\underbrace{f(1)+f(1)+\cdots+f(1)}_m\text{ \textit{ summands }}=\underbrace{1_R+1_R+\cdots+1_R}_m\text{ \textit{ summands }}.$
fong-spivak-invitation-EQ0254	318	■1.000 ●W +0	$\begin{aligned} \left[f, g \right] \colon A \sqcup B &\longrightarrow T \\ \text{apple } 1 &\longmapsto a \end{aligned}$	$\begin{aligned} \left[f, g \right] \colon A \sqcup B &\longrightarrow T \\ \text{apple } 1 &\longmapsto a \end{aligned}$	$\begin{aligned} \left[f, g \right] \colon A \sqcup B &\longrightarrow T \\ \text{apple } 1 &\longmapsto a \end{aligned}$
fong-spivak-invitation-EQ0261	338	■1.000 ●K +0	$U=\bigcup_{i\in I}\left\{x\in\mathbb{R}\mid a_{\mathrm{i}}<x<b_{\mathrm{i}}\right\}$	$U=\bigcup_{i\in I}\{x\in\mathbb{R}\mid a_i<x<b_i\}$	$U=\bigcup_{i\in I}\{x\in\mathbb{R}\mid a_i<x<b_i\}$
fong-spivak-invitation-EQ0262	338	■1.000 ●K +0	$U=\bigcup_{j\in J}\left\{x\in\mathbb{R}\mid a_{\mathrm{j}}<x<b_{\mathrm{j}}\right\}=\bigcup_{j\in J}\left(o_{\left[a_{\mathrm{j}},\,b_{\mathrm{j}}\right]}\cap\mathbb{R}\right)=\left(\bigcup_{j\in J}o_{\left[a_{\mathrm{j}},\,b_{\mathrm{j}}\right]}\right)\cap\mathbb{R}$	$U=\bigcup_{j\in J}\{x\in\mathbb{R}\mid a_j<x<b_j\}=\bigcup_{j\in J}(o_{[a_j,b_j]}\cap\mathbb{R})=\left(\bigcup_{j\in J}o_{[a_j,b_j]}\right)\cap\mathbb{R}$	$U=\bigcup_{j\in J}\{x\in\mathbb{R}\mid a_j<x<b_j\}=\bigcup_{j\in J}(o_{[a_j,b_j]}\cap\mathbb{R})=\left(\bigcup_{j\in J}o_{[a_j,b_j]}\right)\cap\mathbb{R}$
fong-spivak-invitation-EQ0064	114	— ●C +31	—	—	$G := \left[\begin{array}{c} 1 \\ \bullet \end{array} \xrightarrow{a} \begin{array}{c} 2 \\ \bullet \end{array} \xrightarrow{b} \begin{array}{c} 3 \\ \bullet \end{array} \right]$ $H := \begin{array}{c} 4 \\ \bullet \end{array} \xrightarrow[c]{d} \begin{array}{c} 5 \\ \bullet \end{array} \curvearrowright^e$
fong-spivak-invitation-EQ0065	114	— ●C +77	—	—	Arrow source target Vertex $G :=$ a 1 b 2 2 3 3 c 4 d 4 e 5 5

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Inline formulas that did not render (1 of 4061)

Identifier	Page	Conf.	LaTeX source	Rendered
fong-spivak-invitation_F02349	228	—	$\operatorname{cospan} \underline{a} \xrightarrow{\operatorname{id}_{\underline{a}}} \underline{a} \stackrel{\operatorname{id}_{\underline{a}}}{\rightarrow} \underline{a}$	(not rendered)
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.				

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																								
fong-spivak-invitation_Table_068	068	■0.972	$\begin{tabular}{t}{ l l l }\hline\text{Arrow } \text{\(a\)} \& \text{source } \text{\(s(a) \in V\)} \& \text{target } \text{\(t(a) \in V\)} \\\hline\text{\(a\)} \& 1 \& ? \\\hline\text{\(b\)} \& 1 \& 3 \\\hline\text{\(c\)} \& ? \& ? \\\hline d \& ? \& ? \\\hline\text{\(e\)} \& ? \& ? \\\hline\end{tabular}$	<table><tr><th>Arrow a</th><th>source $s(a) \in V$</th><th>target $t(a) \in V$</th></tr><tr><td>a</td><td>1</td><td>?</td></tr><tr><td>b</td><td>1</td><td>3</td></tr><tr><td>c</td><td>?</td><td>?</td></tr><tr><td>d</td><td>?</td><td>?</td></tr><tr><td>e</td><td>?</td><td>?</td></tr></table>	Arrow a	source $s(a) \in V$	target $t(a) \in V$	a	1	?	b	1	3	c	?	?	d	?	?	e	?	?	<table><tr><th>Arrow a</th><th>source $s(a) \in V$</th><th>target $t(a) \in V$</th></tr><tr><td>a</td><td>1</td><td>?</td></tr><tr><td>b</td><td>1</td><td>3</td></tr><tr><td>c</td><td>?</td><td>?</td></tr><tr><td>d</td><td>?</td><td>?</td></tr><tr><td>e</td><td>?</td><td>?</td></tr></table>	Arrow a	source $s(a) \in V$	target $t(a) \in V$	a	1	?	b	1	3	c	?	?	d	?	?	e	?	?																																				
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fong-spivak-invitation_Table_065	065	■1.000	$\begin{tabular}{t}{ l l l l l l }\hline\text{\(\\wedge\)} \& \text{false} \& \text{true} \& * \& 0 \& 1 \\\hline\text{false} \& \text{false} \& \text{false} \& 0 \& 0 \& 0 \\\hline\text{true} \& \text{false} \& \text{true} \& 1 \& 0 \& 1 \\\hline\end{tabular}$	<table><tr><th>$\wedge$</th><th>false</th><th>true</th><th>*</th><th>0</th><th>1</th></tr><tr><td>false</td><td>false</td><td>false</td><td>0</td><td>0</td><td>0</td></tr><tr><td>true</td><td>false</td><td>true</td><td>1</td><td>0</td><td>1</td></tr></table>	$\wedge$	false	true	*	0	1	false	false	false	0	0	0	true	false	true	1	0	1	<table><tr><th>$\wedge$</th><th>false</th><th>true</th><th>*</th><th>0</th><th>1</th></tr><tr><td>false</td><td>false</td><td>false</td><td>0</td><td>0</td><td>0</td></tr><tr><td>true</td><td>false</td><td>true</td><td>1</td><td>0</td><td>1</td></tr></table>	$\wedge$	false	true	*	0	1	false	false	false	0	0	0	true	false	true	1	0	1																																				
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fong-spivak-invitation_Table_071	071	■0.289	$\begin{tabular}{t}{ l l l l l l }\hline\text{\(\\leq\)} \& \text{\(p\)} \& \text{\(q\)} \& \text{\(r\)} \& \text{\(s\)} \& \text{\(t\)} \\\hline\text{true} \& \text{true} \& \text{true} \& \text{true} \& \text{true} \& \text{true} \\\hline\text{\(q\)} \& \text{false} \& \text{true} \& \text{false} \& \text{true} \& \text{true} \\\hline\text{\(r\)} \& \text{false} \& \text{false} \& \text{true} \& \text{true} \& \text{true} \\\hline\text{\(s\)} \& \text{false} \& \text{false} \& \text{false} \& \text{true} \& \text{true} \\\hline\text{\(t\)} \& \text{false} \& \text{false} \& \text{false} \& \text{false} \& \text{true} \\\hline\end{tabular}$	<table><tr><th>$\leq$</th><th>p</th><th>q</th><th>r</th><th>s</th><th>t</th></tr><tr><td>p</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>q</td><td>false</td><td>true</td><td>false</td><td>true</td><td>true</td></tr><tr><td>r</td><td>false</td><td>false</td><td>true</td><td>true</td><td>true</td></tr><tr><td>s</td><td>false</td><td>false</td><td>false</td><td>true</td><td>true</td></tr><tr><td>t</td><td>false</td><td>false</td><td>false</td><td>false</td><td>true</td></tr></table>	$\leq$	p	q	r	s	t	p	true	true	true	true	true	q	false	true	false	true	true	r	false	false	true	true	true	s	false	false	false	true	true	t	false	false	false	false	true	<table><tr><th>$\cdot \leq \cdot$</th><th>p</th><th>q</th><th>r</th><th>s</th><th>t</th></tr><tr><td>p</td><td>true</td><td>true</td><td>true</td><td>true</td><td>true</td></tr><tr><td>q</td><td>false</td><td>true</td><td>false</td><td>true</td><td>true</td></tr><tr><td>r</td><td>false</td><td>false</td><td>true</td><td>true</td><td>true</td></tr><tr><td>s</td><td>false</td><td>false</td><td>false</td><td>true</td><td>true</td></tr><tr><td>t</td><td>false</td><td>false</td><td>false</td><td>false</td><td>true</td></tr></table>	$\cdot \leq \cdot$	p	q	r	s	t	p	true	true	true	true	true	q	false	true	false	true	true	r	false	false	true	true	true	s	false	false	false	true	true	t	false	false	false	false	true
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_Table_075	075	0.623	$\begin{tabular}{t}{ l l l l }\hline \hspace{0.5cm} \nearrow \\ \& \hspace{0.5cm} (X\hspace{0.5cm}) \& \hspace{0.5cm} (y\hspace{0.5cm}) \& \hspace{0.5cm} z \hspace{0.5cm} \\ \hline y \& \hspace{0.5cm} 3 \& \hspace{0.5cm} 0 \& \hspace{0.5cm} 6 \hspace{0.5cm} \\ \hline z \& \hspace{0.5cm} 7 \& \hspace{0.5cm} 4 \& \hspace{0.5cm} 0 \hspace{0.5cm} \\ \hline \end{tabular}$	$\begin{array}{c ccc} d(\nearrow) & X & y & z \\ \hline x & 0 & 4 & 3 \\ \hline y & 3 & 0 & 6 \\ \hline z & 7 & 4 & 0 \end{array}$	$\begin{array}{c ccc} d(\nearrow) & x & y & z \\ \hline x & 0 & 4 & 3 \\ \hline y & 3 & 0 & 6 \\ \hline z & 7 & 4 & 0 \end{array}$
fong-spivak-invitation_Table_075	075	0.578	$\begin{tabular}{t}{ l l l l l }\hline \hspace{0.5cm} \nearrow \\ \& \hspace{0.5cm} (A\hspace{0.5cm}) \& \hspace{0.5cm} (B\hspace{0.5cm}) \& \hspace{0.5cm} (C\hspace{0.5cm}) \& \hspace{0.5cm} (D\hspace{0.5cm}) \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} ? \hspace{0.5cm} \& \hspace{0.5cm} ? \hspace{0.5cm} \& \hspace{0.5cm} ? \hspace{0.5cm} \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline B \& \hspace{0.5cm} 2 \& \hspace{0.5cm} ? \& \hspace{0.5cm} 5 \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline C \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline D \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline \end{tabular}$	$\begin{array}{c ccccc} d(\nearrow) & A & B & C & D \\ \hline A & 0 & ? & ? & ? \\ \hline B & 2 & ? & 5 & ? \\ \hline C & ? & ? & ? & ? \\ \hline D & ? & ? & ? & ? \end{array}$	$\begin{array}{c ccccc} d(\nearrow) & A & B & C & D \\ \hline A & 0 & ? & ? & ? \\ \hline B & 2 & ? & 5 & ? \\ \hline C & ? & ? & ? & ? \\ \hline D & ? & ? & ? & ? \end{array}$
fong-spivak-invitation_Table_076	076	0.600	$\begin{tabular}{t}{ l l l l l l }\hline \hspace{0.5cm} \nearrow \\ \multirrow{2}{*}{\& \hspace{0.5cm} \nearrow \& \hspace{0.5cm} (A\hspace{0.5cm}) \& \hspace{0.5cm} (B\hspace{0.5cm}) \& \hspace{0.5cm} C \& \hspace{0.5cm} D \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} (A\hspace{0.5cm}) \& \hspace{0.5cm} 0 \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} (M_{\hspace{0.5cm}X}\hspace{0.5cm})=\hspace{0.5cm} \hspace{0.5cm} \& \hspace{0.5cm} B \& \hspace{0.5cm} 2 \& \hspace{0.5cm} 0 \& \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} (\infty\hspace{0.5cm}) \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} C \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} D \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \& \hspace{0.5cm} ? \hspace{0.5cm} \\ \hline \end{tabular}$	(not rendered)	$M_X = \begin{array}{c ccccc} \nearrow & A & B & C & D \\ \hline A & 0 & ? & ? & ? \\ \hline B & 2 & 0 & \infty & ? \\ \hline C & ? & ? & ? & ? \\ \hline D & ? & ? & ? & ? \end{array}$
fong-spivak-invitation_Table_081	081	0.535	$\begin{tabular}{t}{ l l l l }\hline X \& \hspace{0.5cm} (A\hspace{0.5cm}) \& \hspace{0.5cm} (B\hspace{0.5cm}) \& \hspace{0.5cm} (C\hspace{0.5cm}) \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} (A\hspace{0.5cm}) \& \hspace{0.5cm} 0 \& \hspace{0.5cm} 2 \& \hspace{0.5cm} 5 \hspace{0.5cm} \\ \hline B \& \hspace{0.5cm} (\infty\hspace{0.5cm}) \& \hspace{0.5cm} 0 \& \hspace{0.5cm} 3 \hspace{0.5cm} \\ \hline C \& \hspace{0.5cm} (\infty\hspace{0.5cm}) \& \hspace{0.5cm} (\infty\hspace{0.5cm}) \& \hspace{0.5cm} 0 \hspace{0.5cm} \\ \hline \end{tabular}$	$\begin{array}{c ccc} X & A & B & C \\ \hline A & 0 & 2 & 5 \\ \hline B & \infty & 0 & 3 \\ \hline C & \infty & \infty & 0 \end{array}$	$\begin{array}{c ccc} X & A & B & C \\ \hline A & 0 & 2 & 5 \\ \hline B & \infty & 0 & 3 \\ \hline C & \infty & \infty & 0 \end{array}$
fong-spivak-invitation_Table_081	081	0.910	$\begin{tabular}{t}{ l l l }\hline y \& \hspace{0.5cm} (p\hspace{0.5cm}) \& \hspace{0.5cm} (q\hspace{0.5cm}) \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} (p\hspace{0.5cm}) \& \hspace{0.5cm} 0 \& \hspace{0.5cm} 5 \hspace{0.5cm} \\ \hline \& \hspace{0.5cm} (q\hspace{0.5cm}) \& \hspace{0.5cm} 8 \& \hspace{0.5cm} 0 \hspace{0.5cm} \\ \hline \end{tabular}$	$\begin{array}{c cc} y & p & q \\ \hline p & 0 & 5 \\ \hline q & 8 & 0 \end{array}$	$\begin{array}{c cc} y & p & q \\ \hline p & 0 & 5 \\ \hline q & 8 & 0 \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																																		
fong-spivak-invitation_1081	081	0.647	$\begin{tabular}{t}{ l l l l l l l }\hline \backslash(x\times y)\&\backslash((A,p)\backslash)\&\backslash((B,p)\backslash)\&\backslash((C,p)\backslash)\&\backslash((A,q)\backslash)\&\backslash((B,q)\backslash)\&\backslash((C,q)\backslash)\backslash\hline(\backslash(A,p)\backslash)\&0\&2\&5\&5\&7\&10\backslash\hline\backslash((B,p)\backslash)\&\backslash(\infty)\&0\&3\&\backslash(\infty)\&5\&8\backslash\hline\backslash((C,p)\backslash)\&\backslash(\infty)\&\backslash(\infty)\&0\&\backslash(\infty)\&5\backslash\hline\backslash((A,q)\backslash)\&8\&10\&13\&0\&2\&5\backslash\hline\backslash((B,q)\backslash)\&\backslash(\infty)\&8\&11\&\backslash(\infty)\&0\&3\backslash\hline\backslash((C,q)\backslash)\&\backslash(\infty)\&\backslash(\infty)\&8\&\backslash(\infty)\&\backslash(\infty)\&0\backslash\hline\end{tabular}$	<table><tr><td>$x\times y$</td><td>(A,p)</td><td>(B,p)</td><td>(C,p)</td><td>(A,q)</td><td>(B,q)</td><td>(C,q)</td></tr><tr><td>(A,p)</td><td>0</td><td>2</td><td>5</td><td>5</td><td>7</td><td>10</td></tr><tr><td>(B,p)</td><td>∞</td><td>0</td><td>3</td><td>∞</td><td>5</td><td>8</td></tr><tr><td>(C,p)</td><td>∞</td><td>∞</td><td>0</td><td>∞</td><td>∞</td><td>5</td></tr><tr><td>(A,q)</td><td>8</td><td>10</td><td>13</td><td>0</td><td>2</td><td>5</td></tr><tr><td>(B,q)</td><td>∞</td><td>8</td><td>11</td><td>∞</td><td>0</td><td>3</td></tr><tr><td>(C,q)</td><td>∞</td><td>∞</td><td>8</td><td>∞</td><td>∞</td><td>0</td></tr></table>	$x\times y$	(A,p)	(B,p)	(C,p)	(A,q)	(B,q)	(C,q)	(A,p)	0	2	5	5	7	10	(B,p)	∞	0	3	∞	5	8	(C,p)	∞	∞	0	∞	∞	5	(A,q)	8	10	13	0	2	5	(B,q)	∞	8	11	∞	0	3	(C,q)	∞	∞	8	∞	∞	0	<table><tr><td>$\mathcal{X}\times\mathcal{Y}$</td><td>$(A,p)$</td><td>$(B,p)$</td><td>$(C,p)$</td><td>$(A,q)$</td><td>$(B,q)$</td><td>$(C,q)$</td></tr><tr><td>$(A,p)$</td><td>0</td><td>2</td><td>5</td><td>5</td><td>7</td><td>10</td></tr><tr><td>(B,p)</td><td>∞</td><td>0</td><td>3</td><td>∞</td><td>5</td><td>8</td></tr><tr><td>(C,p)</td><td>∞</td><td>∞</td><td>0</td><td>∞</td><td>∞</td><td>5</td></tr><tr><td>(A,q)</td><td>8</td><td>10</td><td>13</td><td>0</td><td>2</td><td>5</td></tr><tr><td>(B,q)</td><td>∞</td><td>8</td><td>11</td><td>∞</td><td>0</td><td>3</td></tr><tr><td>(C,q)</td><td>∞</td><td>∞</td><td>8</td><td>∞</td><td>∞</td><td>0</td></tr></table>	$\mathcal{X}\times\mathcal{Y}$	(A,p)	(B,p)	(C,p)	(A,q)	(B,q)	(C,q)	(A,p)	0	2	5	5	7	10	(B,p)	∞	0	3	∞	5	8	(C,p)	∞	∞	0	∞	∞	5	(A,q)	8	10	13	0	2	5	(B,q)	∞	8	11	∞	0	3	(C,q)	∞	∞	8	∞	∞	0
$x\times y$	(A,p)	(B,p)	(C,p)	(A,q)	(B,q)	(C,q)																																																																																																	
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fong-spivak-invitation_1088	088	0.935	$\begin{tabular}{t}{ l l l l }\hline \backslash(M_Y)\&\backslash(x)\&\backslash(y)\&\backslash(z)\backslash\hline\backslash(x)\&0\&4\&3\backslash\hline\backslash(y)\&3\&0\&\infty\backslash\hline\backslash(z)\&\infty\&4\&0\backslash\hline\end{tabular}$	<table><tr><td>M_Y</td><td>x</td><td>y</td><td>z</td></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>∞</td></tr><tr><td>z</td><td>∞</td><td>4</td><td>0</td></tr></table>	M_Y	x	y	z	x	0	4	3	y	3	0	∞	z	∞	4	0	<table><tr><td>M_Y</td><td>x</td><td>y</td><td>z</td></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>∞</td></tr><tr><td>z</td><td>∞</td><td>4</td><td>0</td></tr></table>	M_Y	x	y	z	x	0	4	3	y	3	0	∞	z	∞	4	0																																																																		
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fong-spivak-invitation_1088	088	0.860	$\begin{tabular}{t}{ l l l l }\hline \backslash(d_Y)\&\backslash(x)\&\backslash(y)\&\backslash(z)\backslash\hline\backslash(x)\&0\&4\&3\backslash\hline\backslash(y)\&3\&0\&6\backslash\hline\backslash(z)\&7\&4\&0\backslash\hline\end{tabular}$	<table><tr><td>d_Y</td><td>x</td><td>y</td><td>z</td></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>6</td></tr><tr><td>z</td><td>7</td><td>4</td><td>0</td></tr></table>	d_Y	x	y	z	x	0	4	3	y	3	0	6	z	7	4	0	<table><tr><td>d_Y</td><td>x</td><td>y</td><td>z</td></tr><tr><td>x</td><td>0</td><td>4</td><td>3</td></tr><tr><td>y</td><td>3</td><td>0</td><td>6</td></tr><tr><td>z</td><td>7</td><td>4</td><td>0</td></tr></table>	d_Y	x	y	z	x	0	4	3	y	3	0	6	z	7	4	0																																																																		
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z	7	4	0																																																																																																				
fong-spivak-invitation_1088	088	0.833	$\begin{tabular}{t}{ l l l l l }\hline \backslash\multirow{4}{*}{\backslash(M_Y^2=\backslash)\&\succ\&\backslash(X)\&\backslash(y)\&\backslash(z)\backslash\hline\&\backslash(x)\&0\&4\&3\backslash\hline\&y\&3\&0\&6\backslash\hline\&z\&7\&4\&0\backslash\hline\end{tabular}$	(not rendered)	$M_Y^2 = \begin{array}{c ccc} \nearrow & x & y & z \\ x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$																																																																																																		
fong-spivak-invitation_1088	088	0.509	$\begin{tabular}{t}{ l l l l l }\hline \backslash\multirow{4}{*}{\backslash(M_Y^3=\backslash)\&\succ\&\backslash(X)\&\backslash(y)\&z\backslash\hline\&\backslash(x)\&0\&4\&3\backslash\hline\&y\&3\&0\&6\backslash\hline\&z\&7\&4\&0\backslash\hline\end{tabular}$	(not rendered)	$M_Y^3 = \begin{array}{c ccc} \nearrow & x & y & z \\ x & 0 & 4 & 3 \\ y & 3 & 0 & 6 \\ z & 7 & 4 & 0 \end{array}$																																																																																																		
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fong-spivak-invitation_Table	091	■1.000	$\begin{tabular}[t]{ l l l l }\hline Employee & FName & WorksIn & Mngr \\ \hline 1 & Alan & 101 & 2 \\ \hline 2 & Ruth & 101 & 2 \\ \hline 3 & Kris & 102 & 3 \\ \hline \end{tabular}$	<table><tr><th>Employee</th><th>FName</th><th>WorksIn</th><th>Mngr</th></tr><tr><td>1</td><td>Alan</td><td>101</td><td>2</td></tr><tr><td>2</td><td>Ruth</td><td>101</td><td>2</td></tr><tr><td>3</td><td>Kris</td><td>102</td><td>3</td></tr></table>	Employee	FName	WorksIn	Mngr	1	Alan	101	2	2	Ruth	101	2	3	Kris	102	3	<table><tr><th>Employee</th><th>FName</th><th>WorksIn</th><th>Mngr</th></tr><tr><td>1</td><td>Alan</td><td>101</td><td>2</td></tr><tr><td>2</td><td>Ruth</td><td>101</td><td>2</td></tr><tr><td>3</td><td>Kris</td><td>102</td><td>3</td></tr></table>	Employee	FName	WorksIn	Mngr	1	Alan	101	2	2	Ruth	101	2	3	Kris	102	3																						
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Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																				
fong-spivak-invitation_TAL_Q62	105	■1.000	$\begin{array}{ t l l } \hline \text{Beatle} & \& \text{Played} \\ \hline \text{George} & \& \text{lead guitar} \\ \hline \text{John} & \& \text{rhythm guitar} \\ \hline \text{Paul} & \& \text{bass guitar} \\ \hline \text{Ringo} & \& \text{drums} \\ \hline \end{array}$	<table><tr><td>Beatle</td><td>Played</td></tr><tr><td>George</td><td>lead guitar</td></tr><tr><td>John</td><td>rhythm guitar</td></tr><tr><td>Paul</td><td>bass guitar</td></tr><tr><td>Ringo</td><td>drums</td></tr></table>	Beatle	Played	George	lead guitar	John	rhythm guitar	Paul	bass guitar	Ringo	drums	<table><tr><th>Beatle</th><th>Played</th></tr><tr><td>George</td><td>lead guitar</td></tr><tr><td>John</td><td>rhythm guitar</td></tr><tr><td>Paul</td><td>bass guitar</td></tr><tr><td>Ringo</td><td>drums</td></tr></table>	Beatle	Played	George	lead guitar	John	rhythm guitar	Paul	bass guitar	Ringo	drums																
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fong-spivak-invitation_TAL_Q62	105	■1.000	$\begin{array}{ t l } \hline \text{Rock-and-roll instrument} \\ \hline \text{bass guitar} \\ \hline \text{drums} \\ \hline \text{keyboard} \\ \hline \text{lead guitar} \\ \hline \text{rhythm guitar} \\ \hline \end{array}$	<table><tr><td>Rock-and-roll instrument</td></tr><tr><td>bass guitar</td></tr><tr><td>drums</td></tr><tr><td>keyboard</td></tr><tr><td>lead guitar</td></tr><tr><td>rhythm guitar</td></tr></table>	Rock-and-roll instrument	bass guitar	drums	keyboard	lead guitar	rhythm guitar	<table><tr><th>Rock-and-roll instrument</th><td></td></tr><tr><td>bass guitar</td><td></td></tr><tr><td>drums</td><td></td></tr><tr><td>keyboard</td><td></td></tr><tr><td>lead guitar</td><td></td></tr><tr><td>rhythm guitar</td><td></td></tr></table>	Rock-and-roll instrument		bass guitar		drums		keyboard		lead guitar		rhythm guitar																			
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fong-spivak-invitation_TAL_Q62	109	■1.000	$\begin{array}{ t l l } \hline \text{\(\mathbb{N}_{\geq 2}\)} & \& \text{Smallest prime factor} \\ \hline 2 & \& 2 \\ \hline 3 & \& 3 \\ \hline 4 & \& 2 \\ \hline \text{[U+22EE]} & \& \text{[U+22EE]} \\ \hline 49 & \& 7 \\ \hline 50 & \& 2 \\ \hline 51 & \& 3 \\ \hline \text{[U+22EE]} & \& \text{[U+22EE]} \\ \hline \end{array}$	<table><tr><td>$\mathbb{N}_{\geq 2}$</td><td>Smallest prime factor</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr><tr><td>4</td><td>2</td></tr><tr><td>[U+22EE]</td><td>[U+22EE]</td></tr><tr><td>49</td><td>7</td></tr><tr><td>50</td><td>2</td></tr><tr><td>51</td><td>3</td></tr><tr><td>[U+22EE]</td><td>[U+22EE]</td></tr></table>	$\mathbb{N}_{\geq 2}$	Smallest prime factor	2	2	3	3	4	2	[U+22EE]	[U+22EE]	49	7	50	2	51	3	[U+22EE]	[U+22EE]	<table><tr><th>$\mathbb{N}_{\geq 2}$</th><th>Smallest prime factor</th></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>3</td></tr><tr><td>4</td><td>2</td></tr><tr><td>⋮</td><td>⋮</td></tr><tr><td>49</td><td>7</td></tr><tr><td>50</td><td>2</td></tr><tr><td>51</td><td>3</td></tr><tr><td>⋮</td><td>⋮</td></tr></table>	$\mathbb{N}_{\geq 2}$	Smallest prime factor	2	2	3	3	4	2	⋮	⋮	49	7	50	2	51	3	⋮	⋮
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																
fong-spivak-invitation_TAB_123	113	■0.803	\begin{tabular}[t]{l l l l l }\hline Arrow & source & target & Vertex \\ \hline (a) & 1 & 2 & 1 \\ \hline (b) & 1 & 3 & 2 \\ \hline c & 1 & 3 & 3 \\ \hline (d) & 2 & 2 & 4 \\ \hline (e) & 2 & 3 & \\ \hline \end{tabular}	<table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>a</td><td>1</td><td>2</td><td>1</td></tr><tr><td>b</td><td>1</td><td>3</td><td>2</td></tr><tr><td>c</td><td>1</td><td>3</td><td>3</td></tr><tr><td>d</td><td>2</td><td>2</td><td>4</td></tr><tr><td>e</td><td>2</td><td>3</td><td></td></tr></table>	Arrow	source	target	Vertex	a	1	2	1	b	1	3	2	c	1	3	3	d	2	2	4	e	2	3		<table><tr><th>Arrow</th><th>source</th><th>target</th><th>Vertex</th></tr><tr><td>a</td><td>1</td><td>2</td><td>1</td></tr><tr><td>b</td><td>1</td><td>3</td><td>2</td></tr><tr><td>c</td><td>1</td><td>3</td><td>3</td></tr><tr><td>d</td><td>2</td><td>2</td><td>4</td></tr><tr><td>e</td><td>2</td><td>3</td><td></td></tr></table>	Arrow	source	target	Vertex	a	1	2	1	b	1	3	2	c	1	3	3	d	2	2	4	e	2	3																	
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≥0.9

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Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																													
fong-spivak-invitation_TAL_200	120	■0.956	\begin{tabular}[t]{ l l l l }\hline Migration functor & pronounced & reminiscent of & database idea \\ \Delta & delta & duplicate or destroy & duplicate or destroy tables or columns \\ \Sigma & sigma & sum & union (sum up) data \\ \Pi & pi & product & pair ⁹ and query data \\ \hline \end{tabular}	<table><tr><th>Migration functor</th><th>pronounced</th><th>reminiscent of</th><th>database idea</th></tr><tr><td>Δ</td><td>delta</td><td>duplicate or destroy</td><td>duplicate or destroy tables or columns</td></tr><tr><td>Σ</td><td>sigma</td><td>sum</td><td>union (sum up) data</td></tr><tr><td>Π</td><td>pi</td><td>product</td><td>pair⁹ and query data</td></tr></table>	Migration functor	pronounced	reminiscent of	database idea	Δ	delta	duplicate or destroy	duplicate or destroy tables or columns	Σ	sigma	sum	union (sum up) data	Π	pi	product	pair ⁹ and query data	<table><tr><th>Migration functor</th><th>pronounced</th><th>reminiscent of</th><th>database idea</th></tr><tr><td>Δ</td><td>delta</td><td>duplicate or destroy</td><td>duplicate or destroy tables or destroy tables or columns</td></tr><tr><td>Σ</td><td>sigma</td><td>sum</td><td>union (sum up) data</td></tr><tr><td>Π</td><td>pi</td><td>product</td><td>pair⁹ and query data</td></tr></table>				Migration functor	pronounced	reminiscent of	database idea	Δ	delta	duplicate or destroy	duplicate or destroy tables or destroy tables or columns	Σ	sigma	sum	union (sum up) data	Π	pi	product	pair ⁹ and query data																										
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fong-spivak-invitation_TAL_201	121	■1.000	\begin{tabular}[t]{ l l l l l }\hline Email & sent_by & received_by & Address \\ Em_1 & Bob & Grace & Bob \\ Em_2 & Grace & Pat & Doug \\ Em_3 & Bob & Emmy & Emmy \\ Em_4 & Sue & Doug & Grace \\ Em_5 & Doug & Sue & Pat \\ Em_6 & Bob & Bob & Sue \\ \hline \end{tabular}	(not rendered)	<table><tr><th>Email</th><th>sent_by</th><th>received_by</th><th>Address</th></tr><tr><td>Em_1</td><td>Bob</td><td>Grace</td><td>Bob</td></tr><tr><td>Em_2</td><td>Grace</td><td>Pat</td><td>Doug</td></tr><tr><td>Em_3</td><td>Bob</td><td>Emmy</td><td>Emmy</td></tr><tr><td>Em_4</td><td>Sue</td><td>Doug</td><td>Grace</td></tr><tr><td>Em_5</td><td>Doug</td><td>Sue</td><td>Pat</td></tr><tr><td>Em_6</td><td>Bob</td><td>Bob</td><td>Sue</td></tr></table>				Email	sent_by	received_by	Address	Em_1	Bob	Grace	Bob	Em_2	Grace	Pat	Doug	Em_3	Bob	Emmy	Emmy	Em_4	Sue	Doug	Grace	Em_5	Doug	Sue	Pat	Em_6	Bob	Bob	Sue																														
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LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_TAL_067	137	■ 0.747	$\begin{tabular}{t}{ l l l l }\hline \backslash\Phi\backslash & \backslash(x\backslash) & \backslash(y\backslash) & \backslash(z\backslash) \\ \hline \backslash(A\backslash) & ? & ? & 20 \\ \hline B & 11 & ? & ? \\ \hline C & ? & 17 & ? \\ \hline D & ? & ? & ? \\ \hline \end{tabular}$	$\begin{array}{ c c c c }\hline \Phi & x & y & z \\ \hline A & ? & ? & 20 \\ \hline B & 11 & ? & ? \\ \hline C & ? & 17 & ? \\ \hline D & ? & ? & ? \\ \hline \end{array}$	$\begin{array}{ c c c c c }\hline \Phi & x & y & z \\ \hline A & ? & ? & 20 \\ \hline B & 11 & ? & ? \\ \hline C & ? & 17 & ? \\ \hline D & ? & ? & ? \\ \hline \end{array}$
fong-spivak-invitation_TAL_067	137	■ 0.508	$\begin{tabular}{t}{ l l l l l }\hline \backslash(M_X\backslash) & \backslash(A\backslash) & \backslash(B\backslash) & C & D \\ \hline \backslash(A\backslash) & ? & ? & 0 & \backslash(\infty\backslash) \\ \hline B & 2 & 0 & \backslash(\infty\backslash) & 5 \\ \hline C & \backslash(\infty\backslash) & 3 & 0 & \backslash(\infty\backslash) \\ \hline D & \backslash(\infty\backslash) & 4 & 0 & \backslash(\infty\backslash) \\ \hline \end{tabular}$	$\begin{array}{ c c c c c }\hline M_X & A & B & C & D \\ \hline A & 0 & \infty & 3 & \infty \\ \hline B & 2 & 0 & \infty & 5 \\ \hline C & \infty & 3 & 0 & \infty \\ \hline D & \infty & \infty & 4 & 0 \\ \hline \end{array}$	$\begin{array}{ c c c c c }\hline M_X & A & B & C & D \\ \hline A & 0 & \infty & 3 & \infty \\ \hline B & 2 & 0 & \infty & 5 \\ \hline C & \infty & 3 & 0 & \infty \\ \hline D & \infty & \infty & 4 & 0 \\ \hline \end{array}$
fong-spivak-invitation_TAL_067	137	■ 0.558	$\begin{tabular}{t}{ l l l l }\hline \backslash(M_{\Phi}\backslash) & \backslash(x\backslash) & \backslash(y\backslash) & z \\ \hline \backslash(A\backslash) & \backslash(\infty\backslash) & \backslash(\infty\backslash) & \backslash(\infty\backslash) \\ \hline \backslash(B\backslash) & 11 & \backslash(\infty\backslash) & \backslash(\infty\backslash) \\ \hline C & \backslash(\infty\backslash) & \backslash(\infty\backslash) & \backslash(\infty\backslash) \\ \hline D & \backslash(\infty\backslash) & 9 & \backslash(\infty\backslash) \\ \hline \end{tabular}$	$\begin{array}{ c c c c }\hline M_{\Phi} & x & y & z \\ \hline A & \infty & \infty & \infty \\ \hline B & 11 & \infty & \infty \\ \hline C & \infty & \infty & \infty \\ \hline D & \infty & 9 & \infty \\ \hline \end{array}$	$\begin{array}{ c c c c }\hline M_{\Phi} & x & y & z \\ \hline A & \infty & \infty & \infty \\ \hline B & 11 & \infty & \infty \\ \hline C & \infty & \infty & \infty \\ \hline D & \infty & 9 & \infty \\ \hline \end{array}$
fong-spivak-invitation_TAL_067	137	■ 0.626	$\begin{tabular}{t}{ l l l l }\hline \backslash(M_Y\backslash) & \backslash(x\backslash) & \backslash(y\backslash) & \backslash(z\backslash) \\ \hline \backslash(A\backslash) & 0 & 4 & 3 \\ \hline y & 3 & 0 & \infty \\ \hline z & \infty & 4 & 0 \\ \hline \end{tabular}$	$\begin{array}{ c c c c }\hline M_Y & x & y & z \\ \hline x & 0 & 4 & 3 \\ \hline y & 3 & 0 & \infty \\ \hline z & \infty & 4 & 0 \\ \hline \end{array}$	$\begin{array}{ c c c c }\hline M_Y & x & y & z \\ \hline x & 0 & 4 & 3 \\ \hline y & 3 & 0 & \infty \\ \hline z & \infty & 4 & 0 \\ \hline \end{array}$
fong-spivak-invitation_TAL_066	140	■ 0.738	$\begin{tabular}{t}{ l l l l l l }\hline \backslash(\Phi\backslash) & \backslash(a\backslash) & \backslash(b\backslash) & \backslash(c\backslash) & \backslash(d\backslash) & \backslash(e\backslash) \\ \hline \backslash(N\backslash) & \text{true} & \text{false} & \text{true} & \text{false} & \text{false} \\ \hline \backslash(E\backslash) & \text{true} & \text{false} & \text{true} & \text{false} & \text{true} \\ \hline W & \text{true} & \text{true} & \text{true} & \text{true} & \text{false} \\ \hline S & \text{true} & \text{true} & \text{true} & \text{true} & \text{true} \\ \hline \end{tabular}$	$\begin{array}{ c c c c c c }\hline \Phi & a & b & c & d & e \\ \hline N & \text{true} & \text{false} & \text{true} & \text{false} & \text{false} \\ \hline E & \text{true} & \text{false} & \text{true} & \text{false} & \text{true} \\ \hline W & \text{true} & \text{true} & \text{true} & \text{true} & \text{false} \\ \hline S & \text{true} & \text{true} & \text{true} & \text{true} & \text{true} \\ \hline \end{array}$	$\begin{array}{ c c c c c c }\hline \Phi & a & b & c & d & e \\ \hline N & \text{true} & \text{false} & \text{true} & \text{false} & \text{false} \\ \hline E & \text{true} & \text{false} & \text{true} & \text{false} & \text{true} \\ \hline W & \text{true} & \text{true} & \text{true} & \text{true} & \text{false} \\ \hline S & \text{true} & \text{true} & \text{true} & \text{true} & \text{true} \\ \hline \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_TAL_450	140	0.600	$\begin{tabular}{t}{ l l l }\hline \Psi & (x) & (y) \\ \hline a & \text{false} & \text{true} \\ b & \text{true} & \text{true} \\ c & \text{false} & \text{true} \\ d & \text{true} & \text{true} \\ e & \text{false} & \text{false} \\ \hline \end{tabular}$	$\begin{array}{ c c c }\hline \Psi & x & y \\ \hline a & \text{false} & \text{true} \\ b & \text{true} & \text{true} \\ c & \text{false} & \text{true} \\ d & \text{true} & \text{true} \\ e & \text{false} & \text{false} \\ \hline \end{array}$	$\begin{array}{ c c c }\hline \Psi & x & y \\ \hline a & \text{false} & \text{true} \\ b & \text{true} & \text{true} \\ c & \text{false} & \text{true} \\ d & \text{true} & \text{true} \\ e & \text{false} & \text{false} \\ \hline \end{array}$
fong-spivak-invitation_TAL_451	140	0.568	$\begin{tabular}{t}{ l l l }\hline \Phi_9\Psi & (x) & (y) \\ \hline N & \text{false} & \text{true} \\ E & \text{false} & \text{true} \\ W & \text{true} & \text{true} \\ S & \text{true} & \text{true} \\ \hline \end{tabular}$	$\begin{array}{ c c c }\hline \Phi_9\Psi & x & y \\ \hline N & \text{false} & \text{true} \\ E & \text{false} & \text{true} \\ W & \text{true} & \text{true} \\ S & \text{true} & \text{true} \\ \hline \end{array}$	$\begin{array}{ c c c c }\hline \Phi_9\Psi & x & y \\ \hline N & \text{false} & \text{true} \\ E & \text{false} & \text{true} \\ W & \text{true} & \text{true} \\ S & \text{true} & \text{true} \\ \hline \end{array}$
fong-spivak-invitation_TAL_451	141	0.618	$\begin{tabular}{t}{ l l l l l }\hline \Phi_9\Psi & (p) & (q) & (r) & (s) \\ \hline A & ? & 24 & ? & ? \\ B & ? & ? & ? & ? \\ C & ? & ? & ? & ? \\ D & ? & ? & 9 & ? \\ \hline \end{tabular}$	$\begin{array}{ c c c c c }\hline \Phi_9\Psi & p & q & r & s \\ \hline A & ? & 24 & ? & ? \\ B & ? & ? & ? & ? \\ C & ? & ? & ? & ? \\ D & ? & ? & 9 & ? \\ \hline \end{array}$	$\begin{array}{ c c c c c }\hline \Phi_9\Psi & p & q & r & s \\ \hline A & ? & 24 & ? & ? \\ B & ? & ? & ? & ? \\ C & ? & ? & ? & ? \\ D & ? & ? & 9 & ? \\ \hline \end{array}$
fong-spivak-invitation_TAL_456	146	0.706	$\begin{tabular}{t}{ l l l l l l l l }\hline X & A & B & \Phi & \chi & y & y & \chi & y \\ \hline A & 0 & 2 & A & 5 & 8 & x & 0 & 3 \\ B & \infty & 0 & B & \infty & \infty & y & 4 & 0 \\ \hline \end{tabular}$	$\begin{array}{ c c c c c c c c c }\hline X & A & B & \Phi & \chi & y & y & \chi & y \\ \hline A & 0 & 2 & A & 5 & 8 & x & 0 & 3 \\ B & \infty & 0 & B & \infty & \infty & y & 4 & 0 \\ \hline \end{array}$	$\begin{array}{ c c c c c c c c c }\hline X & A & B & \Phi & \chi & y & y & \chi & y \\ \hline A & 0 & 2 & A & 5 & 8 & x & 0 & 3 \\ B & \infty & 0 & B & \infty & \infty & y & 4 & 0 \\ \hline \end{array}$
fong-spivak-invitation_TAL_456	146	0.848	$\begin{tabular}{t}{ l l l l l }\hline \text{Col}(\Phi) & A & B & x & y \\ \hline A & 0 & 2 & 5 & 8 \\ B & \infty & 0 & \infty & \infty \\ x & 0 & 0 & 0 & 3 \\ y & 0 & 0 & 4 & 0 \\ \hline \end{tabular}$	$\begin{array}{ c c c c c }\hline \text{Col}(\Phi) & A & B & x & y \\ \hline A & 0 & 2 & 5 & 8 \\ B & \infty & 0 & \infty & \infty \\ x & 0 & 0 & 0 & 3 \\ y & 0 & 0 & 4 & 0 \\ \hline \end{array}$	$\begin{array}{ c c c c c }\hline \text{Col}(\Phi) & A & B & x & y \\ \hline A & 0 & 2 & 5 & 8 \\ B & \infty & 0 & \infty & \infty \\ x & 0 & 0 & 0 & 3 \\ y & 0 & 0 & 4 & 0 \\ \hline \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																
fong-spivak-invitation_TAL_442	179	■0.620	$\begin{tabular}{t}{ l l l l l }\hline & \backslash(a\backslash) & \backslash(b\backslash) & & \\ & \backslash \backslash \hline \backslash(x\backslash) & 15 & 3 & \backslash \backslash \left(\begin{array}{c} 15 \\ 0 \end{array}\right\backslash \right.\backslash & 21 \\ & \backslash \backslash \hline \backslash(y\backslash) & 0 & 21 & & \\ & \backslash \backslash \hline \end{tabular}$	<table><tr><td></td><td>a</td><td>b</td><td></td><td></td></tr><tr><td>x</td><td>15</td><td>3</td><td>$\left(\begin{array}{c} 15 \\ 0 \end{array}\right)$</td><td>21</td></tr><tr><td>y</td><td>0</td><td>21</td><td></td><td></td></tr></table>		a	b			x	15	3	$\left(\begin{array}{c} 15 \\ 0 \end{array}\right)$	21	y	0	21			<table><tr><td></td><td>a</td><td>b</td><td>$\left(\begin{array}{cc} 15 & 3 \\ 0 & 21 \end{array}\right)$</td></tr><tr><td>$x$</td><td>15</td><td>3</td><td></td></tr><tr><td>y</td><td>0</td><td>21</td><td></td></tr></table>		a	b	$\left(\begin{array}{cc} 15 & 3 \\ 0 & 21 \end{array}\right)$	x	15	3		y	0	21																						
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x	15	3	$\left(\begin{array}{c} 15 \\ 0 \end{array}\right)$	21																																																	
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fong-spivak-invitation_TAL_442	179	■0.615	$\begin{tabular}{t}{ l l l l l }\hline Generator & Icon & Matrix & Arity \\ \hline amplify by a \in R & \neg \overline{a} & (a) & 1 \rightarrow 1 \\ add & \supset & \begin{pmatrix} 1 \\ 1 \end{pmatrix} & 2 \rightarrow 1 \\ zero & \circ & () & 0 \rightarrow 1 \\ copy & \neg \supset & (1 \ 1) & 1 \rightarrow 2 \\ discard & \rightarrow & () & 1 \rightarrow 0 \\ \hline \end{tabular}$	<table><tr><th>Generator</th><th>Icon</th><th>Matrix</th><th>Arity</th></tr><tr><td>amplify by $a \in R$</td><td>- a -</td><td>(a)</td><td>$1 \rightarrow 1$</td></tr><tr><td>add</td><td>D-</td><td>$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$</td><td>$2 \rightarrow 1$</td></tr><tr><td>zero</td><td>$\bigcirc$-</td><td>$()$</td><td>$0 \rightarrow 1$</td></tr><tr><td>copy</td><td>- C</td><td>$\begin{pmatrix} 1 & 1 \end{pmatrix}$</td><td>$1 \rightarrow 2$</td></tr><tr><td>discard</td><td>$\rightarrow$</td><td>$()$</td><td>$1 \rightarrow 0$</td></tr></table>	Generator	Icon	Matrix	Arity	amplify by $a \in R$	- a -	(a)	$1 \rightarrow 1$	add	D-	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$2 \rightarrow 1$	zero	$\bigcirc$ -	$()$	$0 \rightarrow 1$	copy	- C	$\begin{pmatrix} 1 & 1 \end{pmatrix}$	$1 \rightarrow 2$	discard	$\rightarrow$	$()$	$1 \rightarrow 0$	<table><tr><th>Generator</th><th>Icon</th><th>Matrix</th><th>Arity</th></tr><tr><td>amplify by $a \in R$</td><td>$\neg \overline{a}$</td><td>(a)</td><td>$1 \rightarrow 1$</td></tr><tr><td>add</td><td>$\supset$</td><td>$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$</td><td>$2 \rightarrow 1$</td></tr><tr><td>zero</td><td>$\circ$</td><td>$()$</td><td>$0 \rightarrow 1$</td></tr><tr><td>copy</td><td>$\neg \supset$</td><td>$(1 \ 1)$</td><td>$1 \rightarrow 2$</td></tr><tr><td>discard</td><td>$\rightarrow$</td><td>$()$</td><td>$1 \rightarrow 0$</td></tr></table>	Generator	Icon	Matrix	Arity	amplify by $a \in R$	$\neg \overline{a}$	(a)	$1 \rightarrow 1$	add	$\supset$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$2 \rightarrow 1$	zero	$\circ$	$()$	$0 \rightarrow 1$	copy	$\neg \supset$	$(1 \ 1)$	$1 \rightarrow 2$	discard	$\rightarrow$	$()$	$1 \rightarrow 0$
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fong-spivak-invitation_TAL_442	242	■0.665	$\begin{tabular}{t}{ l l l l }\hline \backslash(P\backslash) & \backslash(Q\backslash) & \backslash(\ulcorner true, true \urcorner(P, Q)\backslash) \\ \hline true & true & true \\ true & false & false \\ false & true & false \\ false & false & false \\ \hline \end{tabular}$	<table><tr><th>P</th><th>Q</th><th>$\ulcorner (true, true) \urcorner(P, Q)$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>false</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$\ulcorner (true, true) \urcorner(P, Q)$	true	true	true	true	false	false	false	true	false	false	false	false	<table><tr><th>P</th><th>Q</th><th>$\ulcorner (true, true) \urcorner(P, Q)$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>false</td></tr><tr><td>false</td><td>true</td><td>false</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$\ulcorner (true, true) \urcorner(P, Q)$	true	true	true	true	false	false	false	true	false	false	false	false																		
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fong-spivak-invitation_TAL_442	242	■1.000	$\begin{tabular}{t}{ l l l l }\hline \backslash(P\backslash) & \backslash(Q\backslash) & \backslash(P \vee Q\backslash) \\ \hline true & true & true \\ true & false & true \\ false & true & true \\ false & false & false \\ \hline \end{tabular}$	<table><tr><th>P</th><th>Q</th><th>$P \vee Q$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$P \vee Q$	true	true	true	true	false	true	false	true	true	false	false	false	<table><tr><th>P</th><th>Q</th><th>$P \vee Q$</th></tr><tr><td>true</td><td>true</td><td>true</td></tr><tr><td>true</td><td>false</td><td>true</td></tr><tr><td>false</td><td>true</td><td>true</td></tr><tr><td>false</td><td>false</td><td>false</td></tr></table>	P	Q	$P \vee Q$	true	true	true	true	false	true	false	true	true	false	false	false																		
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fong-spivak-invitation_TAL_442	243	■1.000	$\begin{tabular}{t}{ l l l l l }\hline \backslash(P\backslash) & \backslash(Q\backslash) & \backslash(P \wedge Q\backslash) & \backslash(P=(P \wedge Q)\backslash) \\ \hline true & true & ? & ? \\ true & false & ? & ? \\ false & true & ? & ? \\ false & false & ? & ? \\ \hline \end{tabular}$	<table><tr><th>P</th><th>Q</th><th>$P \wedge Q$</th><th>$P=(P \wedge Q)$</th></tr><tr><td>true</td><td>true</td><td>?</td><td>?</td></tr><tr><td>true</td><td>false</td><td>?</td><td>?</td></tr><tr><td>false</td><td>true</td><td>?</td><td>?</td></tr><tr><td>false</td><td>false</td><td>?</td><td>?</td></tr></table>	P	Q	$P \wedge Q$	$P=(P \wedge Q)$	true	true	?	?	true	false	?	?	false	true	?	?	false	false	?	?	<table><tr><th>P</th><th>Q</th><th>$P \wedge Q$</th><th>$P=(P \wedge Q)$</th></tr><tr><td>true</td><td>true</td><td>?</td><td>?</td></tr><tr><td>true</td><td>false</td><td>?</td><td>?</td></tr><tr><td>false</td><td>true</td><td>?</td><td>?</td></tr><tr><td>false</td><td>false</td><td>?</td><td>?</td></tr></table>	P	Q	$P \wedge Q$	$P=(P \wedge Q)$	true	true	?	?	true	false	?	?	false	true	?	?	false	false	?	?								
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_742_743	273	0.919	$\begin{tabular}{t l l l l }\hline\text{Arrow } (a) & \text{Source } (s(a) \in V) & \text{Target } (t(a) \in V) \\ \hline a & 1 & 2 \\ b & 1 & 3 \\ c & 1 & 3 \\ d & 2 & 2 \\ e & 2 & 3 \\ \hline\end{tabular}$	$\begin{array}{c c c}\text{Arrow } a & \text{Source } s(a) \in V & \text{Target } t(a) \in V \\ \hline a & 1 & 2 \\ b & 1 & 3 \\ c & 1 & 3 \\ d & 2 & 2 \\ e & 2 & 3 \\ \hline\end{array}$	$\begin{array}{c c c}\text{Arrow } a & \text{Source } s(a) \in V & \text{Target } t(a) \in V \\ \hline a & 1 & 2 \\ b & 1 & 3 \\ c & 1 & 3 \\ d & 2 & 2 \\ e & 2 & 3 \\ \hline\end{array}$
fong-spivak-invitation_742_743	280	0.937	$\begin{tabular}{t l l l l l l l }\hline (p) & (q) & (f(p) \leq ? g(q)) & (p \leq ? g(q)) & \text{Same?} \\ \hline 1 & 1 & yes & yes & yes! \\ 1 & 2 & no & no & yes! \\ 1 & 4 & yes & yes & yes! \\ 2 & 1 & no & no & yes! \\ 2 & 2 & yes & yes & yes! \\ 2 & 4 & yes & yes & yes! \\ 3.9 & 1 & no & no & yes! \\ 3.9 & 2 & no & no & yes! \\ 3.9 & 4 & yes & yes & yes! \\ 4 & 1 & no & no & yes! \\ 4 & 2 & no & no & yes! \\ 4 & 4 & yes & yes & yes! \\ \hline\end{tabular}$	$\begin{array}{c c c c c c c}p & q & f(p) & g(q) & f(p) \leq ? g(q) & p \leq ? g(q) & \text{Same?} \\ \hline 1 & 1 & 1 & 1 & yes & yes & yes! \\ 1 & 2 & 1 & 2 & no & no & yes! \\ 1 & 4 & 1 & 4 & yes & yes & yes! \\ 2 & 1 & 2 & 1 & no & no & yes! \\ 2 & 2 & 2 & 2 & yes & yes & yes! \\ 2 & 4 & 2 & 4 & yes & yes & yes! \\ 3.9 & 1 & 4 & 1 & no & no & yes! \\ 3.9 & 2 & 4 & 2 & no & no & yes! \\ 3.9 & 4 & 4 & 4 & yes & yes & yes! \\ 4 & 1 & 4 & 1 & no & no & yes! \\ 4 & 2 & 4 & 2 & no & no & yes! \\ 4 & 4 & 4 & 4 & yes & yes & yes! \\ \hline\end{array}$	$\begin{array}{c c c c c c c}p & q & f(p) & g(q) & f(p) \leq ? g(q) & p \leq ? g(q) & \text{Same?} \\ \hline 1 & 1 & 1 & 1 & yes & yes & yes! \\ 1 & 2 & 1 & 2 & no & no & yes! \\ 1 & 4 & 1 & 4 & yes & yes & yes! \\ 2 & 1 & 2 & 1 & no & no & yes! \\ 2 & 2 & 2 & 2 & yes & yes & yes! \\ 2 & 4 & 2 & 4 & yes & yes & yes! \\ 3.9 & 1 & 4 & 1 & no & no & yes! \\ 3.9 & 2 & 4 & 2 & no & no & yes! \\ 3.9 & 4 & 4 & 4 & yes & yes & yes! \\ 4 & 1 & 4 & 1 & no & no & yes! \\ 4 & 2 & 4 & 2 & no & no & yes! \\ 4 & 4 & 4 & 4 & yes & yes & yes! \\ \hline\end{array}$
fong-spivak-invitation_742_743	283	1.000	$\begin{tabular}{t l l l l }\hline\text{min} & \text{no} & \text{maybe} & \text{yes} \\ \hline\text{no} & \text{no} & \text{no} & \text{no} \\ \text{maybe} & \text{no} & \text{maybe} & \text{maybe} \\ \text{yes} & \text{no} & \text{maybe} & \text{yes} \\ \hline\end{tabular}$	$\begin{array}{c c c c} \text{min} & \text{no} & \text{maybe} & \text{yes} \\ \hline \text{no} & \text{no} & \text{no} & \text{no} \\ \text{maybe} & \text{no} & \text{maybe} & \text{maybe} \\ \text{yes} & \text{no} & \text{maybe} & \text{yes} \\ \hline \end{array}$	$\begin{array}{c c c c} \text{min} & \text{no} & \text{maybe} & \text{yes} \\ \hline \text{no} & \text{no} & \text{no} & \text{no} \\ \text{maybe} & \text{no} & \text{maybe} & \text{maybe} \\ \text{yes} & \text{no} & \text{maybe} & \text{yes} \\ \hline \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_T285	285	0.780	$\begin{tabular}{t}{ l l l l l }\hline \backslash(d\searrow) & \backslash(A) & \backslash(B) & \backslash(C) & \backslash(D) \\ \hline A & 0 & 6 & 3 & 11 \\ B & 2 & 0 & 5 & 5 \\ C & 5 & 3 & 0 & 8 \\ D & 11 & 9 & 6 & 0 \\ \hline \end{tabular}$	$\begin{array}{c c c c c} d(\nearrow) & A & B & C & D \\ \hline A & 0 & 6 & 3 & 11 \\ B & 2 & 0 & 5 & 5 \\ C & 5 & 3 & 0 & 8 \\ D & 11 & 9 & 6 & 0 \end{array}$	$\begin{array}{c c c c c} d(\nearrow) & A & B & C & D \\ \hline A & 0 & 6 & 3 & 11 \\ B & 2 & 0 & 5 & 5 \\ C & 5 & 3 & 0 & 8 \\ D & 11 & 9 & 6 & 0 \end{array}$
fong-spivak-invitation_T285	285	0.593	$\begin{tabular}{t}{ l l l l l l }\hline \multirrow{2}{*} & \nearrow & \backslash(A) & \backslash(B) & C & \backslash(D) \\ \hline & \backslash(A) & 0 & \infty & 3 & \infty \\ \hline \backslash(M_X=\backslash) & B & 2 & 0 & \infty & 5 \\ & C & \infty & 3 & 0 & \infty \\ & D & \infty & \infty & 6 & 0 \\ \hline \end{tabular}$	(not rendered)	$M_X = \begin{array}{c c c c c} \nearrow & A & B & C & D \\ \hline A & 0 & \infty & 3 & \infty \\ B & 2 & 0 & \infty & 5 \\ C & \infty & 3 & 0 & \infty \\ D & \infty & \infty & 6 & 0 \end{array}$
fong-spivak-invitation_T285	285	0.528	$\begin{tabular}{t}{ l l l l l }\hline X(\nearrow) & \backslash(A) & \backslash(B) & \backslash(C) & \backslash(D) \\ \hline A & M & \{boat\} & \emptyset & \{boat\} \\ B & \emptyset & M & \emptyset & \{boat\} \\ C & \{foot, boat\} & \{boat\} & M & M \\ D & \{foot\} & \emptyset & \{foot, car\} & M \\ \hline \end{tabular}$	$\begin{array}{c c c c c} X(\nearrow) & A & B & C & D \\ \hline A & M & \{boat\} & \emptyset & \{boat\} \\ B & \emptyset & M & \emptyset & \{boat\} \\ C & \{foot, boat\} & \{boat\} & M & M \\ D & \{foot\} & \emptyset & \{foot, car\} & M \end{array}$	$\begin{array}{c c c c c} X(\nearrow) & A & B & C & D \\ \hline A & M & \{boat\} & \emptyset & \{boat\} \\ B & \emptyset & M & \emptyset & \{boat\} \\ C & \{foot, boat\} & \{boat\} & M & M \\ D & \{foot\} & \emptyset & \{foot, car\} & M \end{array}$
fong-spivak-invitation_T286	286	0.689	$\begin{tabular}{t}{ l l l l }\hline \backslash(M\searrow) & \backslash(A) & \backslash(B) & \backslash(C) \\ \hline A & \backslash(A) & \backslash(B) & \backslash(C) \\ B & 6 & \infty & 6 \\ C & 10 & 10 & \infty \\ \hline \end{tabular}$	$\begin{array}{c c c c} M(\nearrow) & A & B & C \\ \hline A & \infty & 10 & 10 \\ B & 6 & \infty & 6 \\ C & 10 & 10 & \infty \end{array}$	$\begin{array}{c c c c} M(\nearrow) & A & B & C \\ \hline A & \infty & 10 & 10 \\ B & 6 & \infty & 6 \\ C & 10 & 10 & \infty \end{array}$
fong-spivak-invitation_T288	288	1.000	$\begin{tabular}{t}{ l l l }\hline \Rightarrow & false & true \\ \hline false & true & true \\ true & false & true \\ \hline \end{tabular}$	$\begin{array}{c c c} \Rightarrow & false & true \\ \hline false & true & true \\ true & false & true \end{array}$	$\begin{array}{c c c} \Rightarrow & false & true \\ \hline false & true & true \\ true & false & true \end{array}$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																																		
fong-spivak-invitation_T295b	290	0.508	$\begin{array}{ l l l l l l l } \hline \nearrow & \& \backslash(v_1\backslash) & \& \backslash(f_1\backslash) & \& \backslash(f_1\{ }^{\circ}\backslash\circ\} f_2\backslash) & \& \backslash(v_2\backslash) & \& \backslash(f_2\backslash) & \& \backslash(v_3\backslash) \\ \hline \backslash(v_1\backslash) & \& \backslash(v_3\backslash) \\ \hline \backslash(f_1\backslash) & \& \backslash(f_1\ ; f_2\backslash) & \& \& \& \& \backslash\hline \backslash(f_1\backslash) & \& \& \& \& \& \& \backslash(f_1\backslash) & \& \backslash(f_1\ \stackrel{\circ}{\rel}\{\circ\} f_2\backslash) & \& \backslash\hline \backslash(f_1\{ }^{\circ}\backslash\circ\} f_2\backslash) \\ \hline \backslash(f_1\{ }^{\circ}\backslash\circ\} f_2\backslash) & \& \& \& \& \& \& \backslash(f_1\{ }^{\circ}\backslash\circ\} f_2\backslash) \\ \hline \backslash(v_2\backslash) & \& \& \& \& \& \& \backslash(v_2\backslash) & \& \backslash(f_2\backslash) & \& \backslash\hline \backslash(f_2\backslash) & \& \& \& \& \& \& \backslash(f_2\backslash) \\ \hline \backslash\hline \backslash(v_3\backslash) & \& \& \& \& \& \& \backslash(v_3\backslash) \\ \hline \end{array}$	<table><tr><td>$\nearrow$</td><td>v_1</td><td>f_1</td><td>$f_1^{\circ}f_2$</td><td>v_2</td><td>f_2</td><td>v_3</td></tr><tr><td>v_1</td><td>v_1</td><td>f_1</td><td>$f_1;f_2$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td></tr><tr><td>f_1</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>f_1</td><td>$f_1\overset{\circ}{9}f_2$</td><td>$\boxtimes$</td></tr><tr><td>$f_1^{\circ}f_2$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$f_1^{\circ}f_2$</td></tr><tr><td>v_2</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>v_2</td><td>f_2</td><td>$\boxtimes$</td></tr><tr><td>f_2</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>f_2</td></tr><tr><td>v_3</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>v_3</td></tr></table>	$\nearrow$	v_1	f_1	$f_1^{\circ}f_2$	v_2	f_2	v_3	v_1	v_1	f_1	$f_1;f_2$	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_1	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_1	$f_1\overset{\circ}{9}f_2$	$\boxtimes$	$f_1^{\circ}f_2$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$f_1^{\circ}f_2$	v_2	$\boxtimes$	$\boxtimes$	$\boxtimes$	v_2	f_2	$\boxtimes$	f_2	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_2	v_3	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	v_3	<table><tr><td>$\nearrow$</td><td>v_1</td><td>f_1</td><td>$f_1\overset{\circ}{9}f_2$</td><td>v_2</td><td>f_2</td><td>v_3</td></tr><tr><td>v_1</td><td>v_1</td><td>f_1</td><td>$f_1\overset{\circ}{9}f_2$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td></tr><tr><td>f_1</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>f_1</td><td>$f_1\overset{\circ}{9}f_2$</td><td>$\boxtimes$</td></tr><tr><td>$f_1^{\circ}f_2$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$f_1^{\circ}f_2$</td></tr><tr><td>v_2</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>v_2</td><td>f_2</td><td>$\boxtimes$</td></tr><tr><td>f_2</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>f_2</td></tr><tr><td>v_3</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>$\boxtimes$</td><td>v_3</td></tr></table>	$\nearrow$	v_1	f_1	$f_1\overset{\circ}{9}f_2$	v_2	f_2	v_3	v_1	v_1	f_1	$f_1\overset{\circ}{9}f_2$	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_1	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_1	$f_1\overset{\circ}{9}f_2$	$\boxtimes$	$f_1^{\circ}f_2$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$f_1^{\circ}f_2$	v_2	$\boxtimes$	$\boxtimes$	$\boxtimes$	v_2	f_2	$\boxtimes$	f_2	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	f_2	v_3	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	$\boxtimes$	v_3
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fong-spivak-invitation_T295b	295	1.000	$\begin{array}{ l l l } \hline \text{Arrow} & \text{source} & \text{target} \\ \hline \text{Mngr} & \text{Employee} & \text{Employee} \\ \hline \text{WorksIn} & \text{Employee} & \text{Department} \\ \hline \text{Secr} & \text{Department} & \text{Employee} \\ \hline \text{FName} & \text{Employee} & \text{string} \\ \hline \text{DName} & \text{Department} & \text{string} \\ \hline \end{array}$	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>Mngr</td><td>Employee</td><td>Employee</td></tr><tr><td>WorksIn</td><td>Employee</td><td>Department</td></tr><tr><td>Secr</td><td>Department</td><td>Employee</td></tr><tr><td>FName</td><td>Employee</td><td>string</td></tr><tr><td>DName</td><td>Department</td><td>string</td></tr></table>	Arrow	source	target	Mngr	Employee	Employee	WorksIn	Employee	Department	Secr	Department	Employee	FName	Employee	string	DName	Department	string	<table><tr><th>Arrow</th><th>source</th><th>target</th></tr><tr><td>Mngr</td><td>Employee</td><td>Employee</td></tr><tr><td>WorksIn</td><td>Employee</td><td>Department</td></tr><tr><td>Secr</td><td>Department</td><td>Employee</td></tr><tr><td>FName</td><td>Employee</td><td>string</td></tr><tr><td>DName</td><td>Department</td><td>string</td></tr></table>	Arrow	source	target	Mngr	Employee	Employee	WorksIn	Employee	Department	Secr	Department	Employee	FName	Employee	string	DName	Department	string																																																														
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fong-spivak-invitation_T296b	296	1.000	$\begin{array}{ l l l l } \hline \multimap\{2\}\{*\}\{ \} & \& \text{State} & \text{next} \\ \hline \& \& 1 & 4 \\ \hline \multimap\{6\}\{*\}\{!\}(\text{https://cdn.mathpix.com/cropped/05b3e583-fc72-4ff5-9914-6f882b9df5b5-296.jpg?height=214\&width=198\&top_left_y=1515\&top_left_x=552}) & \& 2 & 4 \\ \hline \& \& 3 & 5 \\ \hline \& \& 4 & 5 \\ \hline \& \& 5 & 5 \\ \hline \& \& 6 & 7 \\ \hline \& \& 7 & 6 \\ \hline \end{array}$	(not rendered)	Arrow sourcetarget Vertex Gr G State next DDS Set <table><tr><th>State</th><th>next</th></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>5</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>5</td></tr><tr><td>6</td><td>7</td></tr><tr><td>7</td><td>6</td></tr></table>	State	next	1	4	2	4	3	5	4	5	5	5	6	7	7	6																																																																																		
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Conf. MathPix confidence:
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■ 0.5–0.9
■ <0.5
— no value

LaTeX source MathPix’s reading, unchanged;
Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):
 ● C component
● W weak
● S stable
● N noise
● K clean
● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
					Arrowsourcetarget 141 242 353 454 555 676 767
fong-spivak-invitation_T42_966	296	■1.000	$\begin{tabular}{t}{ l l l }\hline\text{Arrow} \ \& \ \text{source} \ \& \ \text{target} \\\hline 1 \ \& \ 4 \ \& \ 1 \\\hline 2 \ \& \ 4 \ \& \ 2 \\\hline 3 \ \& \ 5 \ \& \ 3 \\\hline 4 \ \& \ 5 \ \& \ 4 \\\hline 5 \ \& \ 5 \ \& \ 5 \\\hline 6 \ \& \ 7 \ \& \ 6 \\\hline 7 \ \& \ 6 \ \& \ 7 \\\hline\end{tabular}$	Arrowsourcetarget 141 242 353 454 555 676 767	
fong-spivak-invitation_T42_966	296	■1.000	$\begin{tabular}{t}{ l }\hline\text{Vertex} \\\hline 1 \\\hline 2 \\\hline 3 \\\hline 4 \\\hline 5 \\\hline 6 \\\hline 7 \\\hline\end{tabular}$	Vertex 1 2 3 4 5 6 7	Vertex 1 2 3 4 5 6 7
fong-spivak-invitation_T42_967	297	■1.000	$\begin{tabular}{t}{ l l l l l }\hline\text{Email} \ \& \ \text{sent_by} \ \& \ \text{received_by} \ \& \ \text{Address} \\\hline \text{Em_1} \ \& \ \text{Bob} \ \& \ \text{Grace} \ \& \ \text{Bob} \\\hline \text{Em_2} \ \& \ \text{Grace} \ \& \ \text{Pat} \ \& \ \text{Doug} \\\hline \text{Em_3} \ \& \ \text{Bob} \ \& \ \text{Emmy} \ \& \ \text{Emmy} \\\hline \text{Em_4} \ \& \ \text{Sue} \ \& \ \text{Doug} \ \& \ \text{Grace} \\\hline \text{Em_5} \ \& \ \text{Doug} \ \& \ \text{Sue} \ \& \ \text{Pat} \\\hline \text{Em_6} \ \& \ \text{Bob} \ \& \ \text{Bob} \ \& \ \text{Sue} \\\hline\end{tabular}$	(not rendered)	Emailsent_byreceived_byAddress Em_1BobGraceBob Em_2GracePatDoug Em_3BobEmmyEmmy Em_4SueDougGrace Em_5DougSuePat Em_6BobBobSue
fong-spivak-invitation_T42_968	300	■0.863	$\begin{tabular}{t}{ l l l l l l }\hline\Phi \ \& \ (a \ \& \ (b \ \& \ (c \ \& \ (d \ \& \ (e \\\hline N \ \& \ \text{true} \ \& \ \text{false} \ \& \ \text{true} \ \& \ \text{false} \ \& \ \text{true} \\\hline E \ \& \ \text{true} \ \& \ \text{true} \ \& \ \text{true} \ \& \ \text{true} \ \& \ \text{true} \\\hline W \ \& \ \text{true} \ \& \ \text{false} \ \& \ \text{true} \ \& \ \text{false} \ \& \ \text{true} \\\hline S \ \& \ \text{true} \ \& \ \text{true} \ \& \ \text{true} \ \& \ \text{true} \ \& \ \text{true} \\\hline\end{tabular}$	Φa b c d e Ntruefalsetruefalsetrue Etruetruetruetruetrue Wtruefalsetruefalsetrue Struetruetruetruetrue	Φa b c d e Ntruefalsetruefalsetrue Etruetruetruetruetrue Wtruefalsetruefalsetrue Struetruetruetruetrue
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_TAB_261	301	■ 0.893	$\begin{tabular}{t}{ l l l l l }\hline \backslash\Phi: \Psi\backslash & \backslash(p\backslash) & \backslash(q\backslash) & \backslash(r\backslash) & \backslash(s\backslash) \\\hline \backslash(A\backslash) & 22 & 24 & 20 & 21 & 20 & 21 \\\hline B & 16 & 18 & 14 & 15 \\\hline C & 19 & 21 & 17 & 18 & 21 & 17 & 18 \\\hline D & 11 & 13 & 9 & 10 \\\hline \end{tabular}$	$\begin{array}{c c c c c} \Phi: \Psi & p & q & r & s \\ \hline A & 22 & 24 & 20 & 21 \\ \hline B & 16 & 18 & 14 & 15 \\ \hline C & 19 & 21 & 17 & 18 \\ \hline D & 11 & 13 & 9 & 10 \end{array}$	$\begin{array}{c c c c c c} \Phi \circ \Psi & p & q & r & s \\ \hline A & 22 & 24 & 20 & 21 \\ \hline B & 16 & 18 & 14 & 15 \\ \hline C & 19 & 21 & 17 & 18 \\ \hline D & 11 & 13 & 9 & 10 \end{array}$
fong-spivak-invitation_TAB_265	325	■ 0.987	$\begin{tabular}{t}{ l l l l l l }\hline & r1 & r2 & r3 & c1 & i1 \\\hline \backslash(s(-)\backslash) & dl & ul & ur & ul & dl \\\hline \backslash(t(-)\backslash) & ul & ur & dr & ur & dr \\\hline \backslash(\ell(-)\backslash) & \backslash(1\Omega\backslash) & \backslash(2\Omega\backslash) & \backslash(1\Omega\backslash) & \backslash(3F\backslash) & \backslash(1H\backslash) \\\hline \end{tabular}$	$\begin{array}{c c c c c c} & r1 & r2 & r3 & c1 & i1 \\ \hline s(-) & dl & ul & ur & ul & dl \\ \hline t(-) & ul & ur & dr & ur & dr \\ \hline \ell(-) & 1\Omega & 2\Omega & 1\Omega & 3F & 1H \end{array}$	$\begin{array}{c c c c c c} & r1 & r2 & r3 & c1 & i1 \\ \hline s(-) & dl & ul & ur & ul & dl \\ \hline t(-) & ul & ur & dr & ur & dr \\ \hline \ell(-) & 1\Omega & 2\Omega & 1\Omega & 3F & 1H \end{array}$
fong-spivak-invitation_TAB_265	325	■ 0.879	$\begin{tabular}{t}{ l l l }\hline & 1 & 2 \\\hline \backslash(f^{\backslash\prime}(-)\backslash) & l & d \\\hline \backslash(g^{\backslash\prime}(-)\backslash) & r & r \\\hline \end{tabular}$	$\begin{array}{c c c} & 1 & 2 \\ \hline f'(-) & l & d \\ \hline g'(-) & r & r \end{array}$	$\begin{array}{c c c} & 1 & 2 \\ \hline f'(-) & l & d \\ \hline g'(-) & r & r \end{array}$
fong-spivak-invitation_TAB_265	325	■ 0.948	$\begin{tabular}{t}{ l l l }\hline & r1' & r2' \\\hline \backslash(s(-)\backslash) & l & r \\\hline \backslash(t(-)\backslash) & r & d \\\hline \backslash(\ell(-)\backslash) & 5\Omega & 8\Omega \\\hline \end{tabular}$	$\begin{array}{c c c} & r1' & r2' \\ \hline s(-) & l & r \\ \hline t(-) & r & d \\ \hline \ell(-) & 5\Omega & 8\Omega \end{array}$	$\begin{array}{c c c} & r1' & r2' \\ \hline s(-) & l & r \\ \hline t(-) & r & d \\ \hline \ell(-) & 5\Omega & 8\Omega \end{array}$
fong-spivak-invitation_TAB_265	325	■ 1.000	$\begin{tabular}{t}{ l l l l l l l l }\hline & r1 & r2 & r3 & c1 & i1 & r1' & r2' \\\hline \backslash(s^{\backslash\prime\prime}(-)\backslash) & dl & ul & m & ul & dl & m & r \\\hline \backslash(t^{\backslash\prime\prime}(-)\backslash) & ul & m & dr & m & dr & r & m \\\hline \backslash(\ell^{\backslash\prime\prime}(-)\backslash) & 1\Omega & 2\Omega & 1\Omega & 3F & 1H & 5\Omega & 8\Omega \\\hline \backslash(1\Omega\backslash) & \backslash(3F\backslash) & \backslash(1H\backslash) & \backslash(5\Omega\backslash) & \backslash(8\Omega\backslash) \\\hline \end{tabular}$	$\begin{array}{c c c c c c c c} & r1 & r2 & r3 & c1 & i1 & r1' & r2' \\ \hline s''(-) & dl & ul & m & ul & dl & m & r \\ \hline t''(-) & ul & m & dr & m & dr & r & m \\ \hline \ell''(-) & 1\Omega & 2\Omega & 1\Omega & 3F & 1H & 5\Omega & 8\Omega \end{array}$	$\begin{array}{c c c c c c c c} & r1 & r2 & r3 & c1 & i1 & r1' & r2' \\ \hline s''(-) & dl & ul & m & ul & dl & m & r \\ \hline t''(-) & ul & m & dr & m & dr & r & m \\ \hline \ell''(-) & 1\Omega & 2\Omega & 1\Omega & 3F & 1H & 5\Omega & 8\Omega \end{array}$
fong-spivak-invitation_TAB_361	331	■ 1.000	$\begin{tabular}{t}{ l l l l }\hline \backslash(P\backslash) & \backslash(Q\backslash) & \backslash(P\wedge Q\backslash) & \backslash(P=(P\wedge Q)\backslash) \\\hline true & true & true & true \\ \hline true & false & false & false \\ \hline false & true & false & true \\ \hline false & false & false & true \\\hline \end{tabular}$	$\begin{array}{c c c c} P & Q & P\wedge Q & P=(P\wedge Q) \\ \hline true & true & true & true \\ \hline true & false & false & false \\ \hline false & true & false & true \\ \hline false & false & false & true \end{array}$	$\begin{array}{c c c c} P & Q & P\wedge Q & P=(P\wedge Q) \\ \hline true & true & true & true \\ \hline true & false & false & false \\ \hline false & true & false & true \\ \hline false & false & false & true \end{array}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

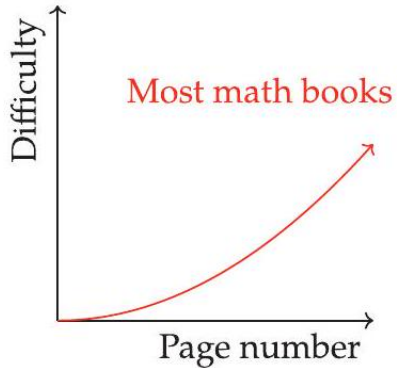
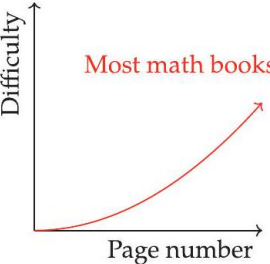
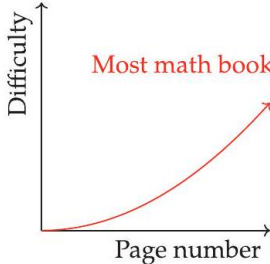
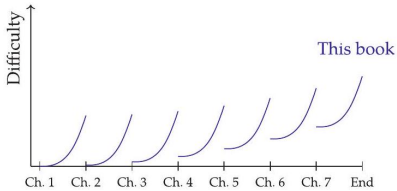
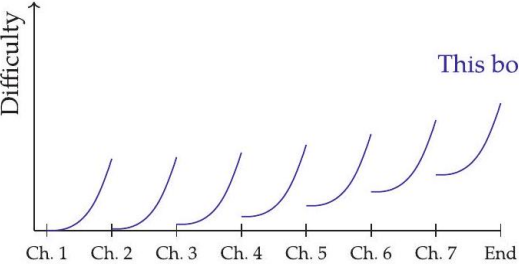
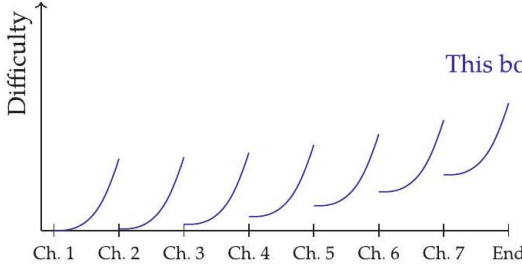
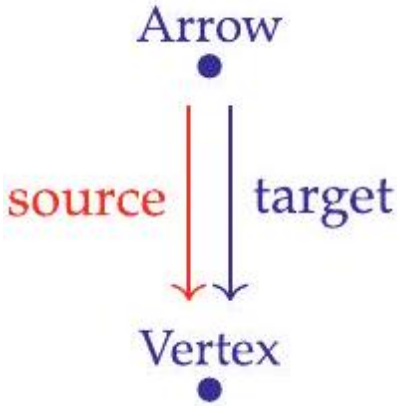
LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
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fong-spivak-invitation_P10_160	013	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-013_348_724_1012_822.jpg			
fong-spivak-invitation_P12_296	296	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_214_198_1515_552.jpg			

contested: 4 rows

share this crop

source

source

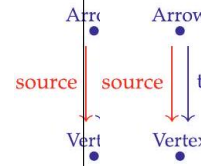
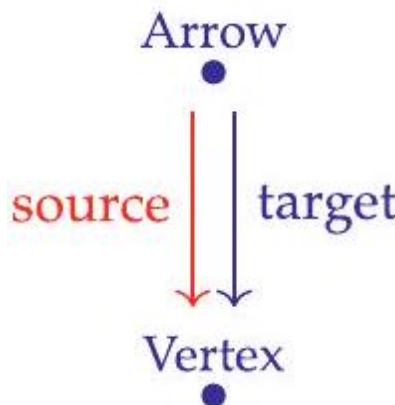
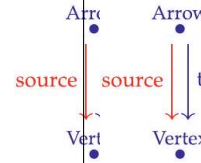
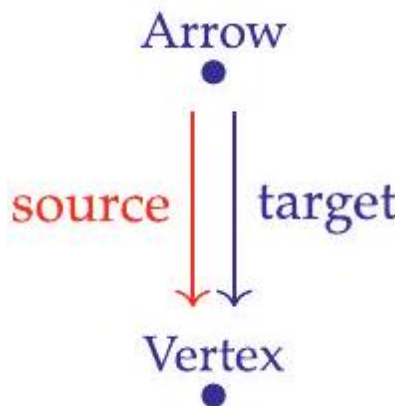
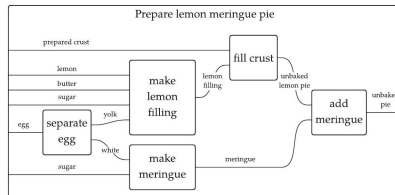
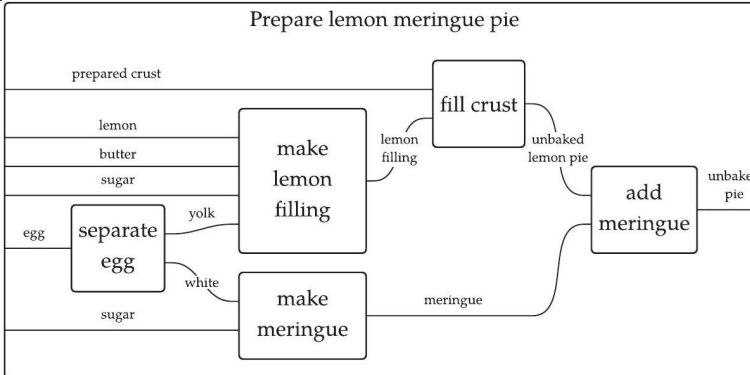
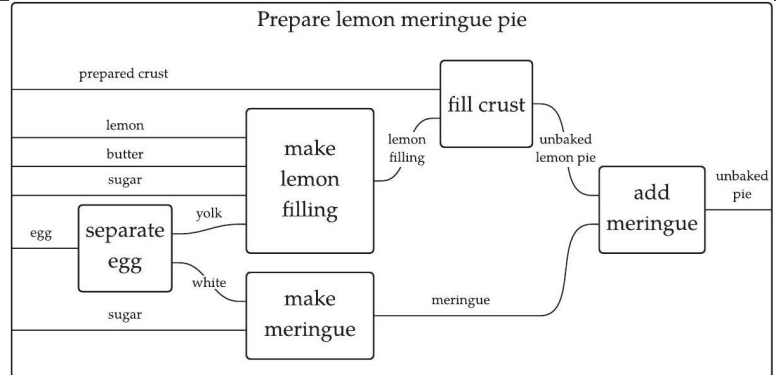
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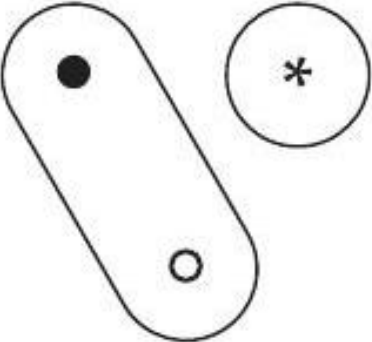
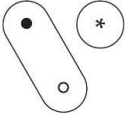
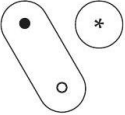
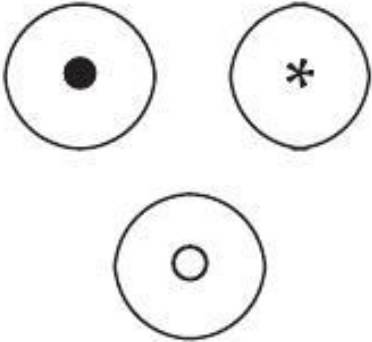
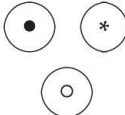
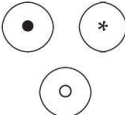
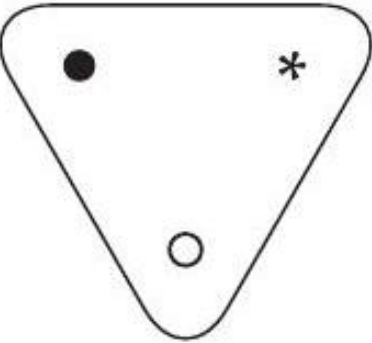
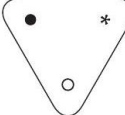
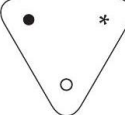
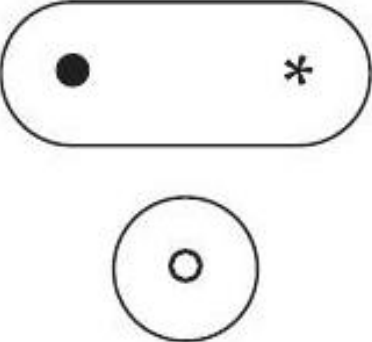
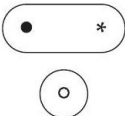
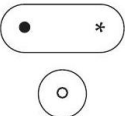
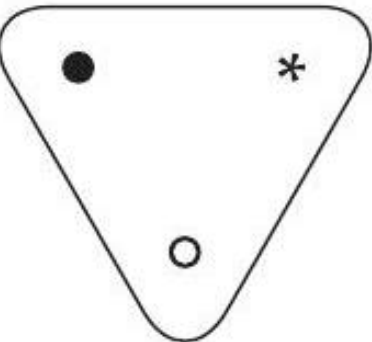
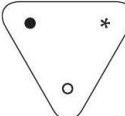
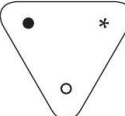
Vert

Vertex

Arrow

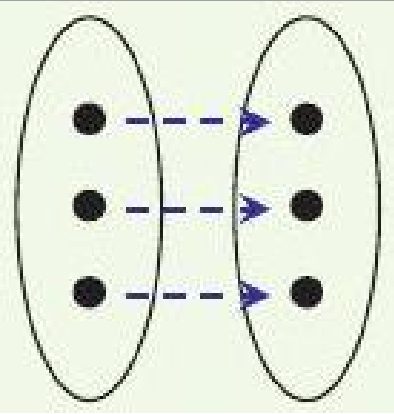
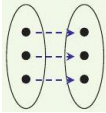
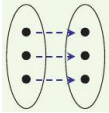
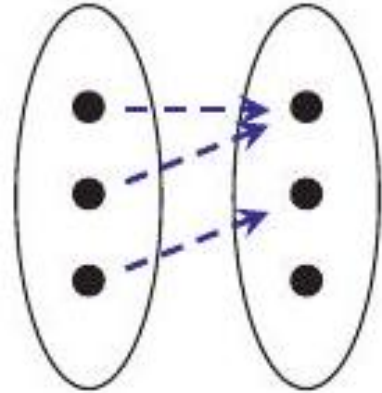
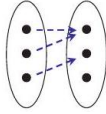
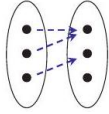
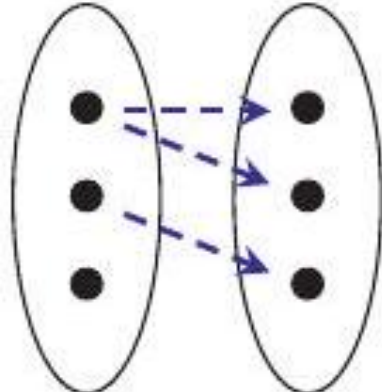
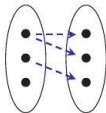
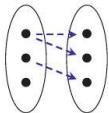
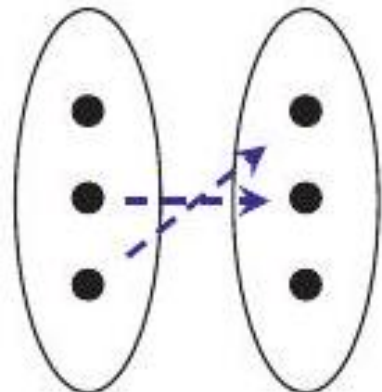
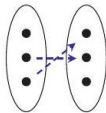
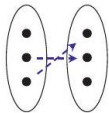
Arrow

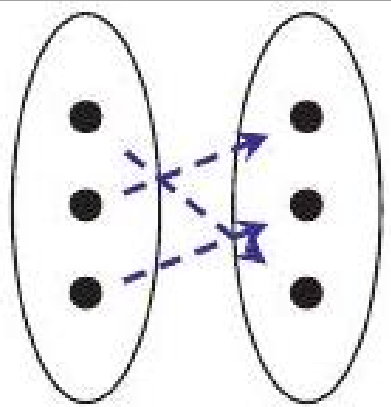
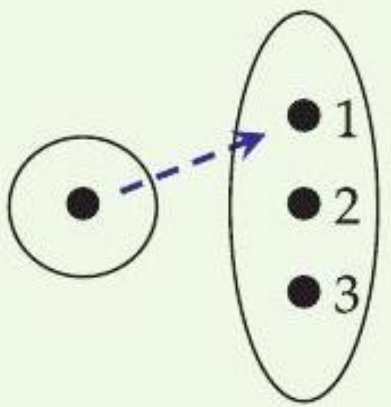
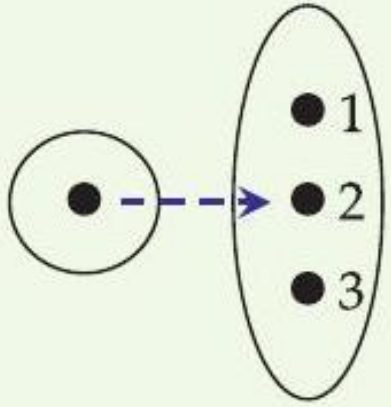
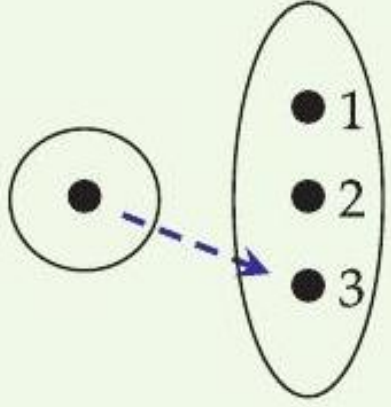
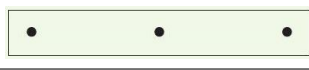
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fong-spivak-invitation_P12_500 contested: 4 rows share this crop	296 — (dup)		05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_214_198_1515_552.jpg			
fong-spivak-invitation_D10_102	012 — (dup)		05b3e583-fc72-4ff5-9914-6f882b9df5b5-012_600_1206_502_338.jpg			

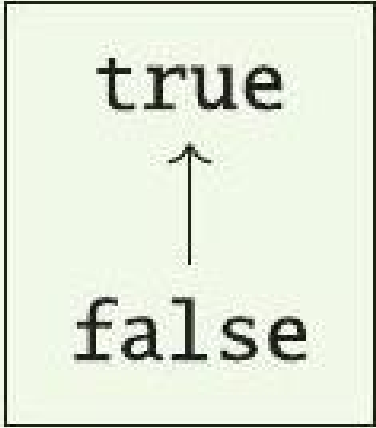
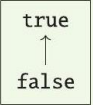
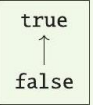
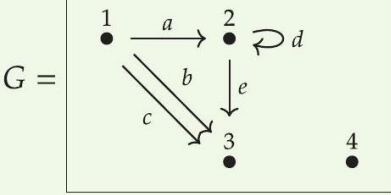
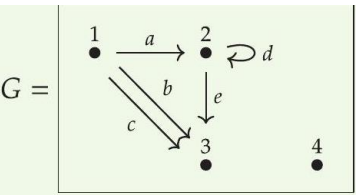
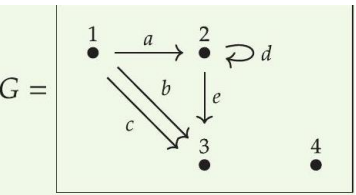
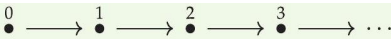
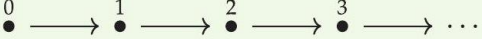
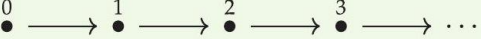
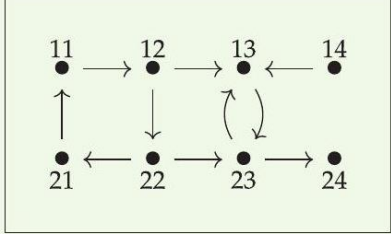
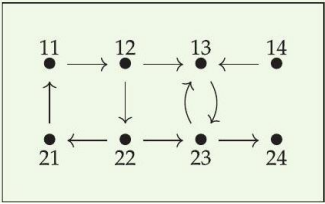
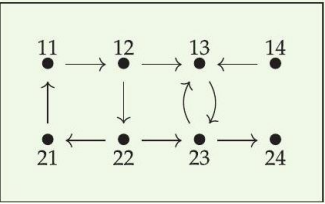
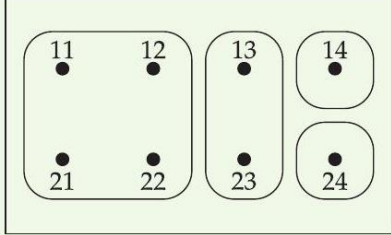
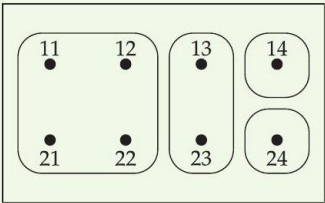
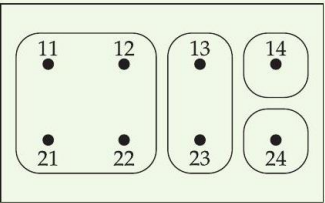
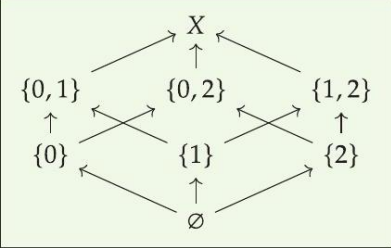
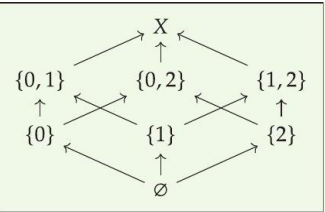
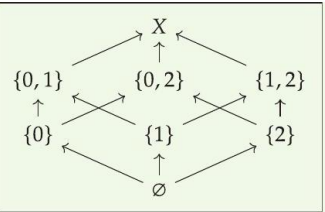
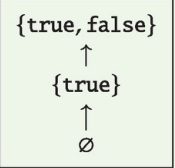
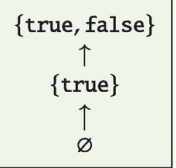
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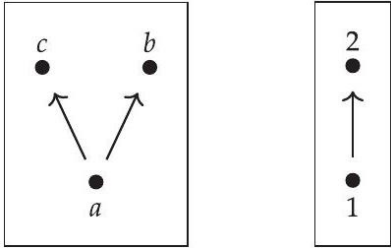
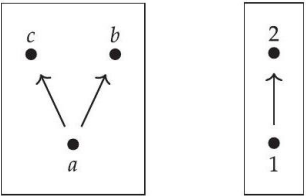
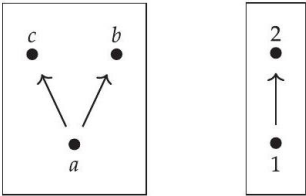
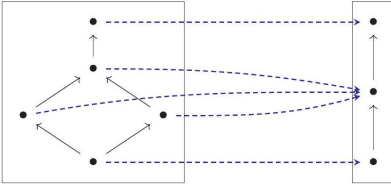
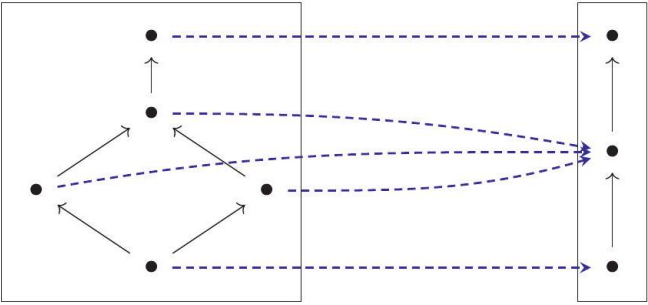
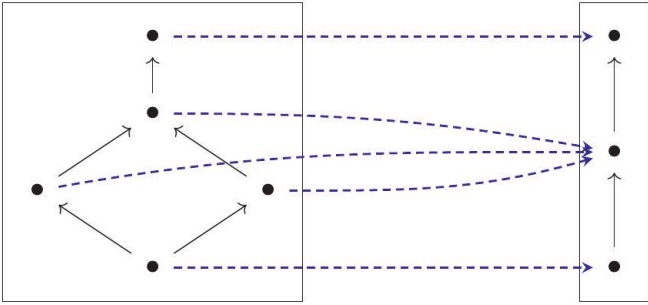
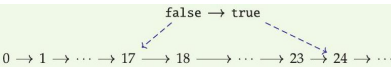
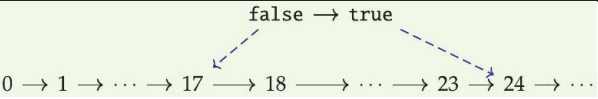
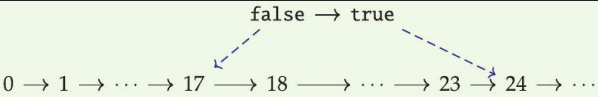
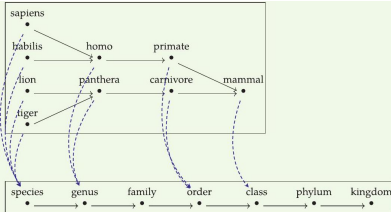
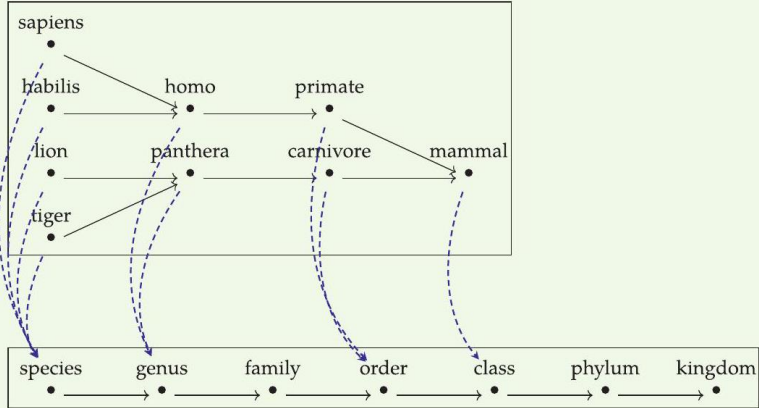
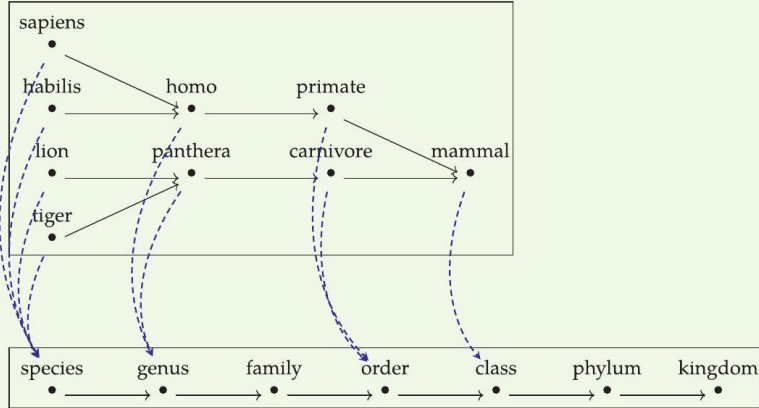
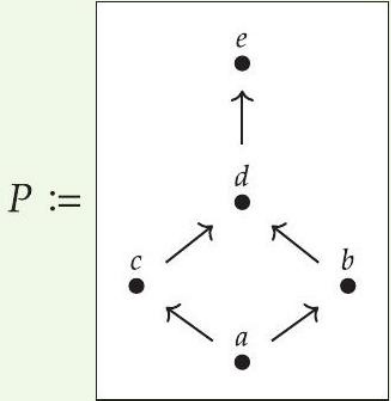
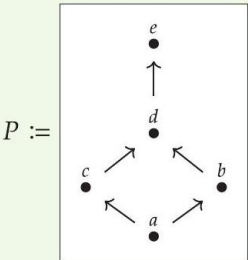
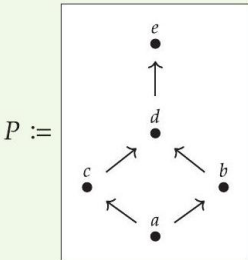
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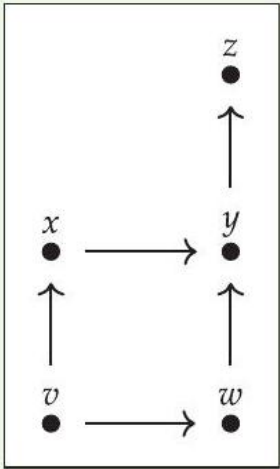
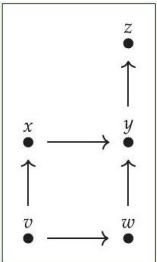
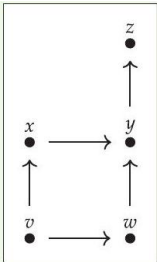
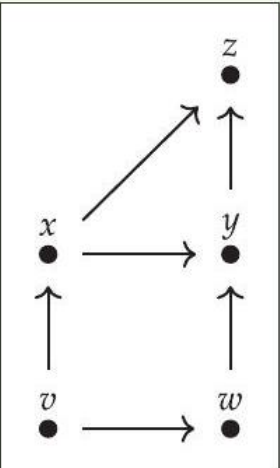
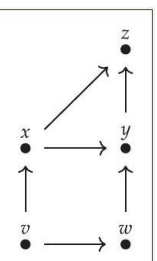
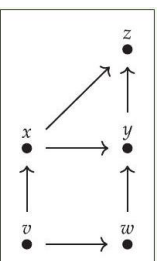
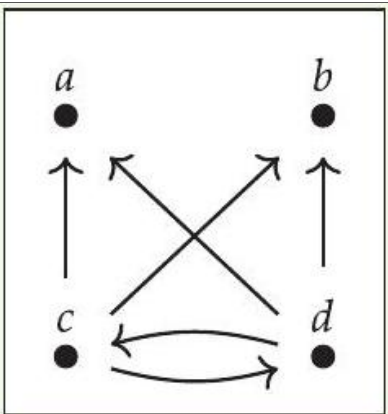
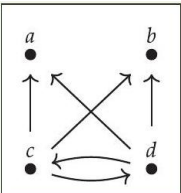
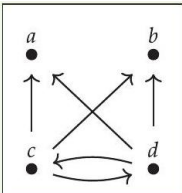
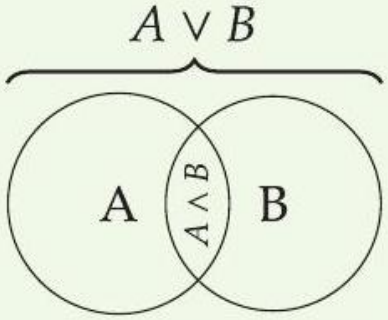
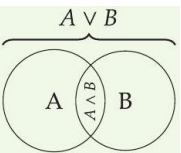
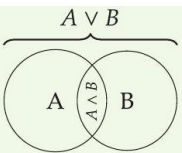
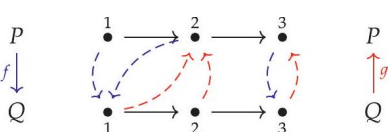
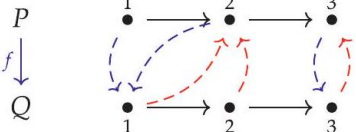
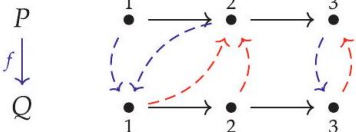
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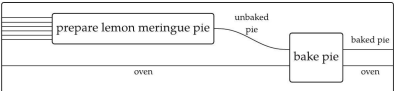
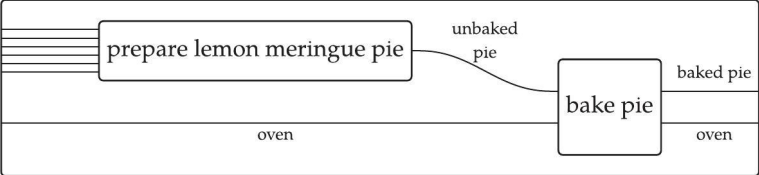
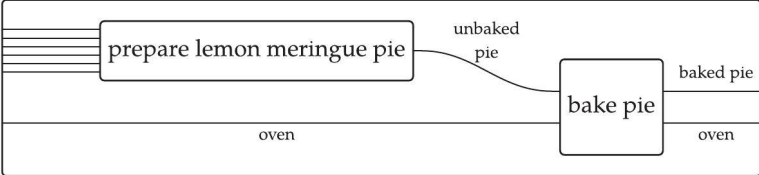
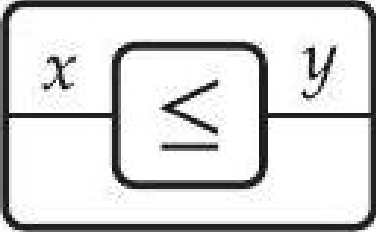
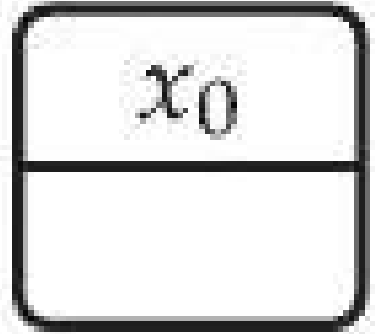
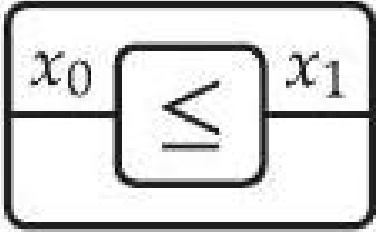
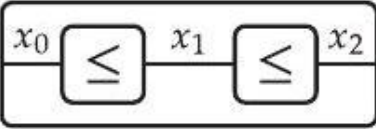
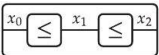
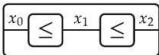
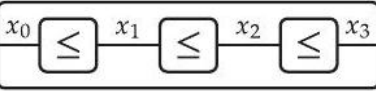
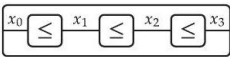
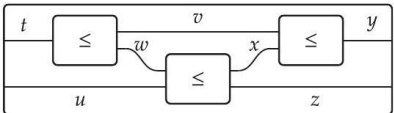
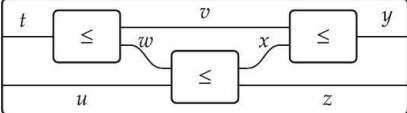
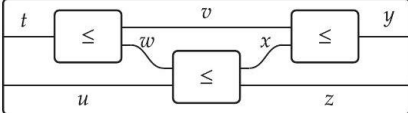
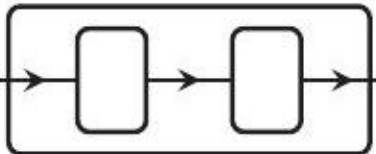
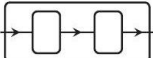
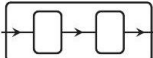
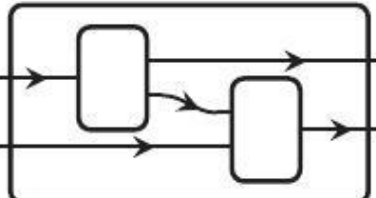
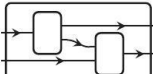
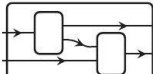
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fong-spivak-invitation_D10_031	031	— (dup)	no image for this region (500 in zip)	(no tex.zip crop for this row)		

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fong-spivak-invitation_D10_033	033	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-033_124_780_1148_548.jpg			
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fong-spivak-invitation_D10_036	036	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-036_344_330_1188_410.jpg			

fong-spivak-invitation_D0_036.jpg	036	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-036_344_286_1188_818.jpg	$Q :=$ 	$Q :=$ 	$Q :=$ 
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fong-spivak-invitation_D0_039.jpg	039	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-039_194_236_1657_820.jpg	$A \vee B$ 	$A \vee B$ 	$A \vee B$ 
fong-spivak-invitation_D0_042.jpg	042	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-042_186_602_996_661.jpg			

fong-spivak-invitation_D0_442	042	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-042_190_604_1364_659.jpg			
fong-spivak-invitation_D0_443	043	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-043_174_228_824_580.jpg			
fong-spivak-invitation_D0_443	043	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-043_180_252_818_1046.jpg			
fong-spivak-invitation_D0_443	043	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-043_310_846_1853_514.jpg			
fong-spivak-invitation_D0_444	044	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-044_266_1178_578_350.jpg			
fong-spivak-invitation_D0_446	046	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-046_312_800_864_540.jpg			
fong-spivak-invitation_D0_050	050	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-050_60_234_474_822.jpg			
fong-spivak-invitation_D0_053	053	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-053_598_1207_270_336.jpg			

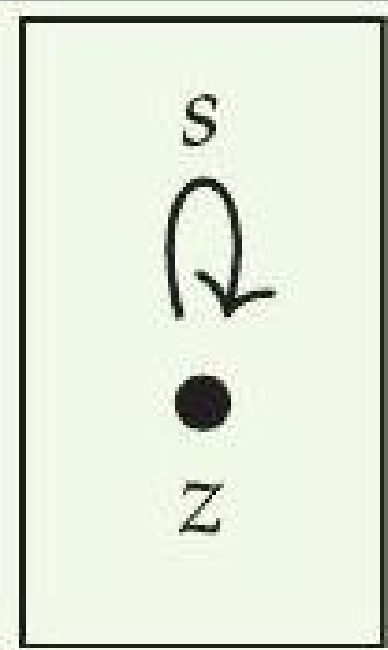
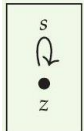
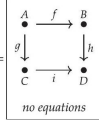
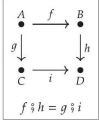
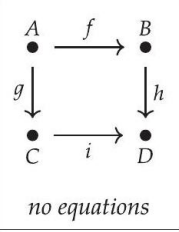
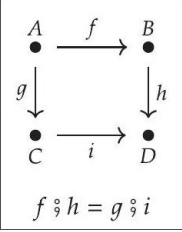
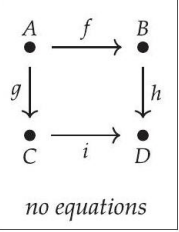
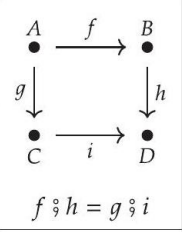
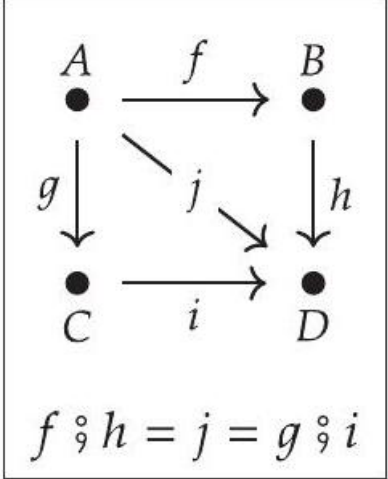
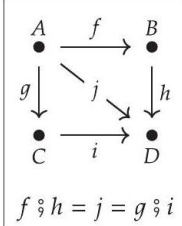
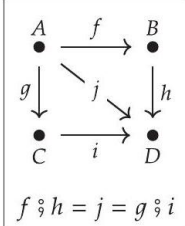
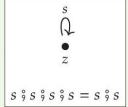
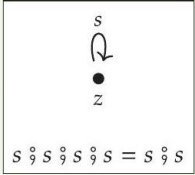
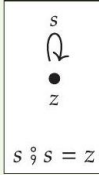
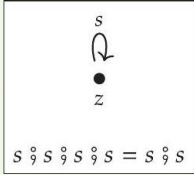
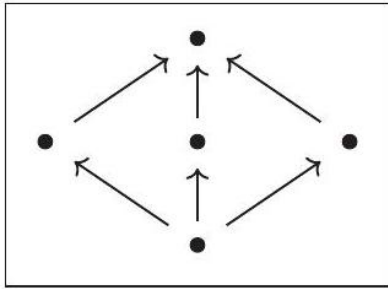
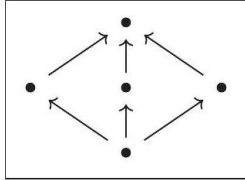
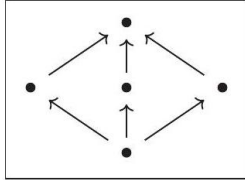
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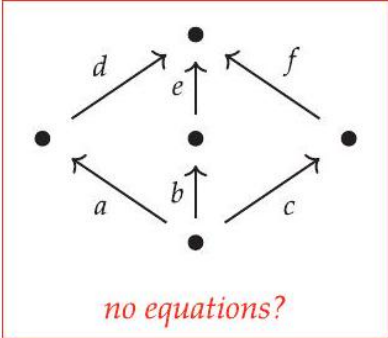
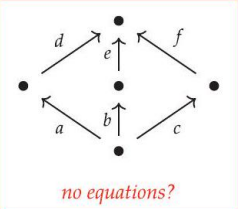
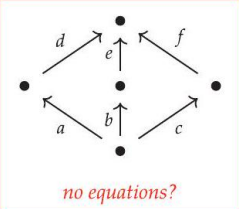
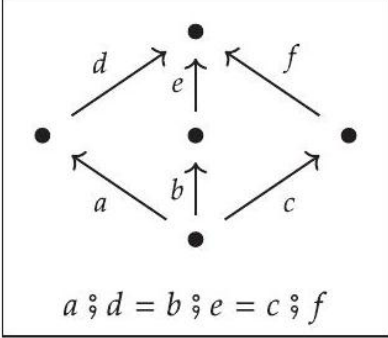
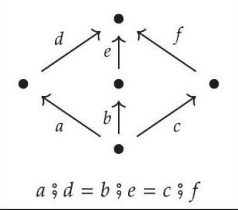
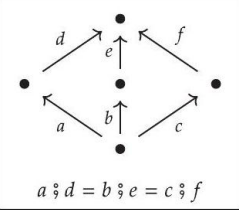
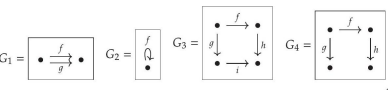
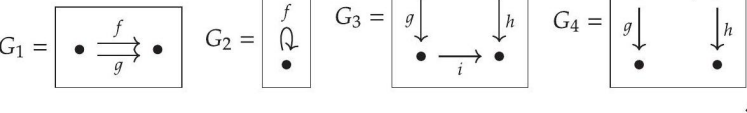
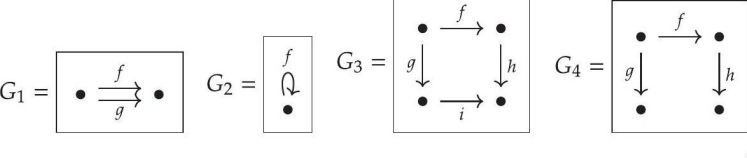
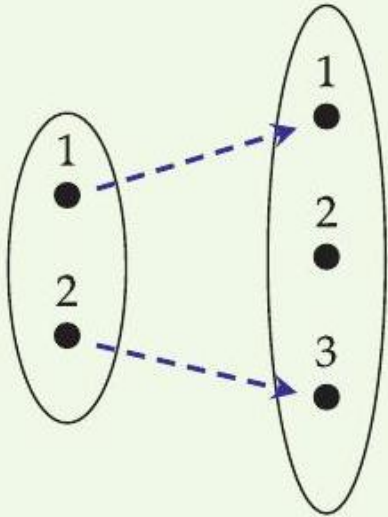
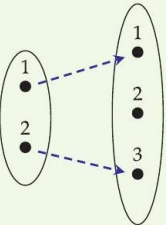
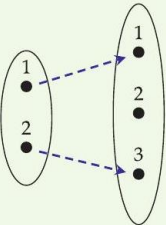
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fong-spivak-invitation_D0_067	058	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-058_66_548_1955_668.jpg			
fong-spivak-invitation_D0_068	059	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-059_200_606_268_634.jpg			
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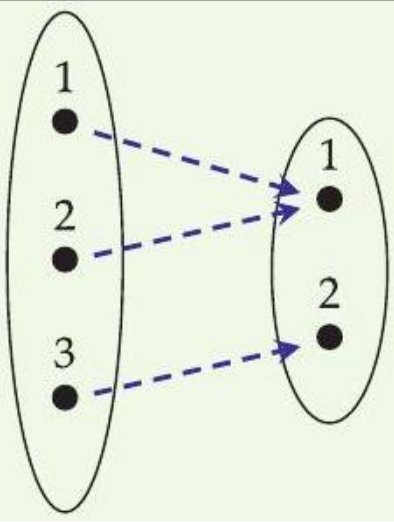
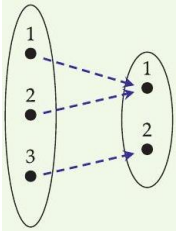
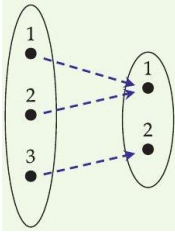
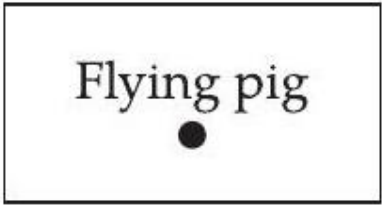
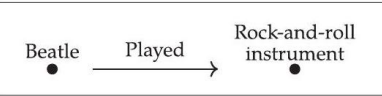
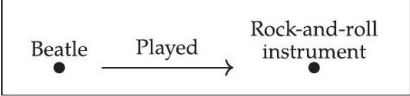
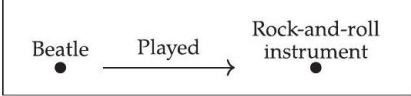
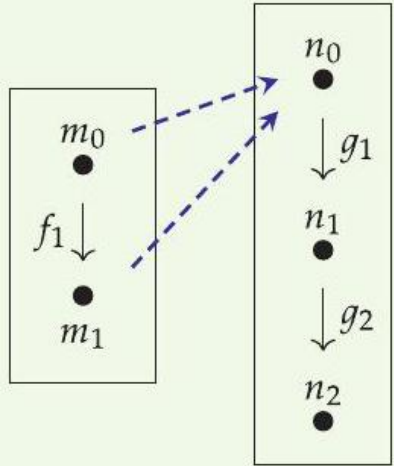
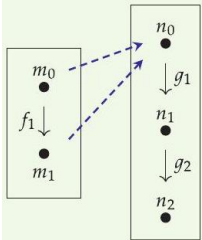
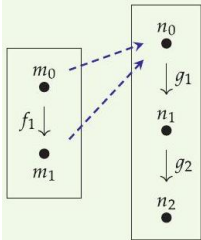
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fong-spivak-invitation_D0_070	070	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-070_262_322_1673_798.jpg			

fong-spivak-invitation_D0_068	073	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-073_203_327_382_775.jpg			
fong-spivak-invitation_D0_068	075	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-075_330_384_602_398.jpg	$X :=$	$X :=$	$X :=$
fong-spivak-invitation_D0_068	075	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-075_330_344_604_1052.jpg			
fong-spivak-invitation_D0_068	080	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-080_116_444_2017_540.jpg	$\mathcal{X} :=$	$\mathcal{X} :=$	$\mathcal{X} :=$
fong-spivak-invitation_D0_068	080	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-080_288_234_1939_1108.jpg			
fong-spivak-invitation_D0_068	081	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-081_328_652_536_610.jpg	$\mathcal{X} \times \mathcal{Y} =$	$\mathcal{X} \times \mathcal{Y} =$	$\mathcal{X} \times \mathcal{Y} =$

fong-spivak-invitation_Dm_088	088	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-088_278_346_670_370.jpg	$Y :=$	$Y :=$	$Y :=$
fong-spivak-invitation_Dm_092	092	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-092_294_1042_814_386.jpg	easySchema :=	easySchema :=	easySchema :=
fong-spivak-invitation_Dm_093	093	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-093_396_1032_258_388.jpg	mySchema :=	mySchema :=	mySchema :=
fong-spivak-invitation_Dm_093	093	— (dup)	no image for this region (500 in zip)	(no tex.zip crop for this row)	schema mySchema = { nodes Employee, Department; attributes DName : Department -> string, FName : Employee -> string; arrows Mngr : Employee -> Employee, WorksIn : Employee -> Department, Secr : Department -> Employee; equations Department.Secr.WorksIn = Department, Employee.Mngr.WorksIn = Employee.WorksIn; }	schema mySchema = { nodes Employee, Department; attributes DName : Department -> string, FName : Employee -> string; arrows Mngr : Employee -> Employee, WorksIn : Employee -> Department, Secr : Department -> Employee; equations Department.Secr.WorksIn = Department, Employee.Mngr.WorksIn = Employee.WorksIn; }
fong-spivak-invitation_Dm_094	094	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-094_488_1194_620_342.jpg	$A :=$	$A :=$	$A :=$

fong-spivak-invitation_D10_097	097	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-097_176_106_962_884.jpg			
fong-spivak-invitation_D10_098	098	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-098_312_984_550_446.jpg	Free_square :=  Comm_square := 	Free_square :=  Comm_square := 	Free_square :=  Comm_square := 
fong-spivak-invitation_D10_098	098	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-098_306_246_1937_820.jpg			
fong-spivak-invitation_D10_099	099	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-099_234_810_452_534.jpg	$\mathcal{C} :=$  $\mathcal{D} :=$ 	$\mathcal{C} :=$  $\mathcal{D} :=$ 	$\mathcal{C} :=$  $\mathcal{D} :=$ 
fong-spivak-invitation_D10_100	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_238_322_314_364.jpg			

fong-spivak-invitation_D1_000	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_282_320_270_780.jpg	 no equations?	 no equations?	 no equations?
fong-spivak-invitation_D1_000	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_282_320_270_1196.jpg	 $a \circ d = b \circ e = c \circ f$	 $a \circ d = b \circ e = c \circ f$	 $a \circ d = b \circ e = c \circ f$
fong-spivak-invitation_D1_000	100	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-100_272_1202_1018_346.jpg			
fong-spivak-invitation_D1_000	103	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-103_296_222_1481_578.jpg			

fong-spivak-invitation_D11_043	103	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-103_298_226_1481_1070.jpg			
fong-spivak-invitation_D11_044	104	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-104_110_238_1509_342.jpg			
fong-spivak-invitation_D11_044	104	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-104_110_262_1509_824.jpg			
fong-spivak-invitation_D11_044	104	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-104_114_210_1505_1331.jpg			
fong-spivak-invitation_D11_045	105	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-105_134_540_626_668.jpg			
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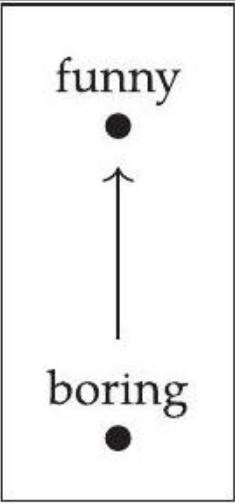
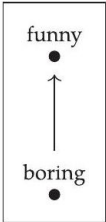
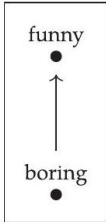
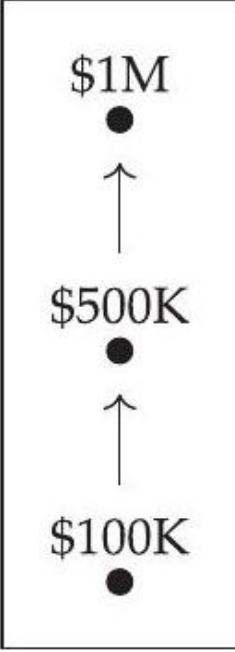
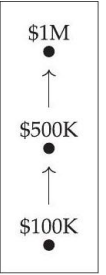
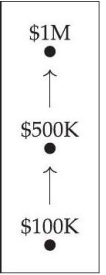
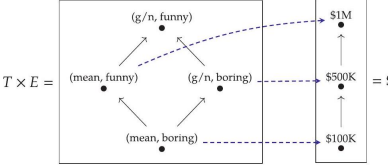
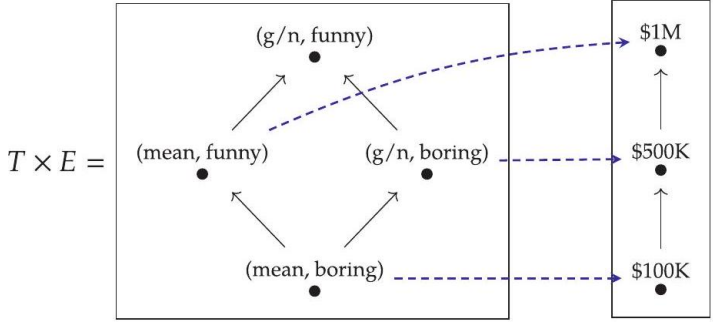
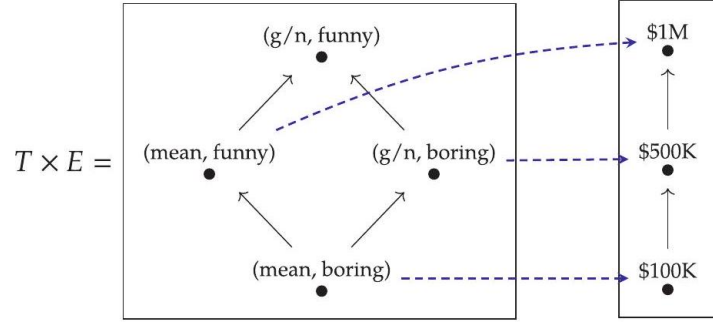
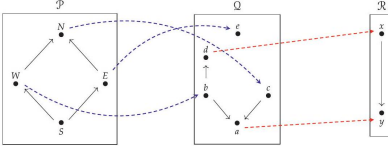
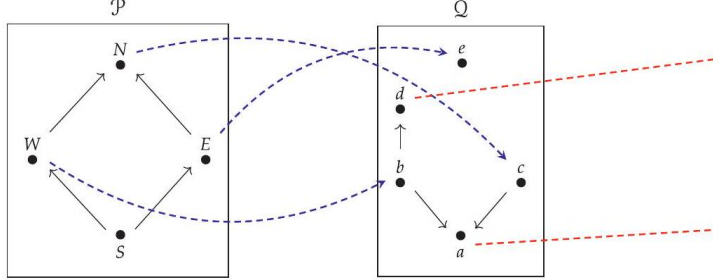
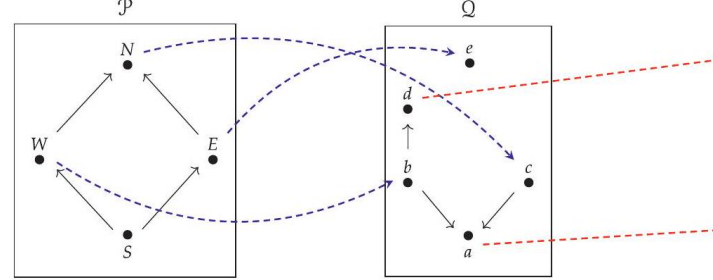
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fong-spivak-invitation_D11_015	105	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-105_308_260_1921_1251.jpg			
fong-spivak-invitation_D11_016	106	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-106_312_1056_954_410.jpg			
fong-spivak-invitation_D11_019	109	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-109_222_132_1048_638.jpg			

fong-spivak-invitation_D1109	109	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-109_174_342_1098_934.jpg	$a \xrightarrow{f} b \xrightarrow{g} c$ $\quad \quad \quad \searrow h$ $f \circ g = f \circ h$	$a \xrightarrow{f} b \xrightarrow{g} c$ $\quad \quad \quad \searrow h$ $f \circ g = f \circ h$	$a \xrightarrow{f} b \xrightarrow{g} c$ $\quad \quad \quad \searrow h$ $f \circ g = f \circ h$
fong-spivak-invitation_D1109	109	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-109_148_230_1937_822.jpg	$\mathcal{C} \xrightarrow{F} \mathcal{D}$ $\quad \downarrow \alpha$ $\mathcal{C} \xrightarrow{G} \mathcal{D}$	$\mathcal{C} \xrightarrow{F} \mathcal{D}$ $\quad \downarrow \alpha$ $\mathcal{C} \xrightarrow{G} \mathcal{D}$	$\mathcal{C} \xrightarrow{F} \mathcal{D}$ $\quad \downarrow \alpha$ $\mathcal{C} \xrightarrow{G} \mathcal{D}$
fong-spivak-invitation_D1110	110	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-110_200_342_1068_770.jpg	$F(c) \xrightarrow{\alpha_c} G(c)$ $F(f) \downarrow \quad \downarrow G(f)$ $F(d) \xrightarrow{\alpha_d} G(d)$	$F(c) \xrightarrow{\alpha_c} G(c)$ $F(f) \downarrow \quad \downarrow G(f)$ $F(d) \xrightarrow{\alpha_d} G(d)$	$F(c) \xrightarrow{\alpha_c} G(c)$ $F(f) \downarrow \quad \downarrow G(f)$ $F(d) \xrightarrow{\alpha_d} G(d)$
fong-spivak-invitation_D1110	110	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-110_276_1192_1801_344.jpg	$\mathcal{C} := \begin{array}{ccc} 1 & \xrightarrow{f} & 2 \\ \downarrow G & & \downarrow G \\ 1 & \xrightarrow{f} & 2 \end{array}$	$\mathcal{C} := \begin{array}{ccc} 1 & \xrightarrow{f} & 2 \\ \downarrow G & & \downarrow G \\ 1 & \xrightarrow{f} & 2 \end{array}$	$\mathcal{C} := \begin{array}{ccc} 1 & \xrightarrow{f} & 2 \\ \downarrow G & & \downarrow G \\ 1 & \xrightarrow{f} & 2 \end{array}$
fong-spivak-invitation_D1114	114	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-114_150_450_388_400.jpg	$G(\text{Arrow}) \xrightarrow{\alpha_{\text{Arrow}}} H(\text{Arrow})$ $G(\text{source}) \downarrow \quad \downarrow H(\text{source})$ $G(\text{Vertex}) \xrightarrow{\alpha_{\text{Vertex}}} H(\text{Vertex})$	$G(\text{Arrow}) \xrightarrow{\alpha_{\text{Arrow}}} H(\text{Arrow})$ $G(\text{source}) \downarrow \quad \downarrow H(\text{source})$ $G(\text{Vertex}) \xrightarrow{\alpha_{\text{Vertex}}} H(\text{Vertex})$	$G(\text{Arrow}) \xrightarrow{\alpha_{\text{Arrow}}} H(\text{Arrow})$ $G(\text{source}) \downarrow \quad \downarrow H(\text{source})$ $G(\text{Vertex}) \xrightarrow{\alpha_{\text{Vertex}}} H(\text{Vertex})$
fong-spivak-invitation_D1114	114	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-114_150_434_388_1042.jpg	$G(\text{Arrow}) \xrightarrow{\alpha_{\text{Arrow}}} H(\text{Arrow})$ $G(\text{target}) \downarrow \quad \downarrow H(\text{target})$ $G(\text{Vertex}) \xrightarrow{\alpha_{\text{Vertex}}} H(\text{Vertex})$	$G(\text{Arrow}) \xrightarrow{\alpha_{\text{Arrow}}} H(\text{Arrow})$ $G(\text{target}) \downarrow \quad \downarrow H(\text{target})$ $G(\text{Vertex}) \xrightarrow{\alpha_{\text{Vertex}}} H(\text{Vertex})$	$G(\text{Arrow}) \xrightarrow{\alpha_{\text{Arrow}}} H(\text{Arrow})$ $G(\text{target}) \downarrow \quad \downarrow H(\text{target})$ $G(\text{Vertex}) \xrightarrow{\alpha_{\text{Vertex}}} H(\text{Vertex})$
fong-spivak-invitation_D1115	115	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-115_204_1112_944_380.jpg	$\text{Gr} := \begin{array}{ccc} \text{Arrow} & \xrightarrow{\text{source}} & \text{Vertex} \\ & \text{target} & \\ & \text{no equations} & \end{array}$	$\text{Gr} := \begin{array}{ccc} \text{Arrow} & \xrightarrow{\text{source}} & \text{Vertex} \\ & \text{target} & \\ & \text{no equations} & \end{array}$	$\text{Gr} := \begin{array}{ccc} \text{Arrow} & \xrightarrow{\text{source}} & \text{Vertex} \\ & \text{target} & \\ & \text{no equations} & \end{array}$
fong-spivak-invitation_D1116	116	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-116_328_542_272_670.jpg	$\text{Gr} \xrightarrow{F} \text{DDS}$	$\text{Gr} \xrightarrow{F} \text{DDS}$	$\text{Gr} \xrightarrow{F} \text{DDS}$
fong-spivak-invitation_D1116	116	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-116_374_768_1869_556.jpg	$\text{Gr} \xrightarrow{F} \text{DDS}$	$\text{Gr} \xrightarrow{F} \text{DDS}$	$\text{Gr} \xrightarrow{F} \text{DDS}$

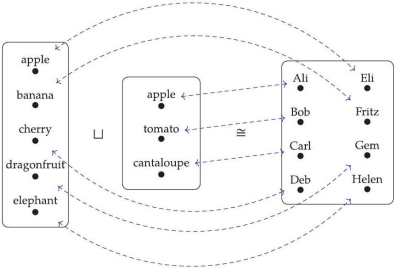
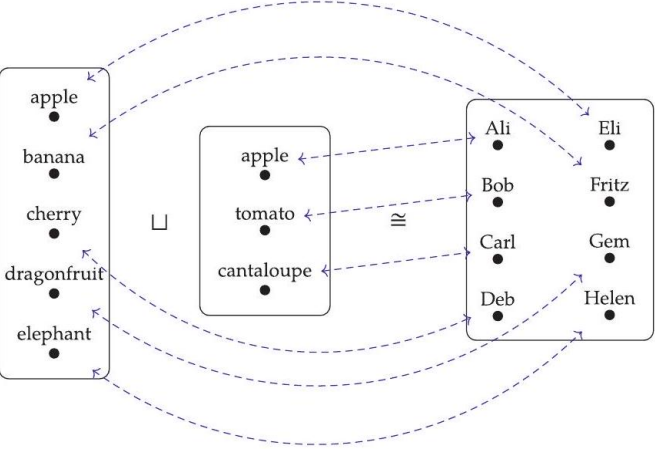
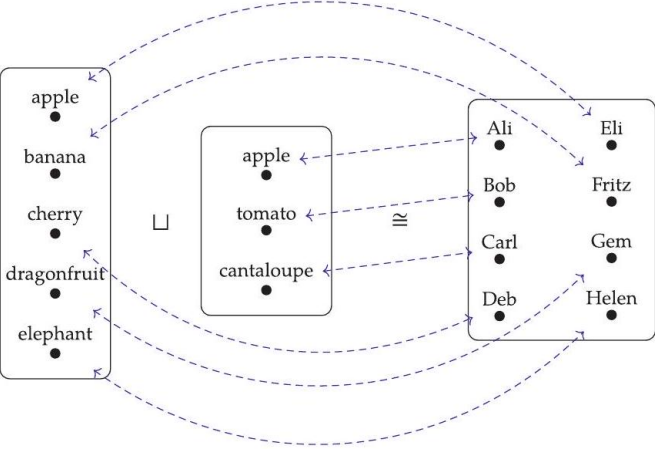
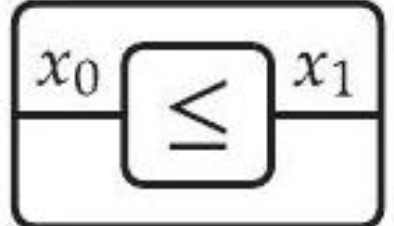
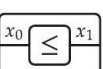
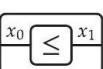
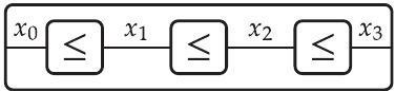
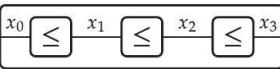
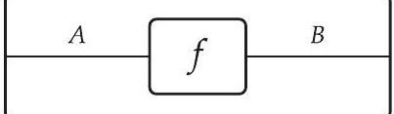
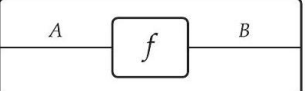
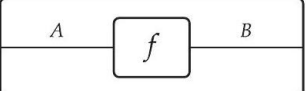
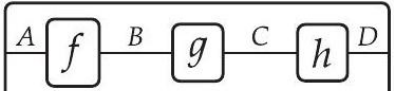
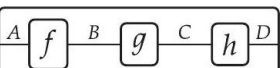
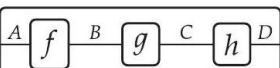
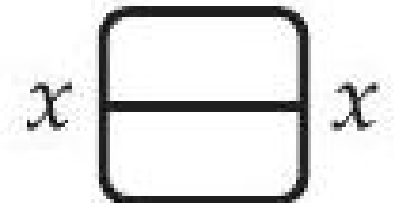
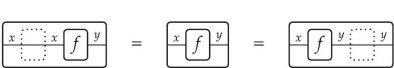
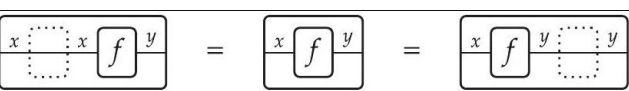
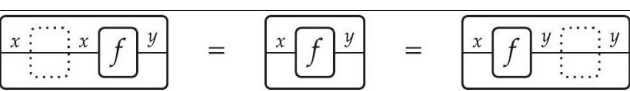
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fong-spivak-invitation_D11_118	118	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-118_206_480_2059_712.jpg			
fong-spivak-invitation_D11_119	119	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-119_172_498_2115_690.jpg			
fong-spivak-invitation_D11_120	120	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-120_368_1215_1217_334.jpg			
fong-spivak-invitation_D11_124	124	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-124_184_350_712_766.jpg			
fong-spivak-invitation_D11_124	124	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-124_562_760_1665_560.jpg			
fong-spivak-invitation_D11_125	125	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-125_220_354_1893_762.jpg			

fong-spivak-invitation_D1_126	126	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-126_118_318_1357_782.jpg			
fong-spivak-invitation_D1_126	126	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-126_268_524_1625_370.jpg			
fong-spivak-invitation_D1_126	126	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-126_266_522_1627_988.jpg			
fong-spivak-invitation_D1_128	128	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-128_206_226_654_560.jpg			
fong-spivak-invitation_D1_128	128	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-128_208_294_652_1026.jpg			
fong-spivak-invitation_D1_128	128	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-128_512_786_1180_546.jpg			
fong-spivak-invitation_D1_132	132	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-132_462_1347_652_206.jpg			

fong-spivak-invitation_D11_014	134	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-134_156_512_1072_560.jpg	$\mathcal{X} :=$ 	$\mathcal{X} :=$ 	$\mathcal{X} :=$
fong-spivak-invitation_D11_014	134	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-134_152_200_1076_1116.jpg	$\mathcal{Y} :=$ 	$\mathcal{Y} :=$ 	$\mathcal{Y} :=$
fong-spivak-invitation_D11_014	136	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-136_486_1038_310_422.jpg	$X :=$ 	$X :=$ 	$X :=$
fong-spivak-invitation_D11_014	137	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-137_366_1014_460_398.jpg	$X :=$ 	$X :=$ 	$X :=$
fong-spivak-invitation_D11_014	138	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-138_194_360_716_760.jpg			
fong-spivak-invitation_D11_014	138	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-138_112_192_1745_844.jpg	T E 	T E 	T E
fong-spivak-invitation_D11_014	138	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-138_290_318_1979_450.jpg	$T :=$ 	$T :=$ 	$T :=$

fong-spivak-invitation_D1_138	138	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-138_290_226_1981_846.jpg	$E :=$ 	$E :=$ 	$E :=$ 
fong-spivak-invitation_D1_139	139	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-139_354_214_1949_1213.jpg	$\$:=$ 	$\$:=$ 	$\$:=$ 
fong-spivak-invitation_D1_140	140	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-140_406_1090_258_362.jpg		$T \times E =$ 	$T \times E =$ 
fong-spivak-invitation_D1_140	140	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-140_406_1090_258_362.jpg			

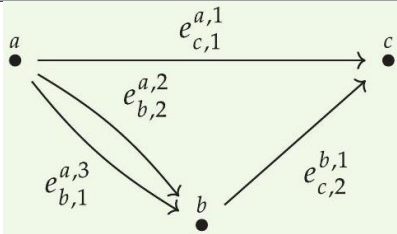
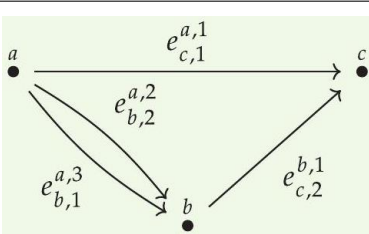
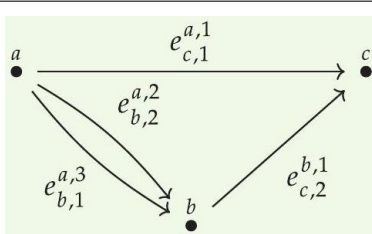
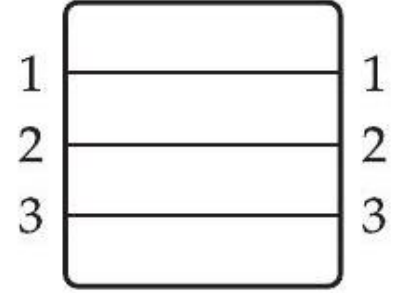
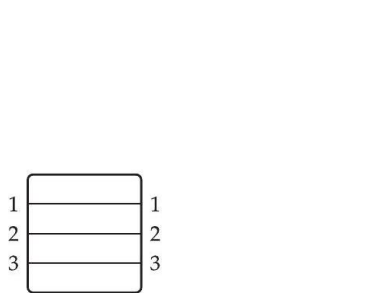
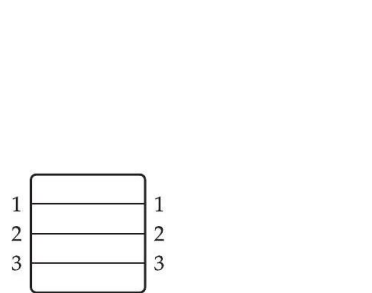
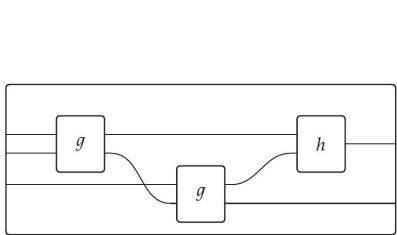
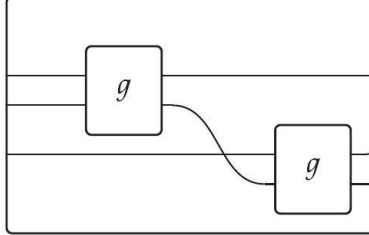
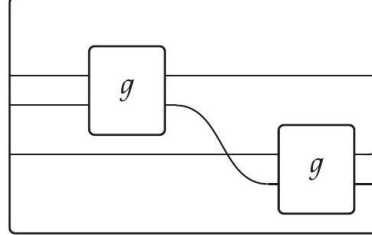
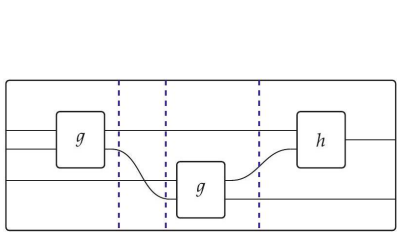
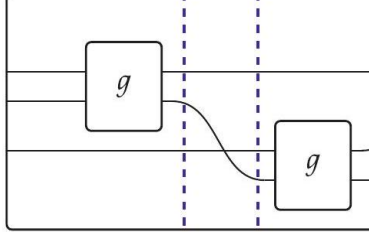
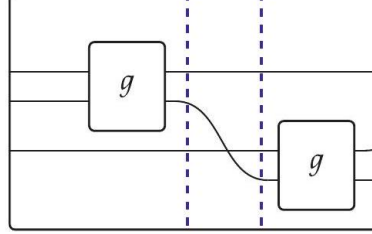
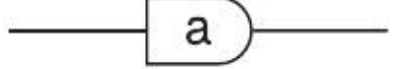
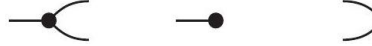
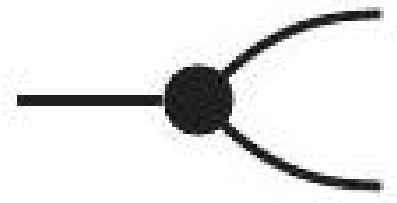
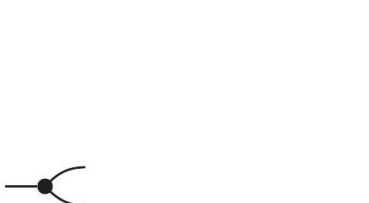
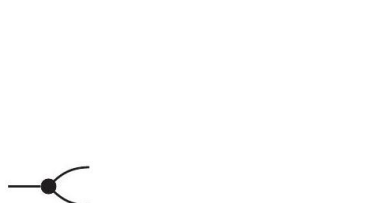
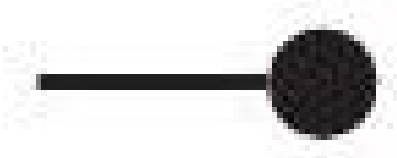
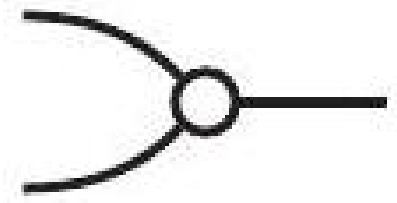
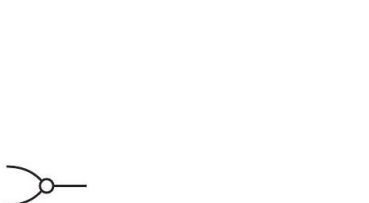
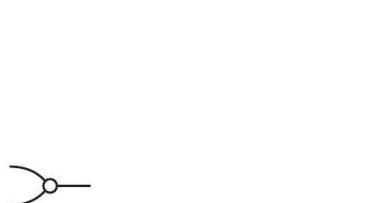
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fong-spivak-invitation_D11_413.jpg	143	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-143_104_700_1144_588.jpg			
fong-spivak-invitation_D11_414.jpg	143	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-143_418_1221_1439_330.jpg			
fong-spivak-invitation_D11_415.jpg	144	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-144_312_900_1581_488.jpg			
fong-spivak-invitation_D11_416.jpg	146	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-146_312_758_792_562.jpg			
fong-spivak-invitation_D11_417.jpg	146	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-146_342_944_1969_468.jpg			

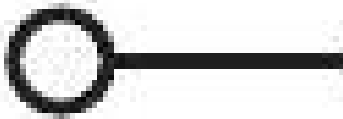
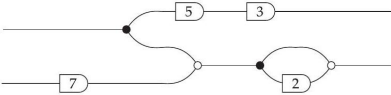
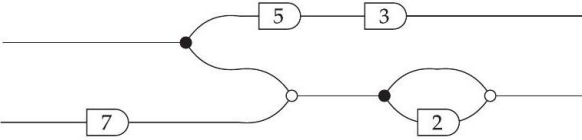
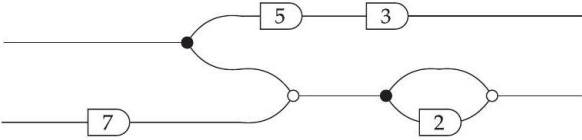
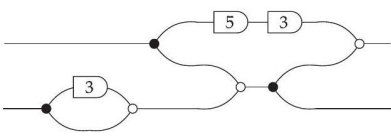
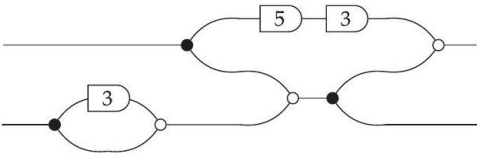
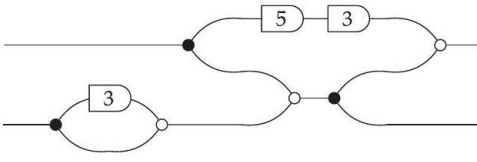
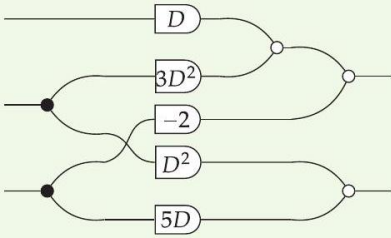
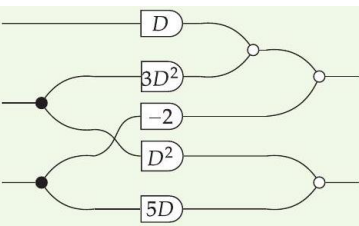
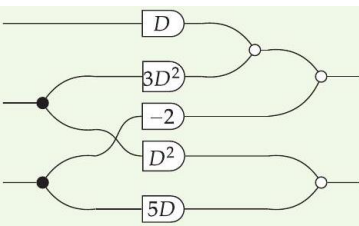
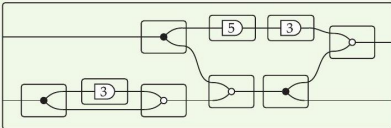
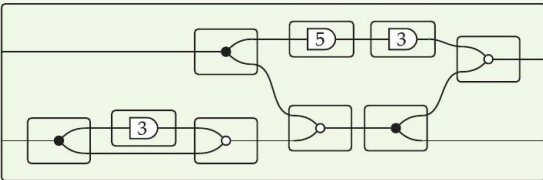
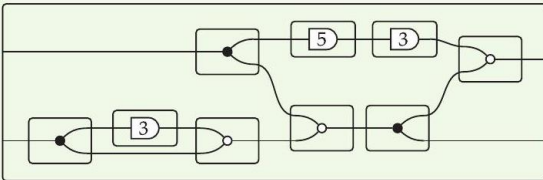
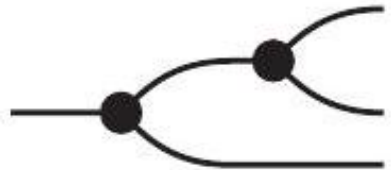
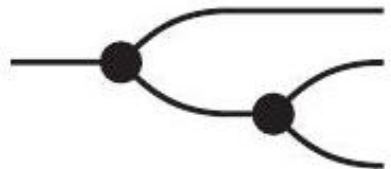
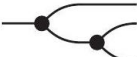
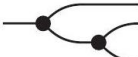
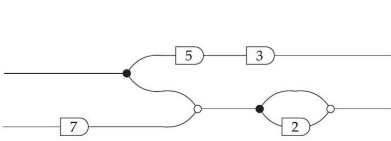
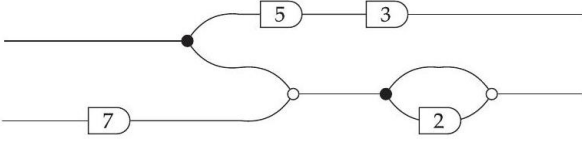
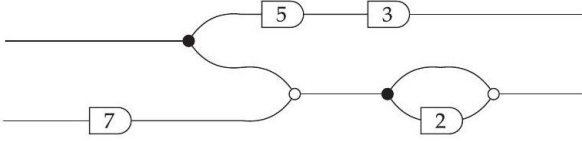
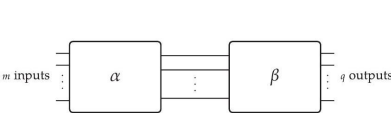
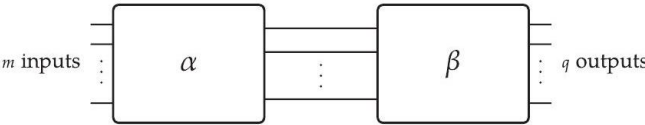
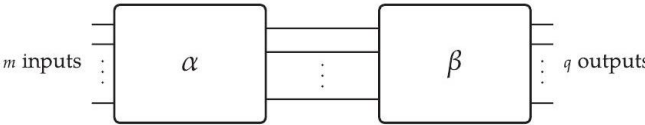
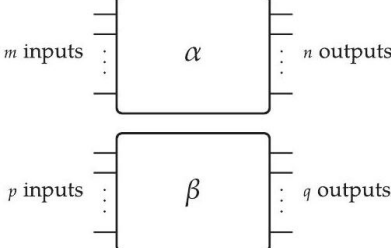
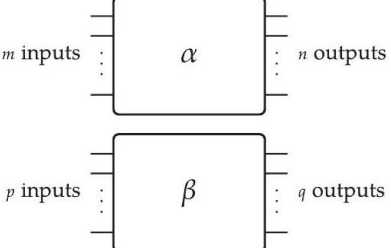
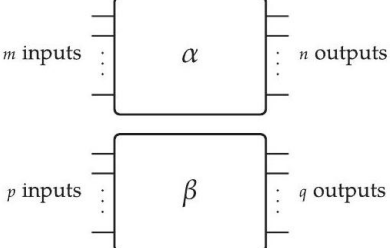
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fong-spivak-invitation_D11_4183	148	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-148_90_142_972_870.jpg			
fong-spivak-invitation_D11_4183	148	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-148_88_376_1200_752.jpg			
fong-spivak-invitation_D11_4183	148	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-148_136_404_1621_738.jpg			
fong-spivak-invitation_D11_4183	148	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-148_104_374_1933_754.jpg			
fong-spivak-invitation_D11_4183	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_64_104_476_542.jpg			
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fong-spivak-invitation_D11_4183	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_30_104_492_1231.jpg			
fong-spivak-invitation_D11_4183	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_106_828_730_524.jpg			

fong-spivak-invitation_D11_416	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_210_404_1211_736.jpg			
fong-spivak-invitation_D11_416	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_212_404_1617_736.jpg			
fong-spivak-invitation_D11_416	149	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-149_260_596_1981_640.jpg			
fong-spivak-invitation_D11_416	152	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-152_264_596_776_642.jpg			
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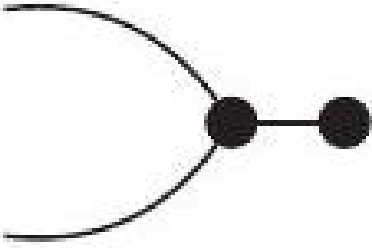
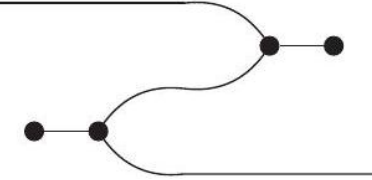
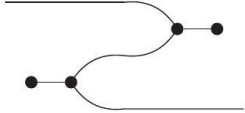
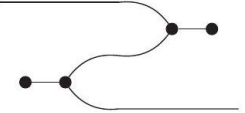
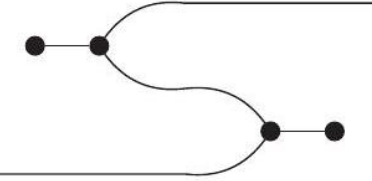
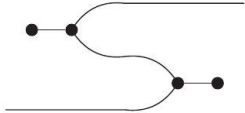
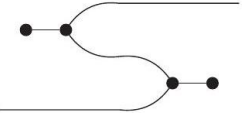
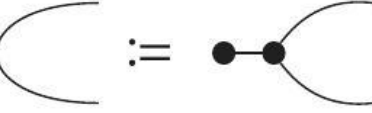
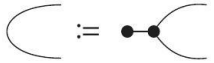
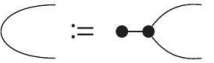
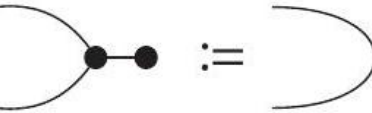
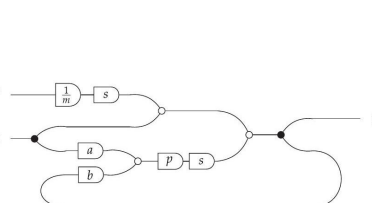
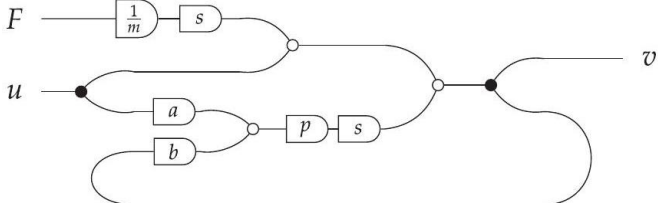
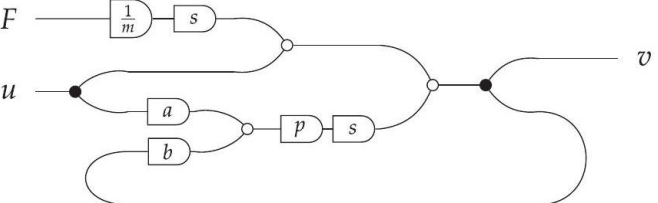
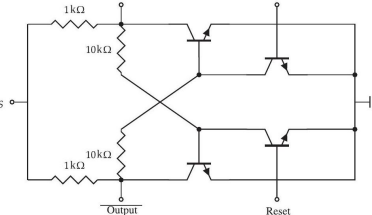
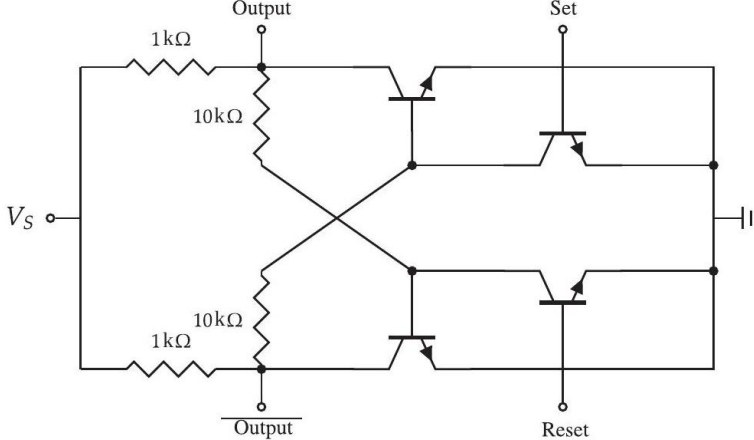
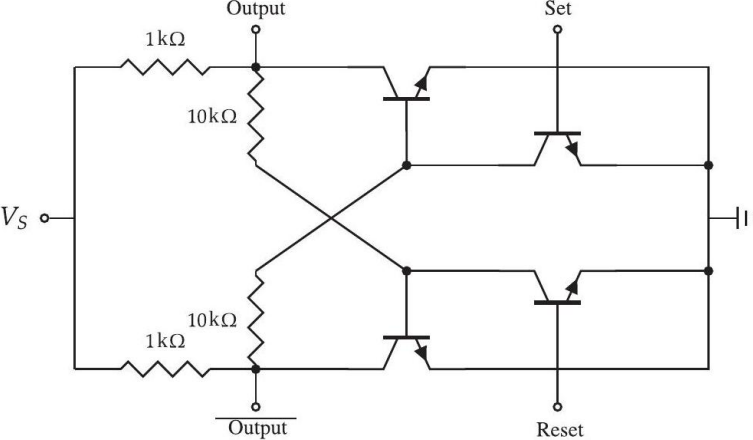
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fong-spivak-invitation_D1_161.jpg	161	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-161_186_767_1826_556.jpg			
fong-spivak-invitation_D1_162.jpg	162	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-162_224_770_542_556.jpg			
fong-spivak-invitation_D1_164.jpg	164	(dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-164_192_806_1402_540.jpg			
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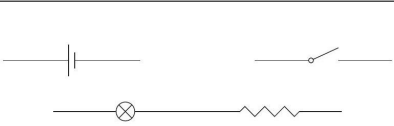
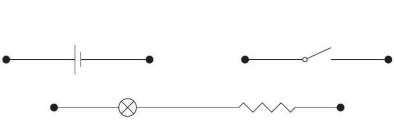
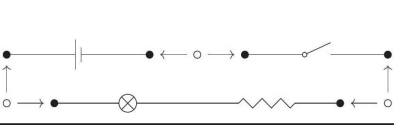
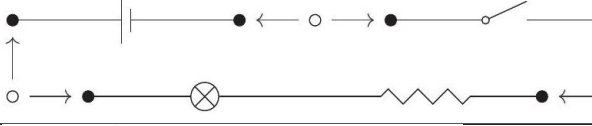
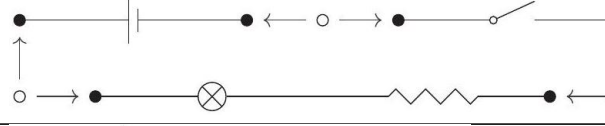
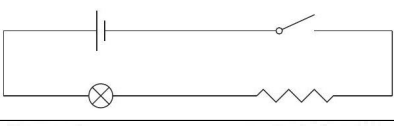
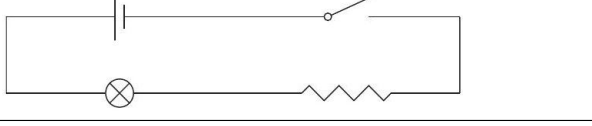
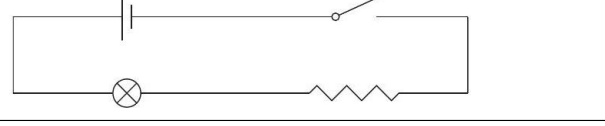
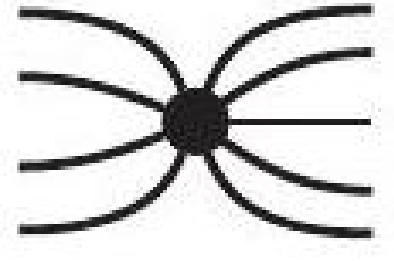
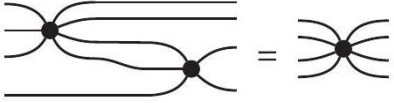
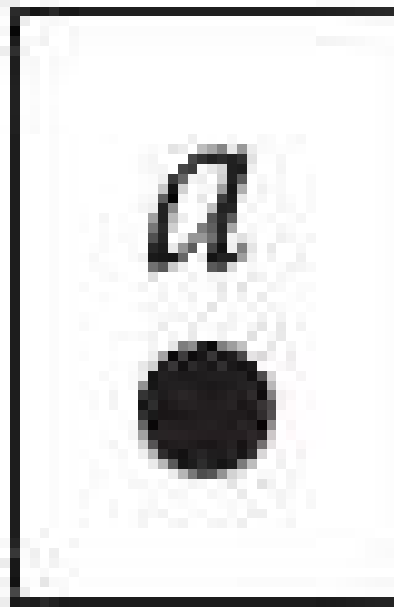
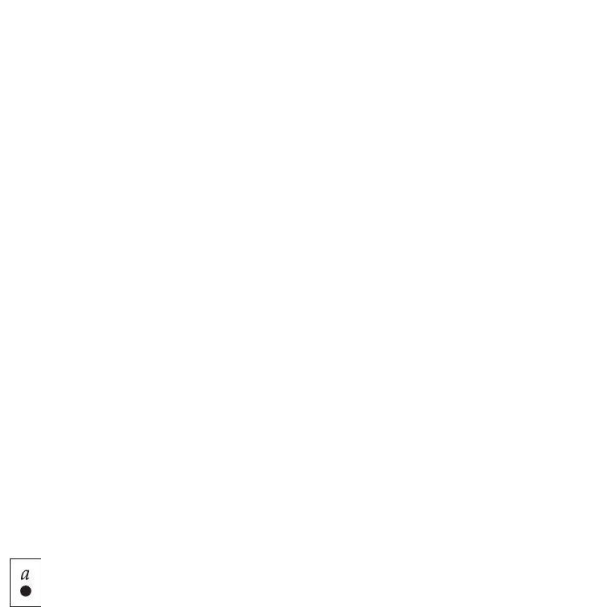
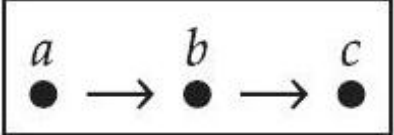
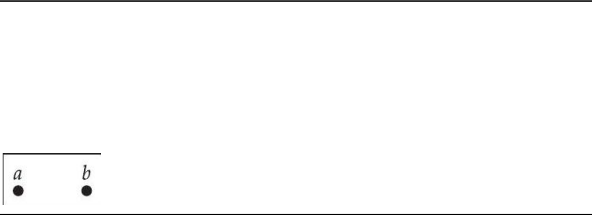
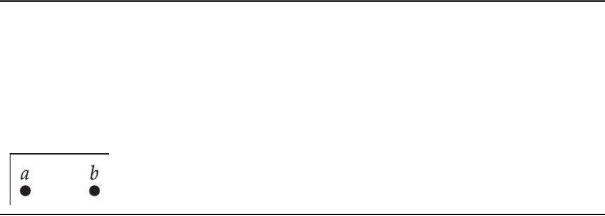
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fong-spivak-invitation_D1_0170	170	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-170_316_808_438_536.jpg			
fong-spivak-invitation_D1_0172	172	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-172_314_808_422_536.jpg			
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fong-spivak-invitation_D1_0175	175	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-175_62_764_470_556.jpg			
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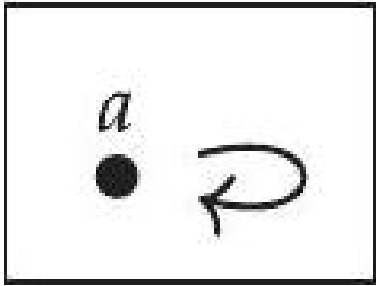
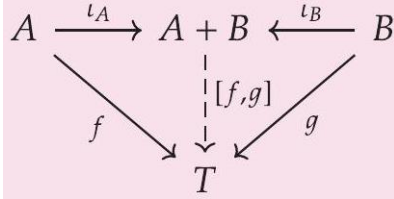
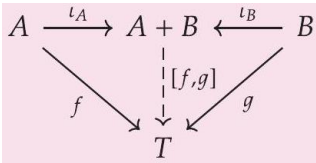
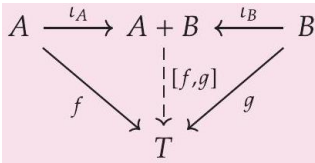
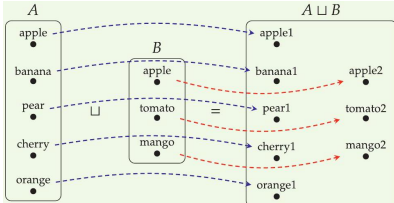
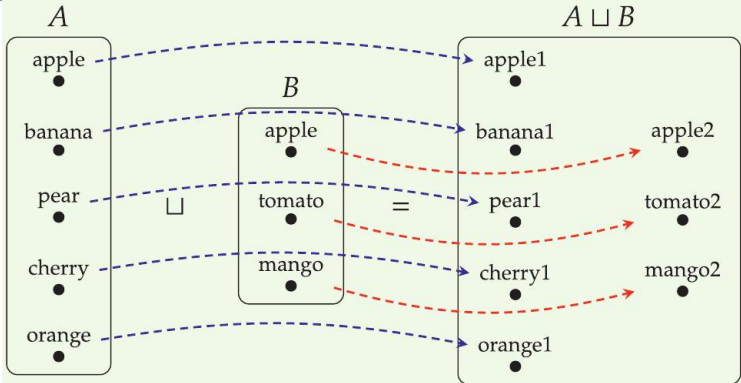
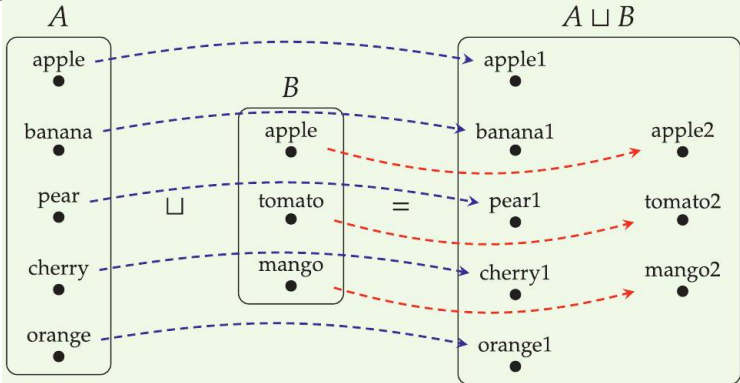
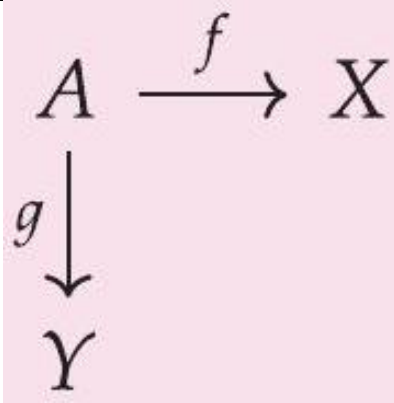
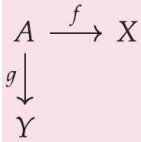
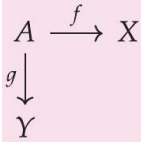
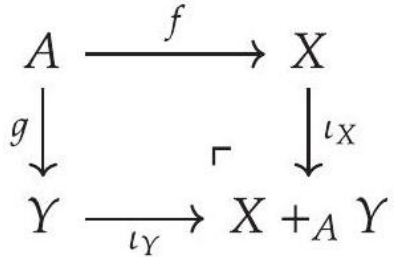
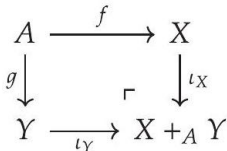
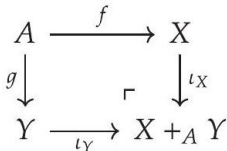
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fong-spivak-invitation_D11_016	177	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-177_236_716_1028_580.jpg			
fong-spivak-invitation_D11_019	177	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-177_88_186_1793_648.jpg			
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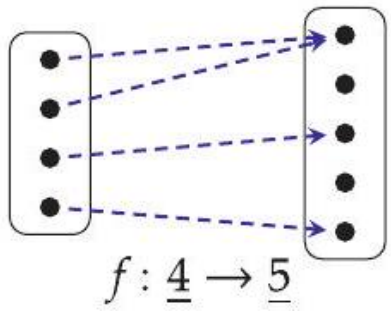
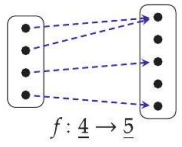
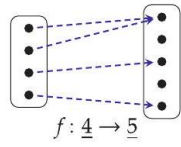
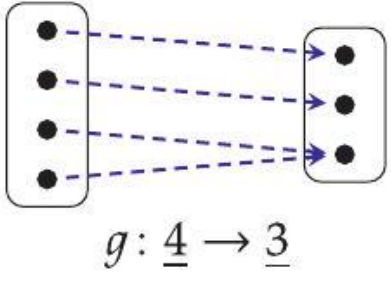
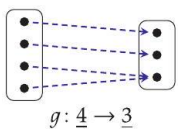
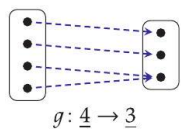
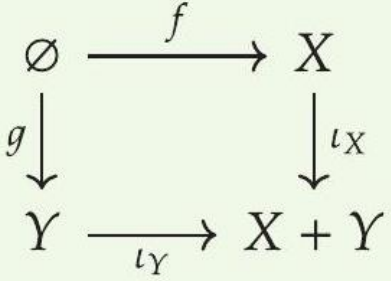
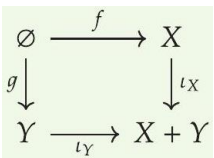
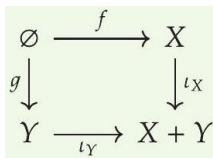
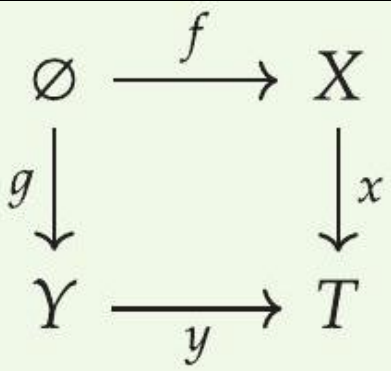
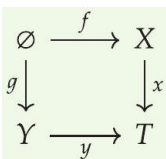
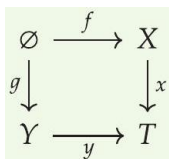
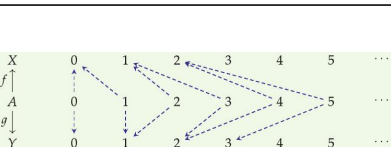
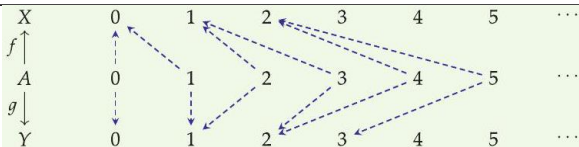
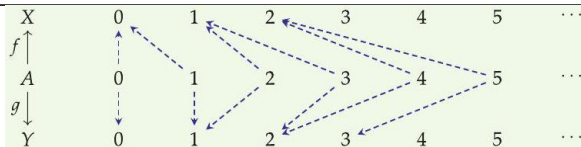
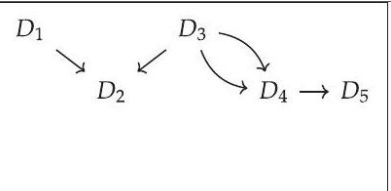
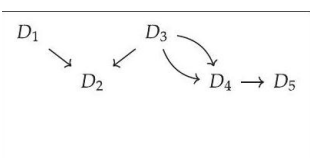
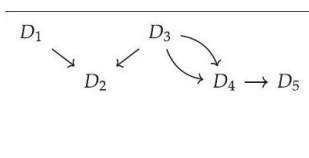
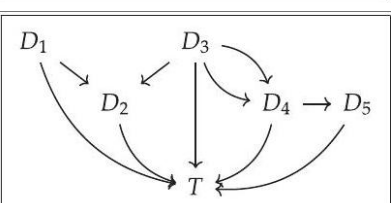
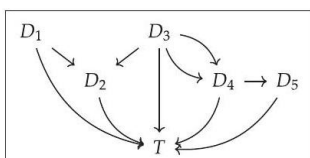
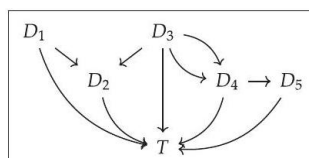
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fong-spivak-invitation_D1_186	186	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-186_50_274_2157_396.jpg			
fong-spivak-invitation_D1_186	186	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-186_50_276_2223_396.jpg			
fong-spivak-invitation_D1_188	188	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-188_66_114_1194_886.jpg			
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fong-spivak-invitation_D1_190	190	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-190_146_612_624_636.jpg			
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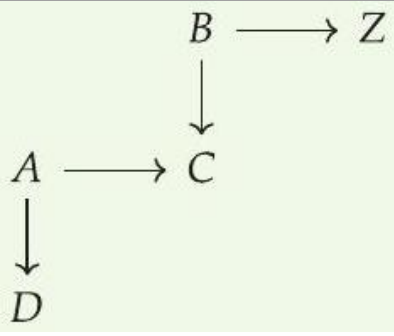
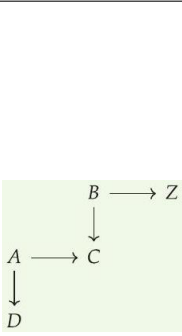
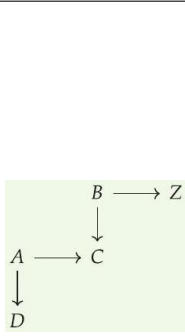
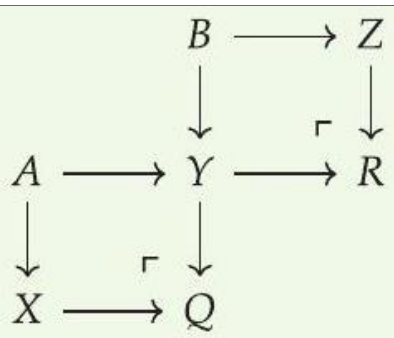
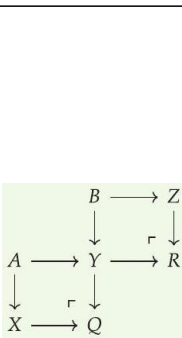
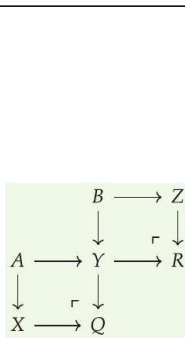
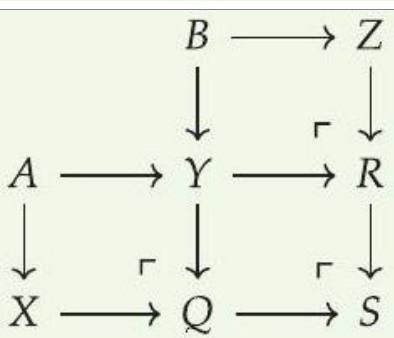
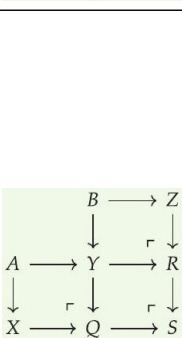
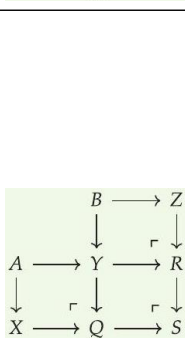
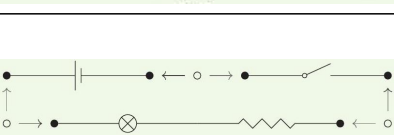
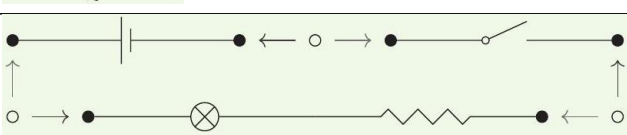
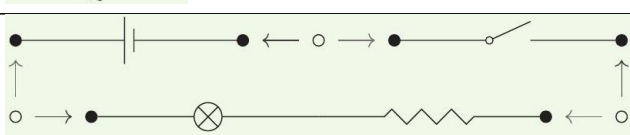
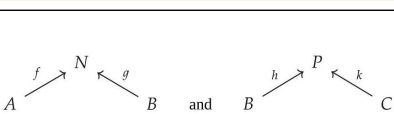
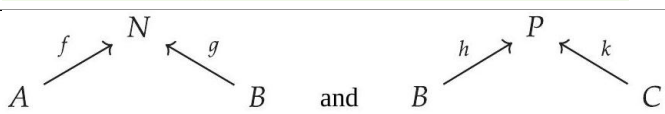
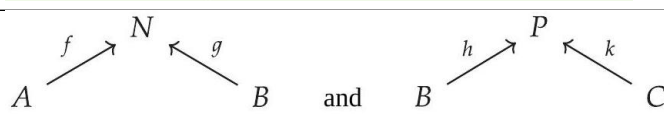
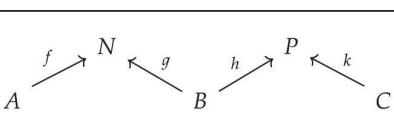
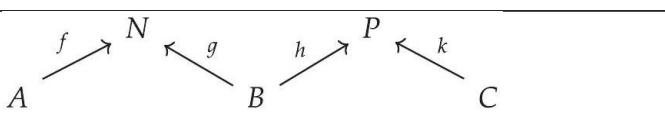
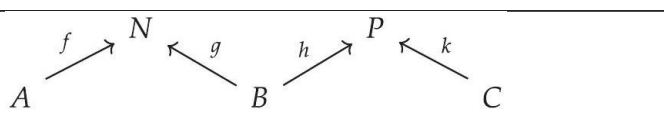
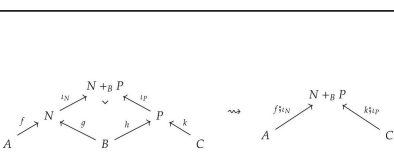
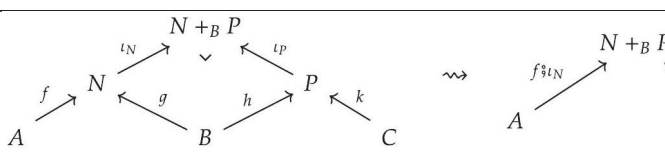
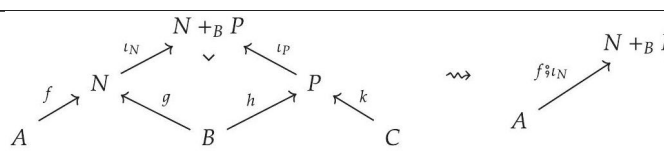
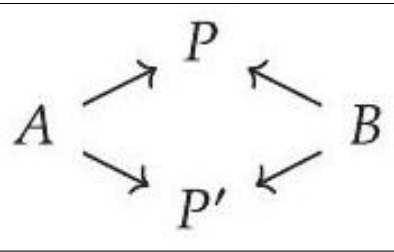
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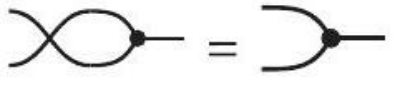
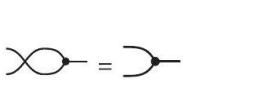
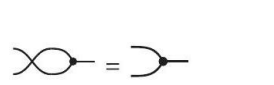
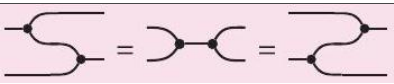
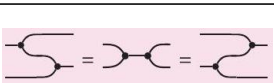
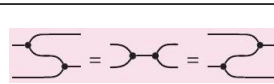
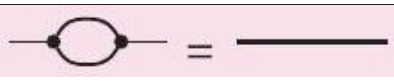
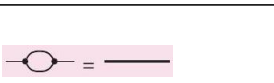
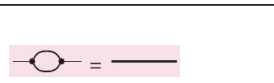
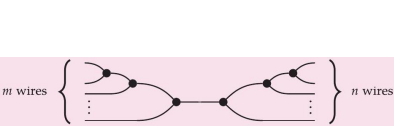
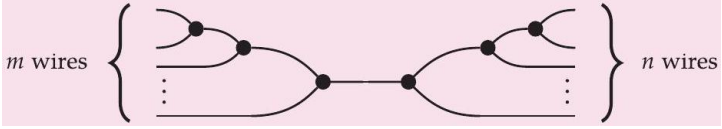
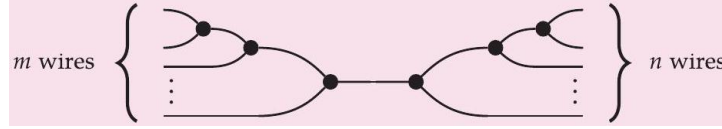
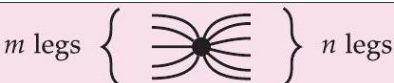
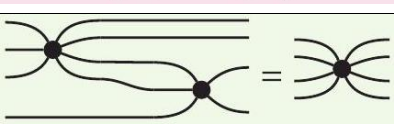
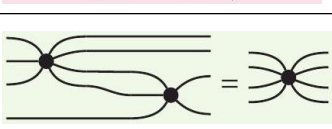
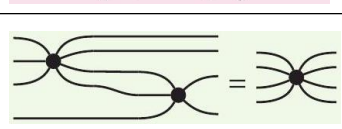
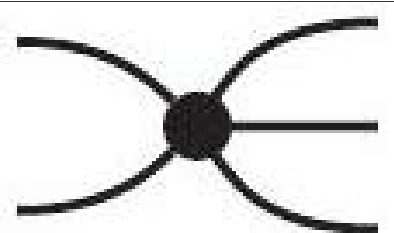
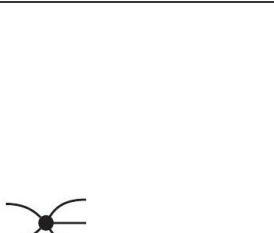
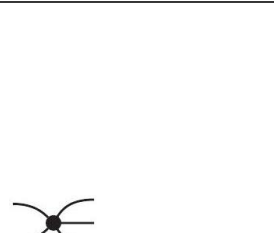
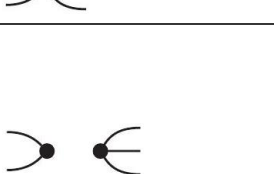
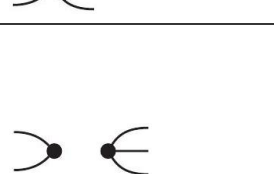
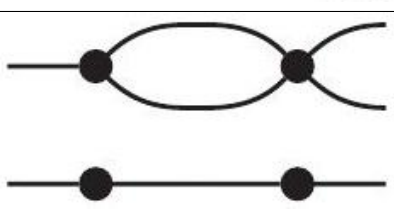
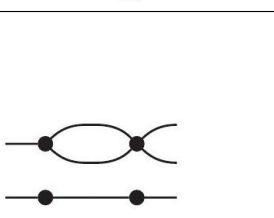
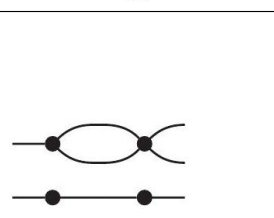
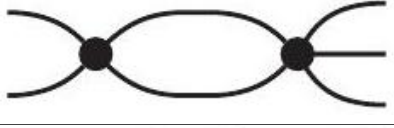
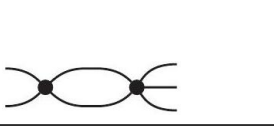
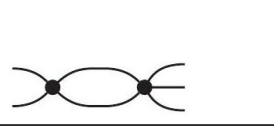
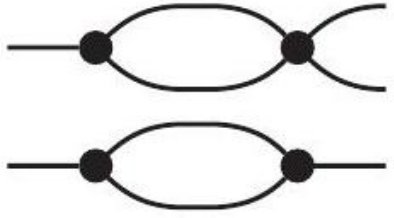
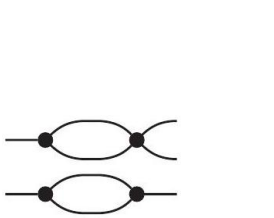
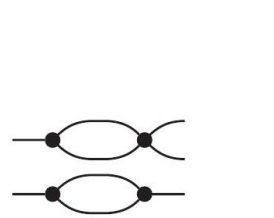
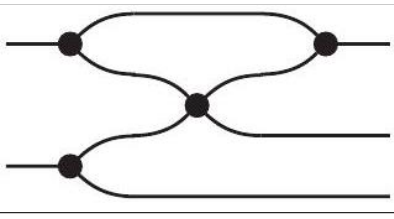
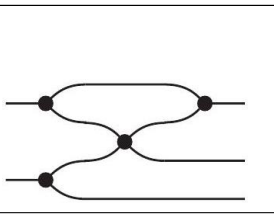
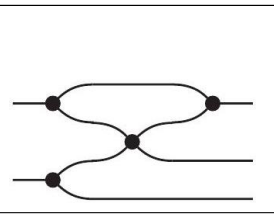
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fong-spivak-invitation_D1_200	200	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-200_504_964_448_424.jpg			
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fong-spivak-invitation_D1_201	201	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-201_202_298_1661_818.jpg			

fong-spivak-invitation_D12_021	201	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-201_204_214_2023_860.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ g \downarrow & & \downarrow x \\ Y & \xrightarrow{y} & T \end{array}$
fong-spivak-invitation_D12_022	202	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-202_294_400_256_768.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ g \downarrow & & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X +_A Y \end{array} \quad \begin{array}{c} \curvearrowright x \\ \searrow t \\ T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ g \downarrow & & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X +_A Y \end{array} \quad \begin{array}{c} \curvearrowright x \\ \searrow t \\ T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ g \downarrow & & \downarrow \iota_X \\ Y & \xrightarrow{\iota_Y} & X +_A Y \end{array} \quad \begin{array}{c} \curvearrowright x \\ \searrow t \\ T \end{array}$
fong-spivak-invitation_D12_023	202	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-202_210_220_1217_832.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & \ulcorner & \downarrow j \\ A' & \xrightarrow{f'} & X' \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & \ulcorner & \downarrow j \\ A' & \xrightarrow{f'} & X' \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & \ulcorner & \downarrow j \\ A' & \xrightarrow{f'} & X' \end{array}$
fong-spivak-invitation_D12_024	202	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-202_198_216_1565_832.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & & \downarrow x \\ A' & \xrightarrow{a} & T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & & \downarrow x \\ A' & \xrightarrow{a} & T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & & \downarrow x \\ A' & \xrightarrow{a} & T \end{array}$
fong-spivak-invitation_D12_025	202	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-202_284_316_1913_784.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & & \downarrow j \\ A' & \xrightarrow{f'} & X' \end{array} \quad \begin{array}{c} \curvearrowright x \\ \searrow x' \\ T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & & \downarrow j \\ A' & \xrightarrow{f'} & X' \end{array} \quad \begin{array}{c} \curvearrowright x \\ \searrow x' \\ T \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & X \\ i \downarrow & & \downarrow j \\ A' & \xrightarrow{f'} & X' \end{array} \quad \begin{array}{c} \curvearrowright x \\ \searrow x' \\ T \end{array}$

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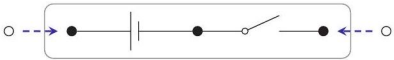
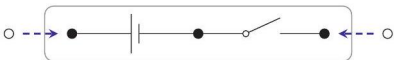
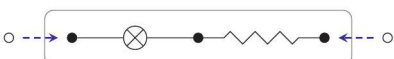
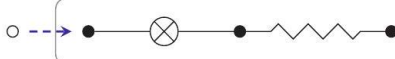
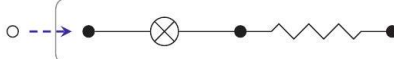
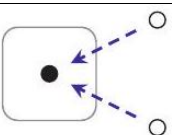
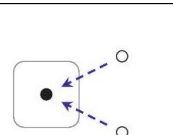
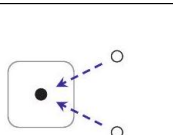
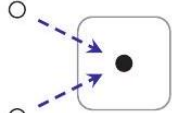
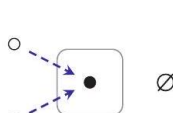
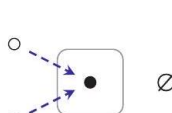
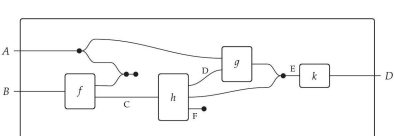
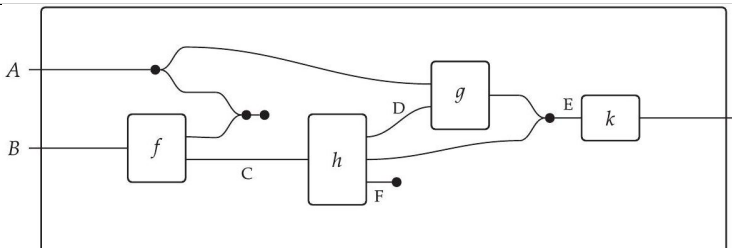
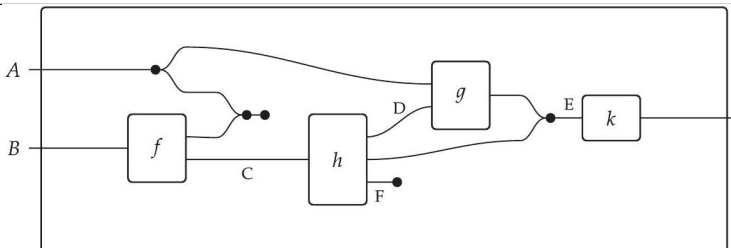
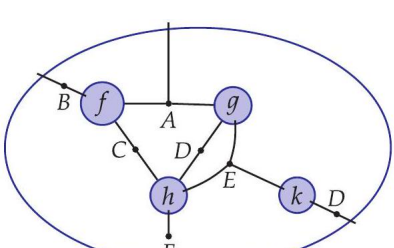
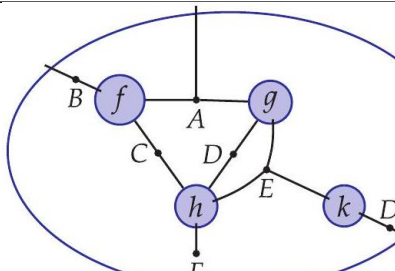
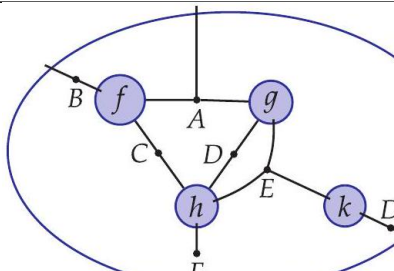
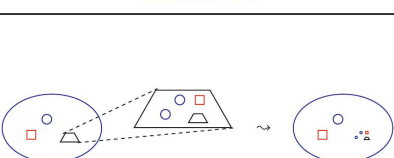
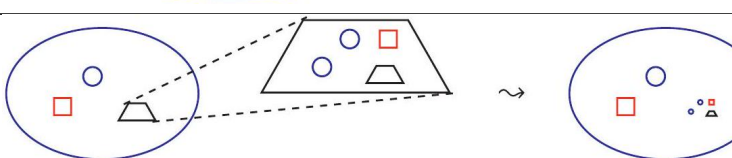
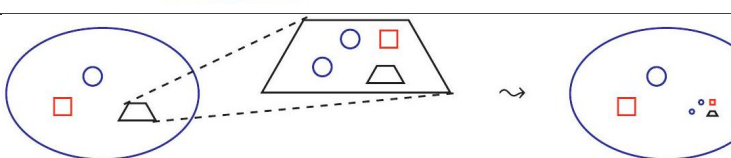
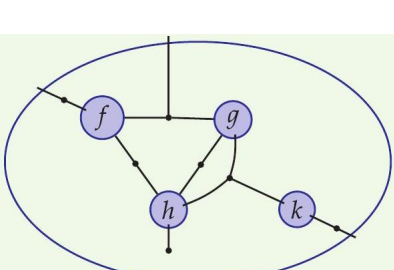
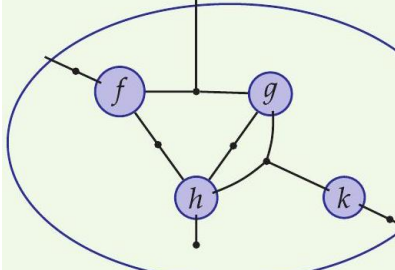
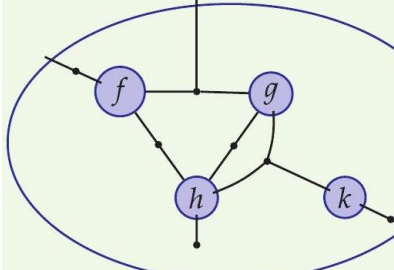
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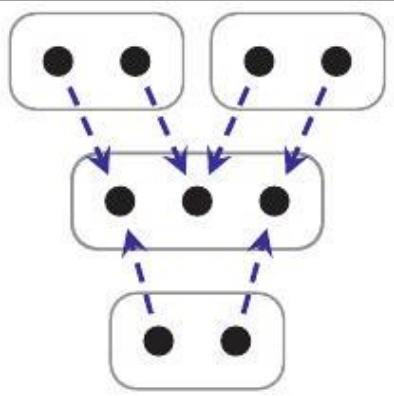
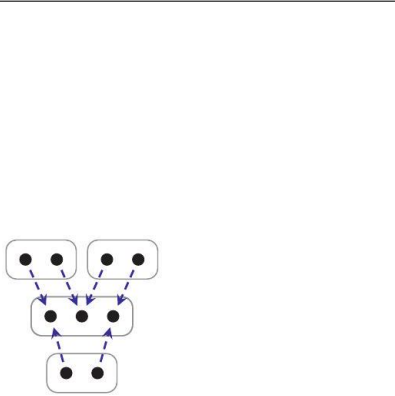
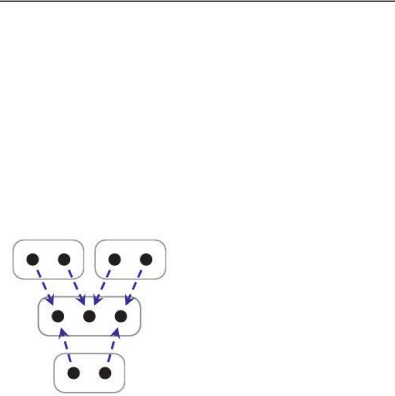
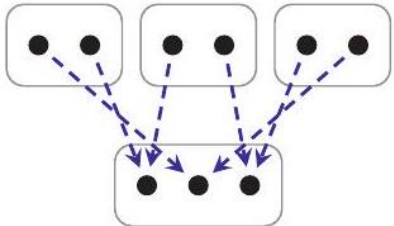
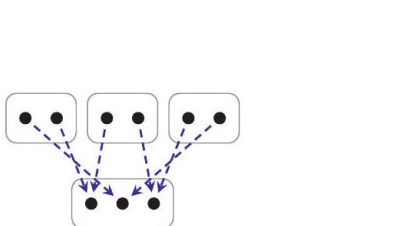
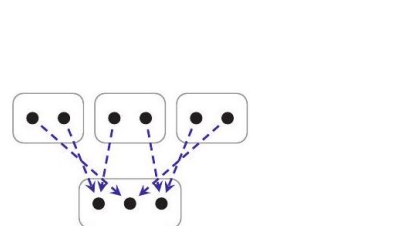
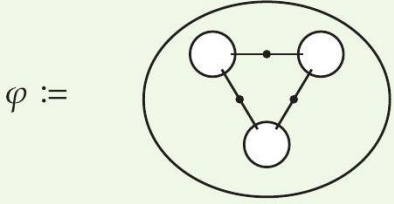
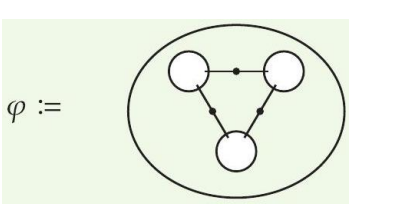
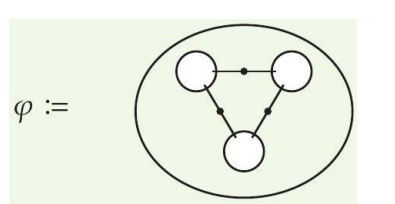
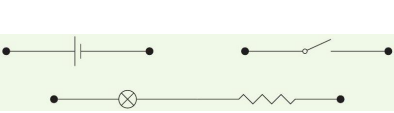
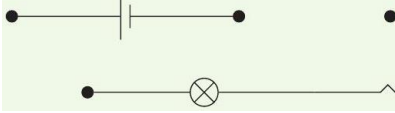
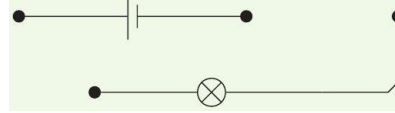
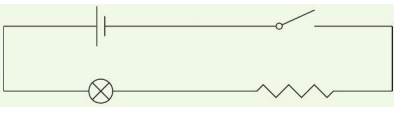
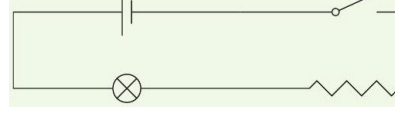
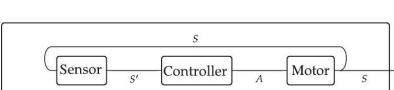
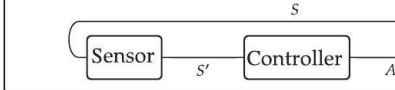
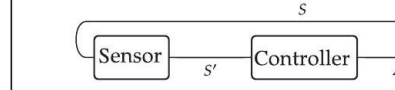
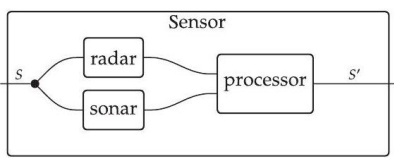
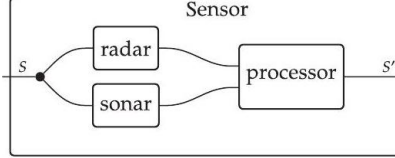
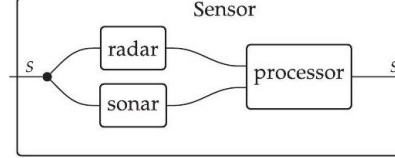
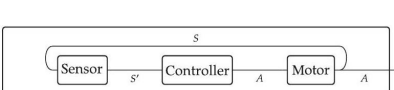
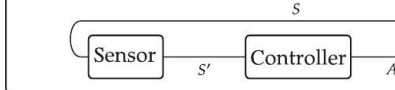
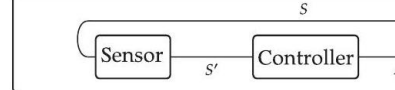
fong-spivak-invitation_D12_026	208	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-208_106_176_2013_1060.jpg			
fong-spivak-invitation_D12_026	209	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-209_224_224_1437_576.jpg			
fong-spivak-invitation_D12_026	209	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-209_198_322_1455_980.jpg			
fong-spivak-invitation_D12_026	210	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-210_224_732_310_576.jpg			
fong-spivak-invitation_D12_026	210	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-210_260_998_1076_408.jpg			
fong-spivak-invitation_D12_026	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_44_80_686_779.jpg			
fong-spivak-invitation_D12_026	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_19_51_699_1081.jpg			
fong-spivak-invitation_D12_026	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_70_272_895_452.jpg			
fong-spivak-invitation_D12_026	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_53_266_902_806.jpg			

fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_51_238_904_1157.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_70_352_1999_635.jpg			
fong-spivak-invitation_D12_121	211	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-211_40_222_2014_1054.jpg			
fong-spivak-invitation_D12_122	212	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-212_164_936_620_472.jpg			
fong-spivak-invitation_D12_122	212	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-212_92_416_968_732.jpg			
fong-spivak-invitation_D12_122	212	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-212_120_430_1921_726.jpg			
fong-spivak-invitation_D12_123	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_72_114_324_904.jpg			
fong-spivak-invitation_D12_123	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_72_184_480_868.jpg			
fong-spivak-invitation_D12_123	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_116_232_634_844.jpg			
fong-spivak-invitation_D12_123	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_70_232_836_844.jpg			
fong-spivak-invitation_D12_123	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_132_232_986_844.jpg			
fong-spivak-invitation_D12_123	213	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-213_160_320_1245_800.jpg			

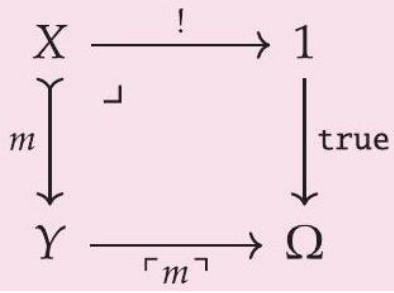
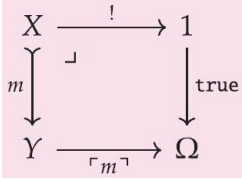
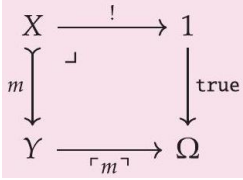
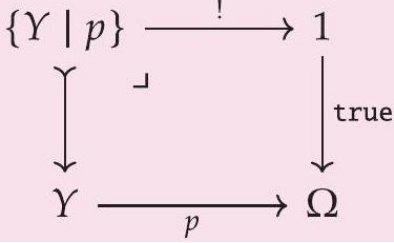
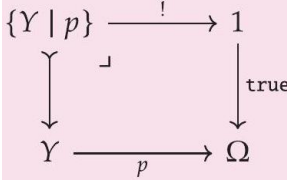
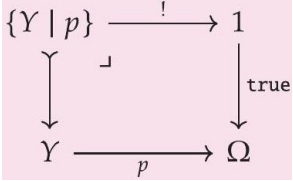
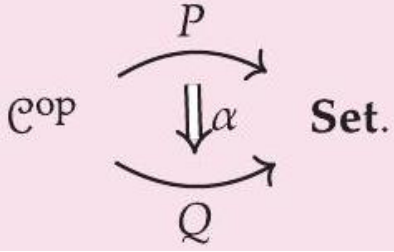
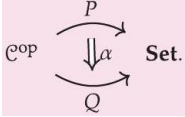
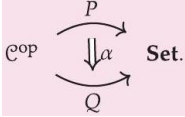
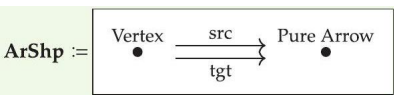
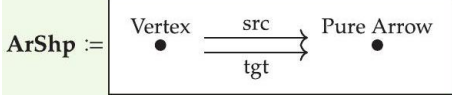
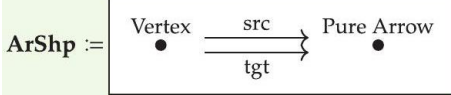
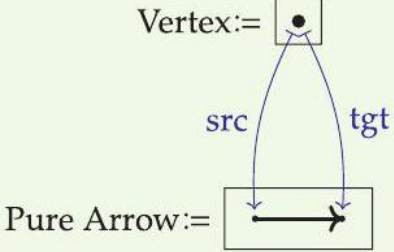
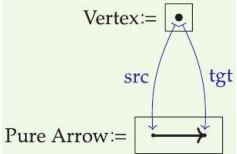
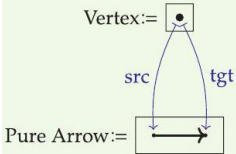
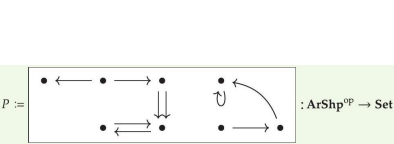
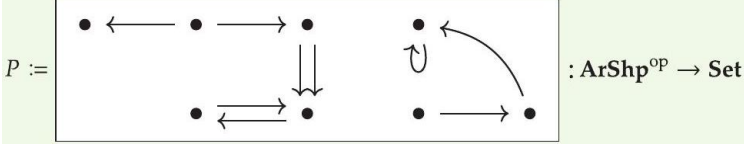
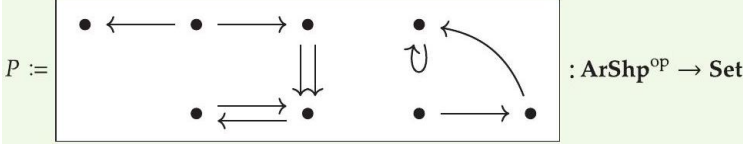
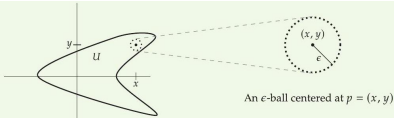
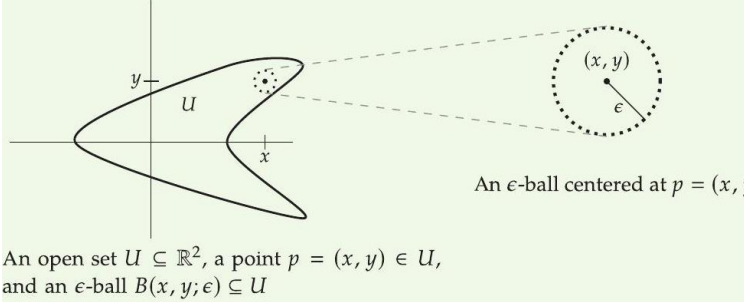
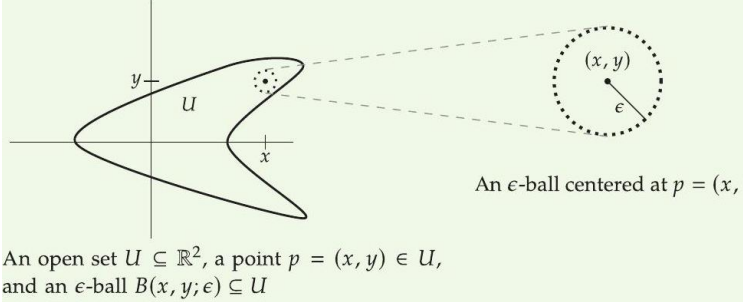
fong-spivak-invitation_D12_126	214	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-214_304_894_1096_494.jpg			
fong-spivak-invitation_D12_126	215	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-215_324_924_420_478.jpg			
fong-spivak-invitation_D12_126	216	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-216_84_272_954_488.jpg			
fong-spivak-invitation_D12_126	216	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-216_84_270_954_1122.jpg			
fong-spivak-invitation_D12_126	216	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-216_352_866_1200_510.jpg			
fong-spivak-invitation_D12_126	217	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-217_210_582_1673_648.jpg	$\begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array}$	$\begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array}$	$\begin{array}{ccc} P(S) \times P(T) & \xrightarrow{\varphi_{S,T}} & P(S \times T) \\ \text{im}_f \times \text{im}_g \downarrow & & \downarrow \text{im}_f \times g \\ P(S') \times P(T') & \xrightarrow{\varphi_{S',T'}} & P(S' \times T') \end{array}$
fong-spivak-invitation_D12_126	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_234_320_366_780.jpg			
fong-spivak-invitation_D12_126	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_286_618_962_632.jpg			
fong-spivak-invitation_D12_126	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_194_626_1625_628.jpg			

fong-spivak-invitation_D14_220	218	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-218_226_878_2021_502.jpg			
fong-spivak-invitation_D14_220	219	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-219_336_1138_396_370.jpg			
fong-spivak-invitation_D14_220	220	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-220_110_224_1927_828.jpg			
fong-spivak-invitation_D14_220	221	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-221_108_520_1068_678.jpg			
fong-spivak-invitation_D14_220	221	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-221_192_520_1269_678.jpg			
fong-spivak-invitation_D14_220	221	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-221_70_810_2049_534.jpg	$b := \text{---} \text{---}$ and $s := \text{---} / \text{---}$	$b := \text{---} \text{---}$ and $s := \text{---} / \text{---}$	$b := \text{---} \text{---}$ and $s := \text{---} / \text{---}$
fong-spivak-invitation_D14_220	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_26_86_672_898.jpg			
fong-spivak-invitation_D14_220	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_94_420_1068_728.jpg			
fong-spivak-invitation_D14_220	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_94_420_1703_728.jpg			
fong-spivak-invitation_D14_220	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_96_420_2147_396.jpg			
fong-spivak-invitation_D14_220	222	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-222_96_420_2149_1064.jpg			
fong-spivak-invitation_D14_220	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_260_814_394_530.jpg			
fong-spivak-invitation_D14_220	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_94_420_764_728.jpg			
fong-spivak-invitation_D14_220	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_70_616_1068_632.jpg			
fong-spivak-invitation_D14_220	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_68_420_1393_728.jpg			

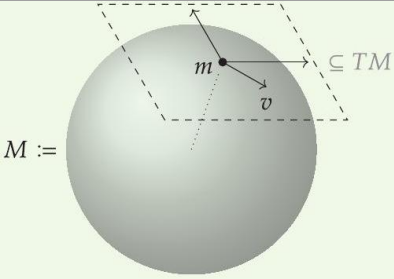
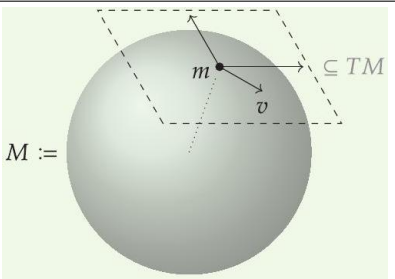
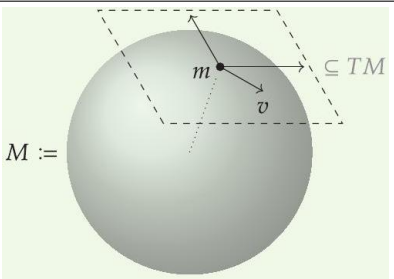
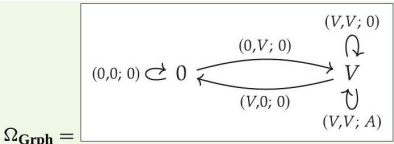
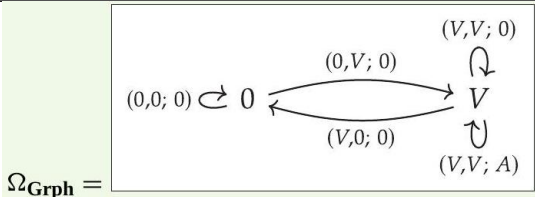
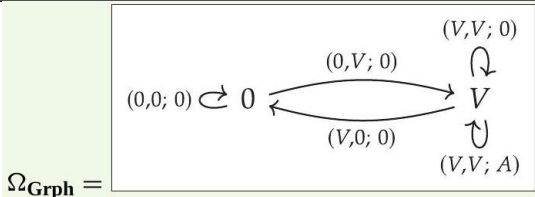
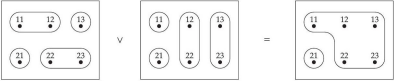
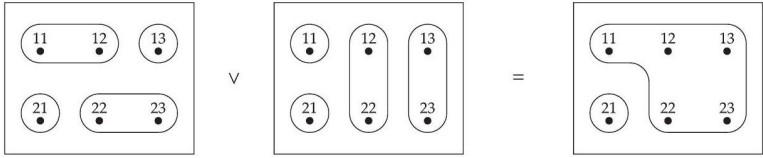
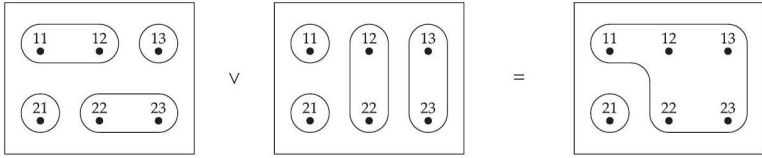
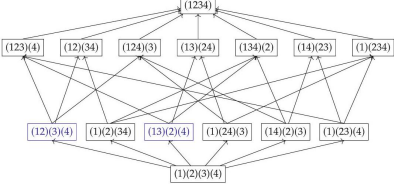
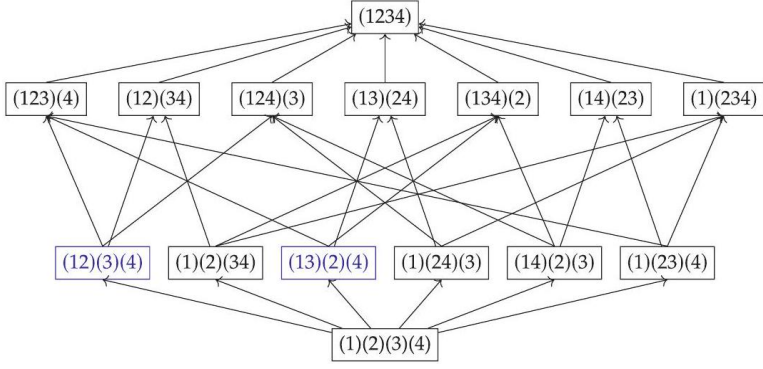
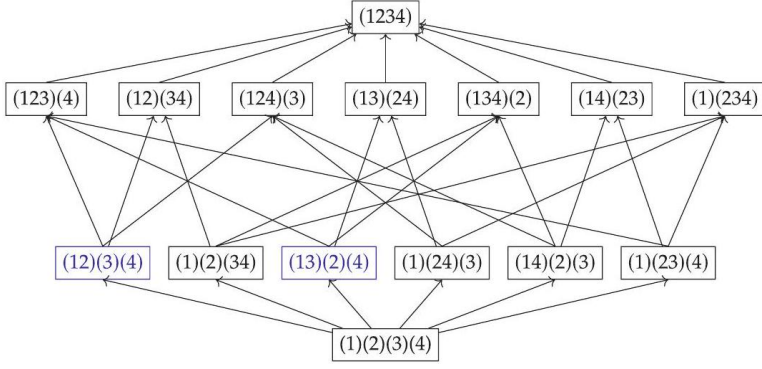
fong-spivak-invitation_D12_223	223	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-223_94_618_1579_630.jpg			
fong-spivak-invitation_D12_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_96_616_382_632.jpg			
fong-spivak-invitation_D12_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_96_616_552_632.jpg			
fong-spivak-invitation_D12_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_194_618_746_632.jpg			
fong-spivak-invitation_D12_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_128_360_1204_460.jpg	$\eta := \emptyset$ 	$\eta := \emptyset$ 	$\eta := \emptyset$ 
fong-spivak-invitation_D12_224	224	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-224_122_364_1207_1056.jpg	 $\emptyset =: \epsilon$	 $\emptyset =: \epsilon$	 $\emptyset =: \epsilon$
fong-spivak-invitation_D12_225	225	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-225_410_1223_1519_330.jpg			
fong-spivak-invitation_D12_226	226	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-226_382_592_258_644.jpg			
fong-spivak-invitation_D12_227	227	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-227_222_1130_884_374.jpg			
fong-spivak-invitation_D12_228	228	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-228_380_592_1893_644.jpg			

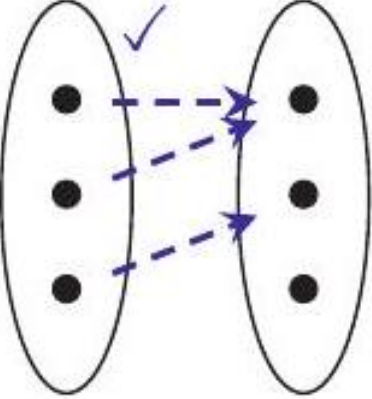
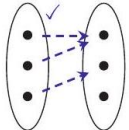
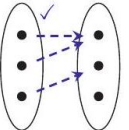
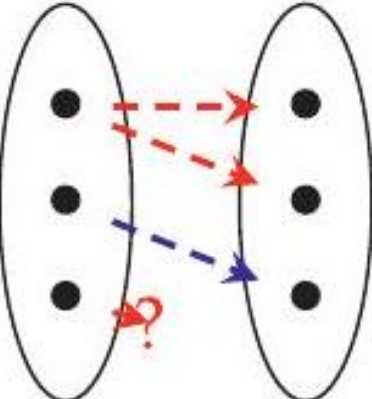
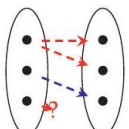
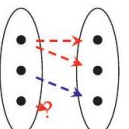
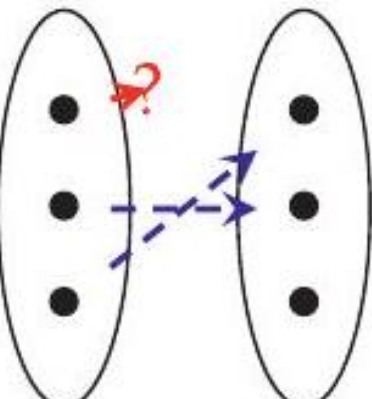
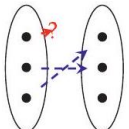
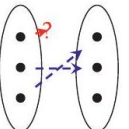
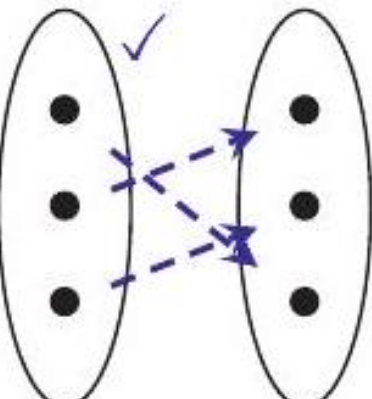
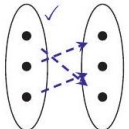
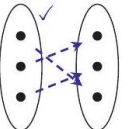
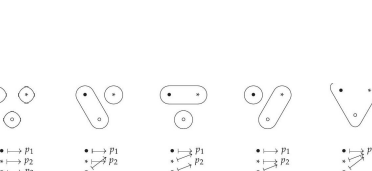
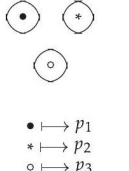
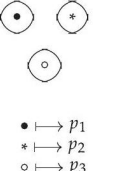
fong-spivak-invitation_D14_003	229	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-229_208_208_1024_856.jpg			
fong-spivak-invitation_D14_004	229	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-229_184_312_1477_804.jpg			
fong-spivak-invitation_D14_005	231	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-231_240_452_298_712.jpg	$\varphi :=$ 	$\varphi :=$ 	$\varphi :=$ 
fong-spivak-invitation_D14_006	231	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-231_158_814_864_532.jpg			
fong-spivak-invitation_D14_007	231	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-231_156_600_1116_638.jpg			
fong-spivak-invitation_D14_008	233	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-233_148_770_1661_552.jpg			
fong-spivak-invitation_D14_009	234	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-234_214_560_710_660.jpg			
fong-spivak-invitation_D14_010	235	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-235_138_772_1475_550.jpg			
fong-spivak-invitation_D14_011	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_112_232_652_612.jpg	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & & \downarrow & \lrcorner & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$
fong-spivak-invitation_D14_012	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_112_232_652_1034.jpg	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ D & \longrightarrow & E & \longrightarrow & F \end{array}$

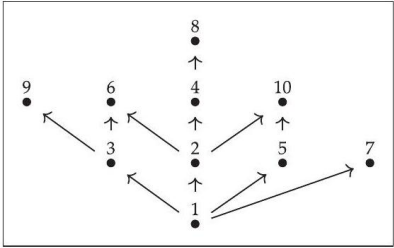
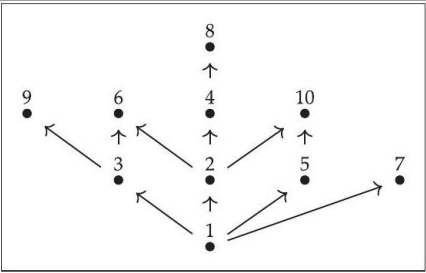
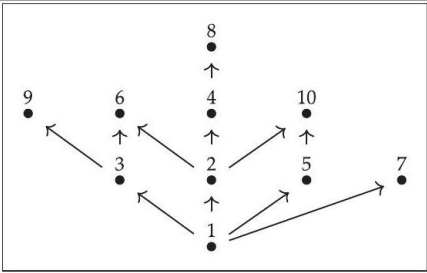
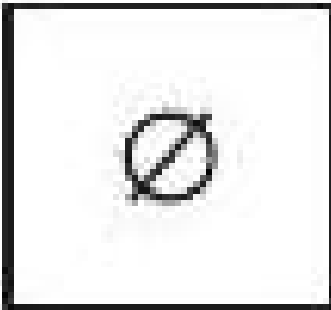
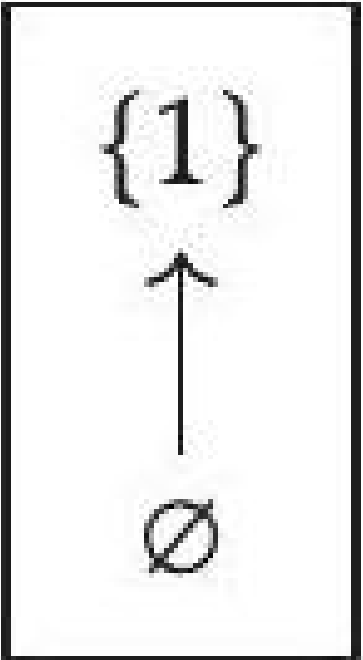
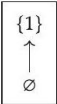
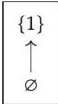
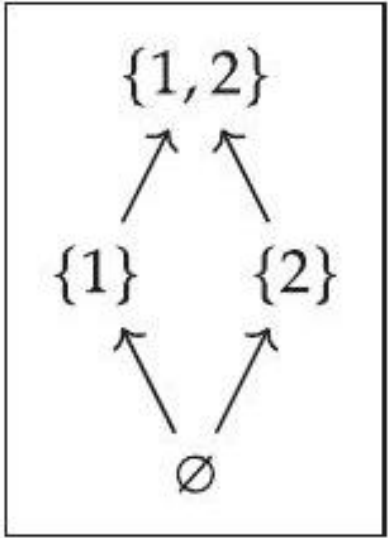
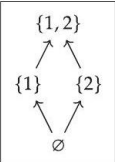
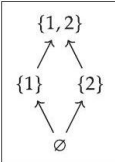
fong-spivak-invitation_D12_0871	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_114_250_1128_814.jpg	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow \lrcorner & & \downarrow \lrcorner & & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow \lrcorner & & \downarrow \lrcorner & & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$	$\begin{array}{ccccc} A & \longrightarrow & B & \longrightarrow & C \\ \downarrow \lrcorner & & \downarrow \lrcorner & & \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' \end{array}$
fong-spivak-invitation_D12_0872	237	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-237_208_604_1945_636.jpg	$\begin{array}{ccc} A \xrightarrow{\text{id}_A} A & A \xrightarrow{f} B \\ \text{id}_A \downarrow \lrcorner \downarrow f & f \downarrow \lrcorner \downarrow \text{id}_B \\ A \xrightarrow{f} B & B \xrightarrow{\text{id}_B} B \end{array}$	$\begin{array}{ccc} A \xrightarrow{\text{id}_A} A & A \xrightarrow{f} B \\ \text{id}_A \downarrow \lrcorner \downarrow f & f \downarrow \lrcorner \downarrow \text{id}_B \\ A \xrightarrow{f} B & B \xrightarrow{\text{id}_B} B \end{array}$	$\begin{array}{ccc} A \xrightarrow{\text{id}_A} A & A \xrightarrow{f} B \\ \text{id}_A \downarrow \lrcorner \downarrow f & f \downarrow \lrcorner \downarrow \text{id}_B \\ A \xrightarrow{f} B & B \xrightarrow{\text{id}_B} B \end{array}$
fong-spivak-invitation_D12_0883	238	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-238_212_220_744_852.jpg	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong \lrcorner \cong \downarrow i & & \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong \lrcorner \cong \downarrow i & & \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ i' \downarrow \cong \lrcorner \cong \downarrow i & & \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-invitation_D12_0884	238	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-238_208_188_1084_868.jpg	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel \lrcorner \parallel & & \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel \lrcorner \parallel & & \\ A & \xrightarrow{f} & B \end{array}$	$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \parallel \lrcorner \parallel & & \\ A & \xrightarrow{f} & B \end{array}$
fong-spivak-invitation_D12_0885	238	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-238_194_232_1489_824.jpg	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow \lrcorner \downarrow f & & \\ B' & \xrightarrow{h} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow \lrcorner \downarrow f & & \\ B' & \xrightarrow{h} & B \end{array}$	$\begin{array}{ccc} A' & \xrightarrow{g} & A \\ f' \downarrow \lrcorner \downarrow f & & \\ B' & \xrightarrow{h} & B \end{array}$
fong-spivak-invitation_D12_0886	239	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-239_186_276_370_800.jpg			

fong-spivak-invitation_D12_450	240	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-240_232_312_1076_474.jpg			
fong-spivak-invitation_D12_450	240	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-240_236_376_1070_1028.jpg			
fong-spivak-invitation_D12_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_154_238_426_820.jpg			
fong-spivak-invitation_D12_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_134_592_806_640.jpg			
fong-spivak-invitation_D12_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_202_304_1036_786.jpg			
fong-spivak-invitation_D12_450	245	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-245_192_968_1421_456.jpg			
fong-spivak-invitation_D12_450	247	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-247_400_998_694_442.jpg			

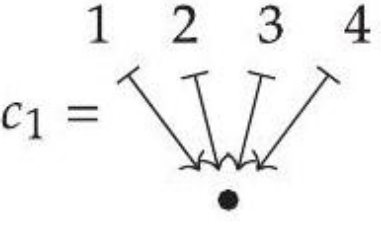
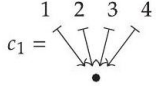
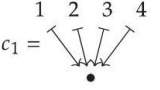
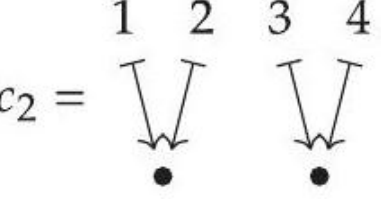
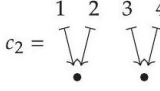
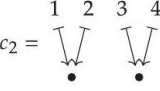
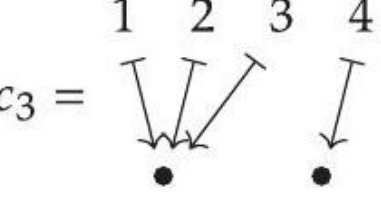
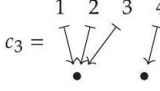
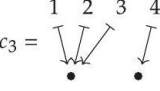
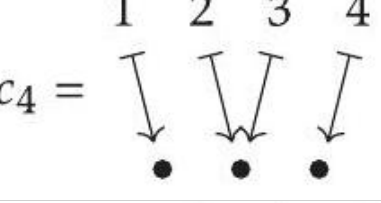
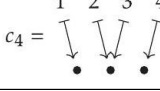
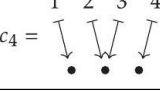
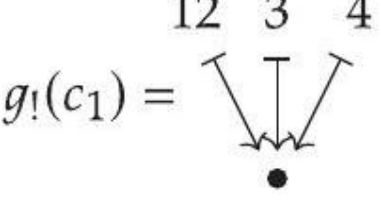
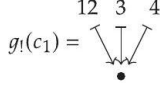
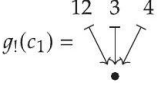
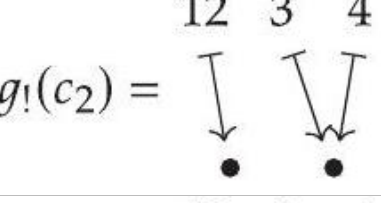
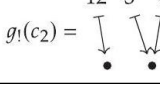
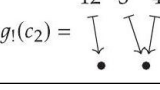
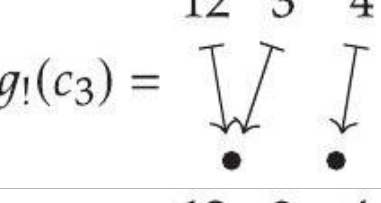
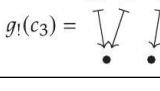
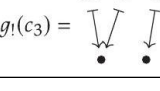
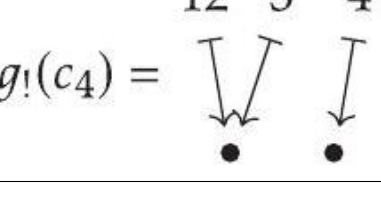
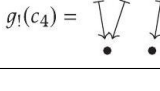
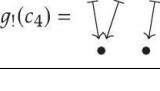
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fong-spivak-invitation_D12_251.jpg	251	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-251_242_626_1787_624.jpg			
fong-spivak-invitation_D12_252.jpg	252	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-252_530_516_564_700.jpg			
fong-spivak-invitation_D12_252.jpg	252	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-252_484_264_1317_808.jpg			

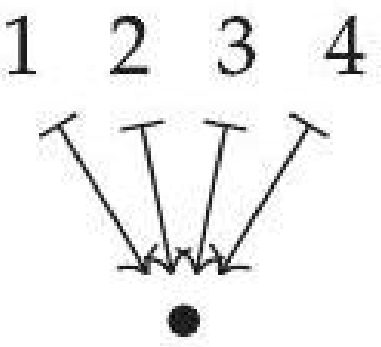
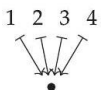
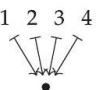
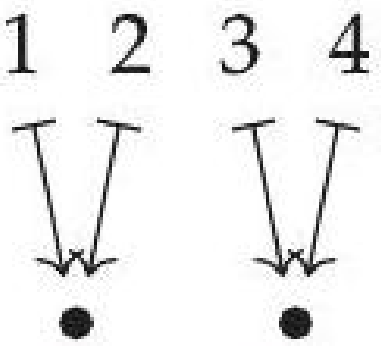
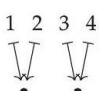
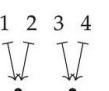
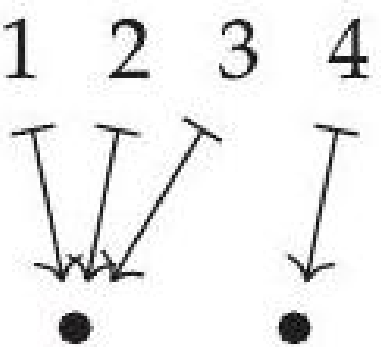
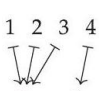
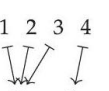
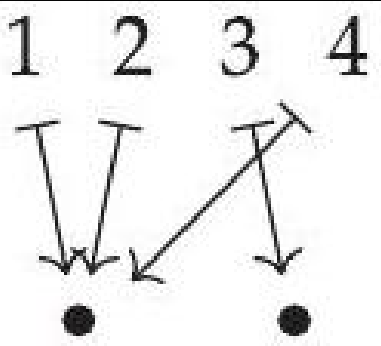
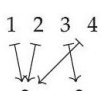
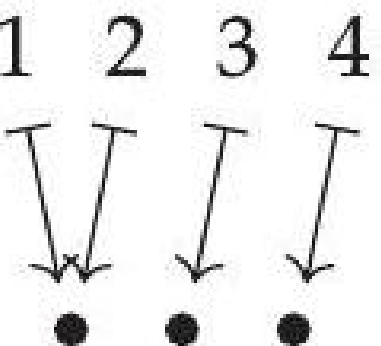
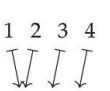
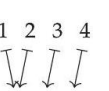
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fong-spivak-invitation_D12_0350	257	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-257_262_696_482_590.jpg	$\Omega_{\text{Grph}} =$ 	$\Omega_{\text{Grph}} =$ 	$\Omega_{\text{Grph}} =$ 
fong-spivak-invitation_D12_0351	257	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-257_124_1010_1413_434.jpg	$\boxed{A \longrightarrow B \quad C} \subseteq \boxed{A \xrightleftharpoons[g]{f} B \xrightarrow{h} C \xrightarrow{i} D}$	$\boxed{A \longrightarrow B \quad C} \subseteq \boxed{A \xrightleftharpoons[g]{f} B \xrightarrow{h} C \xrightarrow{i} D}$	$\boxed{A \longrightarrow B \quad C} \subseteq \boxed{A \xrightleftharpoons[g]{f} B \xrightarrow{h} C \xrightarrow{i} D}$
fong-spivak-invitation_D12_0352	262	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-262_206_304_378_788.jpg	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow \lrcorner & & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow \lrcorner & & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$	$\begin{array}{ccc} \forall_t p & \longrightarrow & 1 \\ \downarrow \lrcorner & & \downarrow \text{true}^T \\ S & \xrightarrow{p'} & \Omega^T \end{array}$
fong-spivak-invitation_D12_0352	262	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-262_176_384_1643_748.jpg	$\begin{array}{ccc} \{S \times T \mid p\} & \xrightarrow{\quad} & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \xrightarrow{\quad} & S \end{array}$	$\begin{array}{ccc} \{S \times T \mid p\} & \xrightarrow{\quad} & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \xrightarrow{\quad} & S \end{array}$	$\begin{array}{ccc} \{S \times T \mid p\} & \xrightarrow{\quad} & S \times T \\ \downarrow & & \downarrow \pi_S \\ \exists_t p & \xrightarrow{\quad} & S \end{array}$
fong-spivak-invitation_D12_0353	270	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-270_250_1219_1635_334.jpg			
fong-spivak-invitation_D12_0353	270	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-270_180_236_2065_844.jpg	$\begin{array}{c} (12) = \text{[box with dot and *]} \\ \uparrow \\ (1)(2) = \text{[circle with dot]} \text{ [circle with *]} \end{array}$	$\begin{array}{c} (12) = \text{[box with dot and *]} \\ \uparrow \\ (1)(2) = \text{[circle with dot]} \text{ [circle with *]} \end{array}$	$\begin{array}{c} (12) = \text{[box with dot and *]} \\ \uparrow \\ (1)(2) = \text{[circle with dot]} \text{ [circle with *]} \end{array}$
fong-spivak-invitation_D12_0356	271	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-271_512_1070_332_422.jpg			

fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_174_164_2071_436.jpg			
fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_174_162_2071_732.jpg			
fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_174_162_2071_1028.jpg			
fong-spivak-invitation_D14_0320	272	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-272_174_162_2071_1323.jpg			
fong-spivak-invitation_D14_0320	273	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-273_286_1202_794_338.jpg			

fong-spivak-invitation_D1_036	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_354_558_866_660.jpg			
fong-spivak-invitation_D1_036	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_58_64_1661_682.jpg			
fong-spivak-invitation_D1_036	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_140_80_1621_852.jpg			
fong-spivak-invitation_D1_036	274	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-274_218_156_1581_1040.jpg			

fong-spivak-invitation_D12_736	275	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-275_360_252_382_336.jpg			
fong-spivak-invitation_D12_736	275	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-275_458_894_384_642.jpg			
fong-spivak-invitation_D12_736	275	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-275_324_732_1096_572.jpg			
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fong-spivak-invitation_D12_736	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_152_472_616_724.jpg			
fong-spivak-invitation_D12_736	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_156_472_1044_724.jpg			

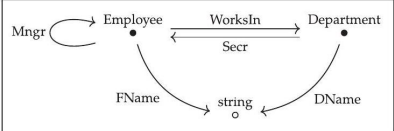
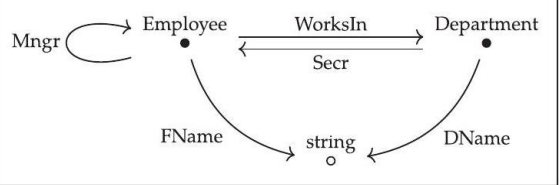
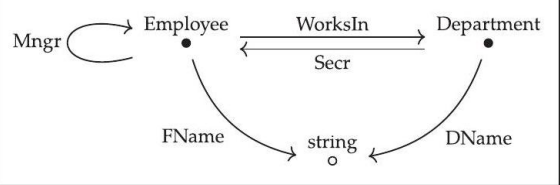
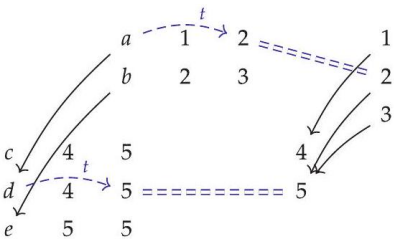
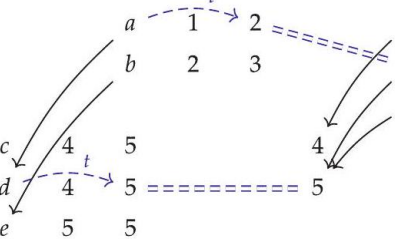
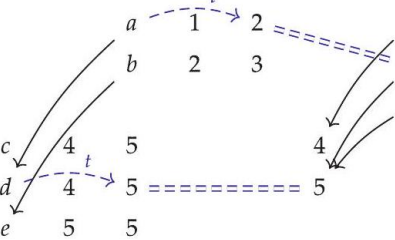
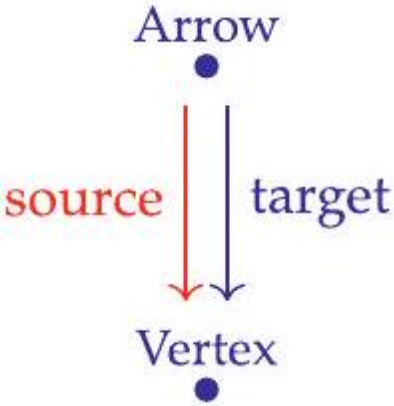
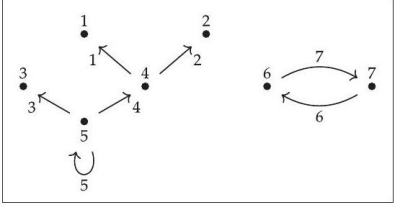
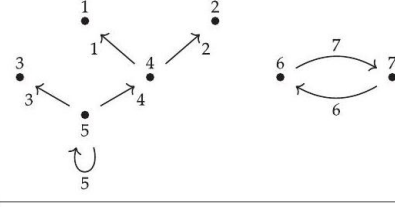
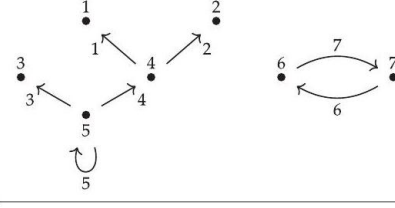
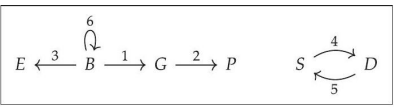
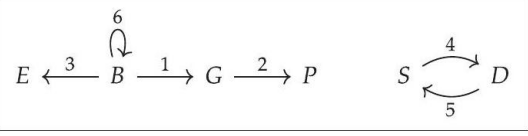
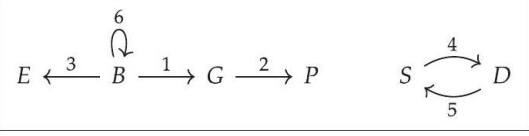
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_118_200_1877_382.jpg	$c_1 =$ 	$c_1 =$ 	$c_1 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_120_220_1877_668.jpg	$c_2 =$ 	$c_2 =$ 	$c_2 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_118_218_1879_974.jpg	$c_3 =$ 	$c_3 =$ 	$c_3 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_220_1881_1279.jpg	$c_4 =$ 	$c_4 =$ 	$c_4 =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_210_2111_376.jpg	$g_!(c_1) =$ 	$g_!(c_1) =$ 	$g_!(c_1) =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_222_2113_670.jpg	$g_!(c_2) =$ 	$g_!(c_2) =$ 	$g_!(c_2) =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_220_2113_976.jpg	$g_!(c_3) =$ 	$g_!(c_3) =$ 	$g_!(c_3) =$ 
fong-spivak-invitation_D12_438	278	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-278_116_222_2113_1281.jpg	$g_!(c_4) =$ 	$g_!(c_4) =$ 	$g_!(c_4) =$ 

fong-spivak-invitation_D14_#384	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_118_128_426_456.jpg			
fong-spivak-invitation_D14_#384	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_126_428_666.jpg			
fong-spivak-invitation_D14_#384	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_126_428_876.jpg			
fong-spivak-invitation_D14_#384	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_128_430_1084.jpg			
fong-spivak-invitation_D14_#384	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_124_430_1297.jpg			

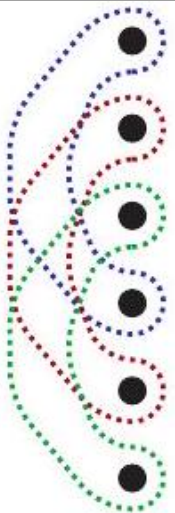
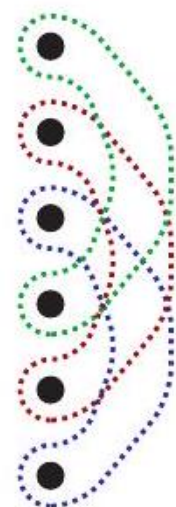
fong-spivak-invitation_D12_#388	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_206_804_644.jpg	$c = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \nearrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$c = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \nearrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$c = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \nearrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$
fong-spivak-invitation_D12_#389	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_230_804_1044.jpg	$g_!(c) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$g_!(c) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$g_!(c) = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$
fong-spivak-invitation_D12_#390	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_112_174_1056_872.jpg	$d = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \nearrow \\ \bullet \end{array}$	$d = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \nearrow \\ \bullet \end{array}$	$d = \begin{array}{c} 12 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \nearrow \\ \bullet \end{array}$
fong-spivak-invitation_D12_#391	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_186_1309_866.jpg	$e = \begin{array}{c} 12 \quad 3 \quad 4 \\ \downarrow \quad \swarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$e = \begin{array}{c} 12 \quad 3 \quad 4 \\ \downarrow \quad \swarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$e = \begin{array}{c} 12 \quad 3 \quad 4 \\ \downarrow \quad \swarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$
fong-spivak-invitation_D12_#392	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_114_232_1539_668.jpg	$g^*(d) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \swarrow \quad \searrow \\ \bullet \end{array}$	$g^*(d) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \swarrow \quad \searrow \\ \bullet \end{array}$	$g^*(d) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \swarrow \quad \searrow \\ \bullet \end{array}$
fong-spivak-invitation_D12_#393	279	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-279_116_250_1539_998.jpg	$g^*(e) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \swarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$g^*(e) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \swarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$	$g^*(e) = \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \swarrow \quad \downarrow \quad \swarrow \quad \searrow \\ \bullet \qquad \bullet \end{array}$
fong-spivak-invitation_D12_#394	280	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-280_320_788_1156_548.jpg	$P := \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \uparrow \\ 3.9 \\ \swarrow \quad \searrow \\ 1 \quad 2 \end{array}} \xrightleftharpoons[f]{g} \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \swarrow \quad \nearrow \\ 1 \quad 2 \end{array}} =: Q \end{array}$	$P := \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \uparrow \\ 3.9 \\ \swarrow \quad \searrow \\ 1 \quad 2 \end{array}} \xrightleftharpoons[f]{g} \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \swarrow \quad \nearrow \\ 1 \quad 2 \end{array}} =: Q \end{array}$	$P := \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \uparrow \\ 3.9 \\ \swarrow \quad \searrow \\ 1 \quad 2 \end{array}} \xrightleftharpoons[f]{g} \begin{array}{c} \boxed{\begin{array}{c} 4 \\ \swarrow \quad \nearrow \\ 1 \quad 2 \end{array}} =: Q \end{array}$
fong-spivak-invitation_D12_#395	281	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-281_290_366_374_754.jpg	$X := \boxed{\begin{array}{c} c_2 \\ \bullet \\ a_1 \end{array}} \xrightarrow{f} \boxed{\begin{array}{c} a \quad b \quad c \\ \bullet \quad \bullet \quad \bullet \end{array}} =: Y$	$X := \boxed{\begin{array}{c} c_2 \\ \bullet \\ a_1 \end{array}} \xrightarrow{f} \boxed{\begin{array}{c} a \quad b \quad c \\ \bullet \quad \bullet \quad \bullet \end{array}} =: Y$	$X := \boxed{\begin{array}{c} c_2 \\ \bullet \\ a_1 \end{array}} \xrightarrow{f} \boxed{\begin{array}{c} a \quad b \quad c \\ \bullet \quad \bullet \quad \bullet \end{array}} =: Y$

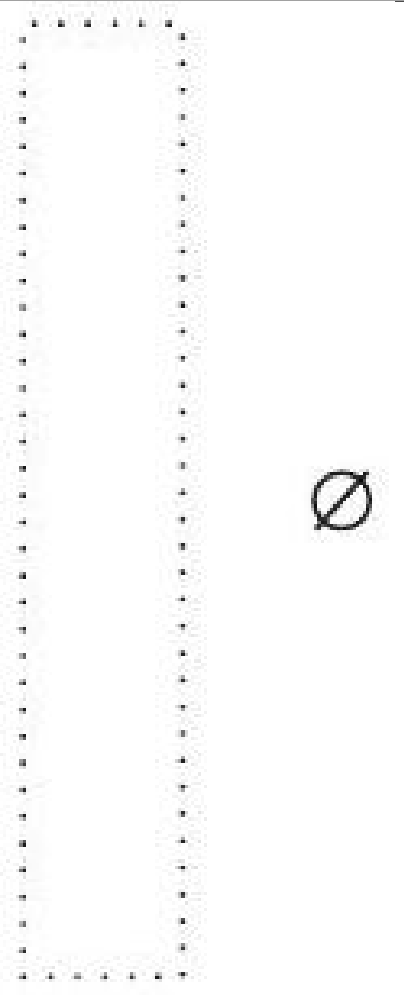
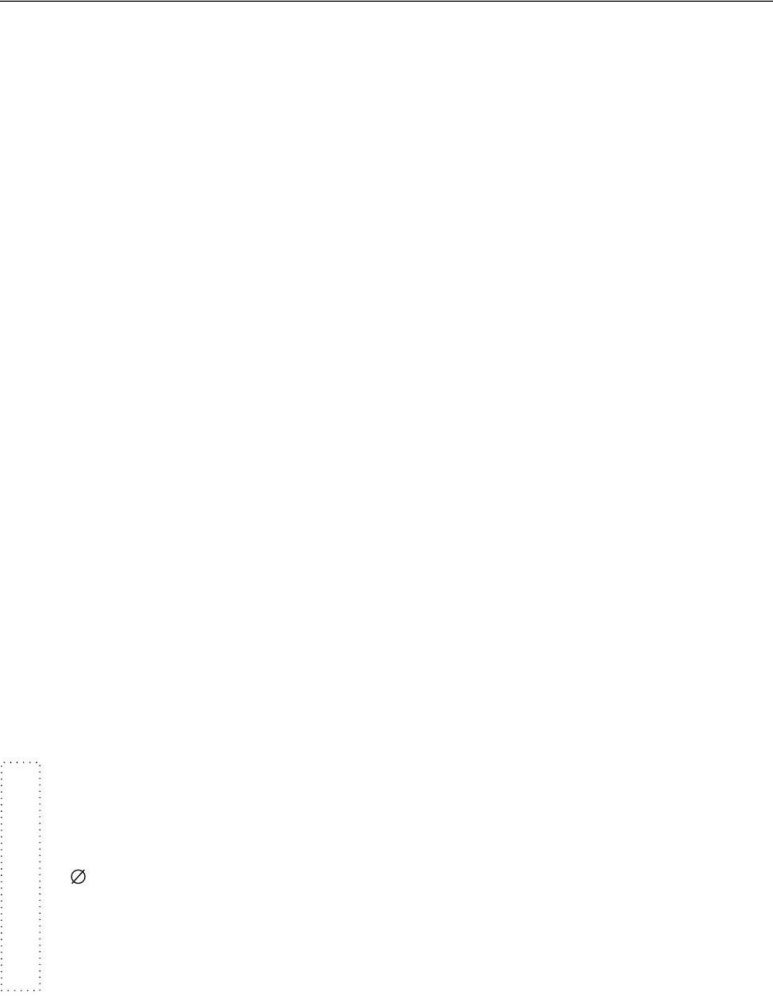
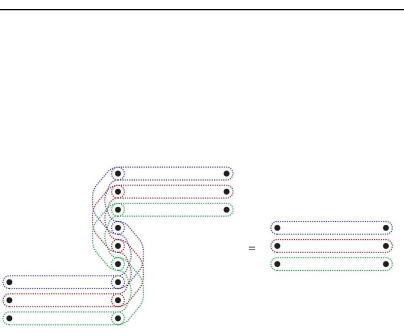
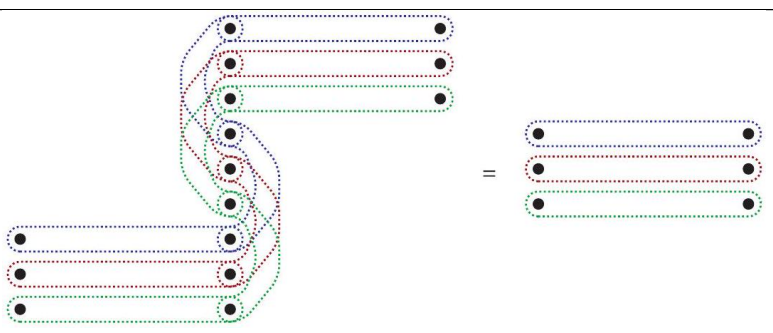
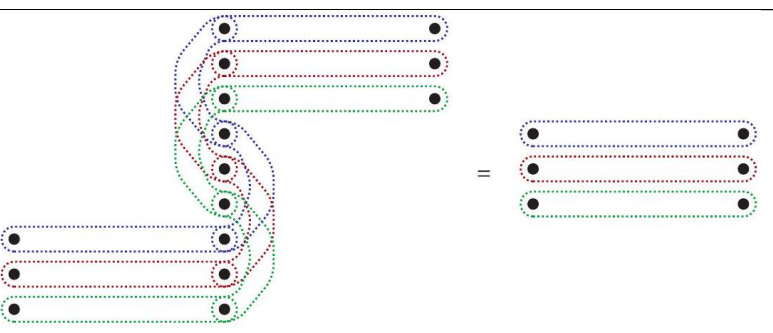
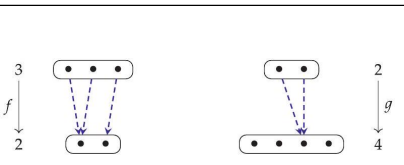
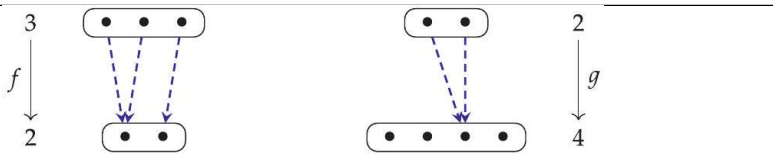
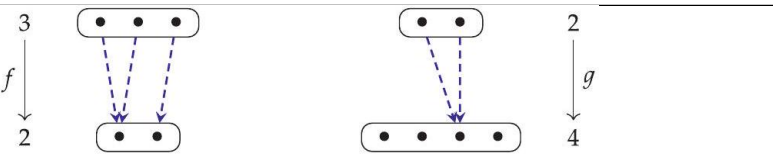
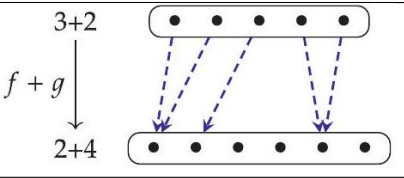
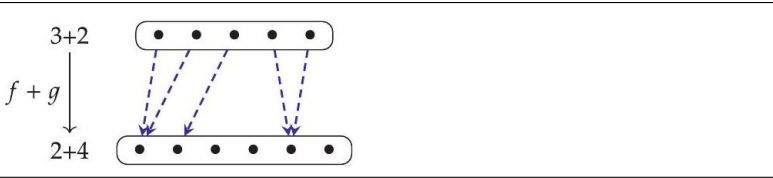
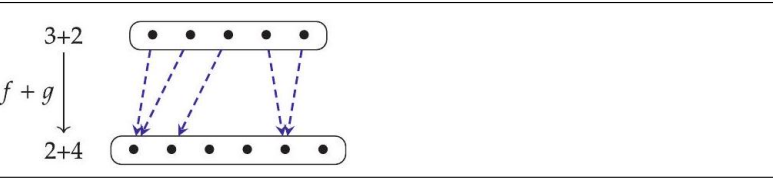
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fong-spivak-invitation_D12_085	285	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-285_326_548_1627_684.jpg			
fong-spivak-invitation_D12_086	286	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-286_228_314_718_804.jpg			
fong-spivak-invitation_D12_086	286	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-286_94_382_1757_750.jpg			
fong-spivak-invitation_D12_088	291	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-291_274_428_480_722.jpg			
fong-spivak-invitation_D12_092	292	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-292_306_258_1797_336.jpg			

fong-spivak-invitation_D14_292	292	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-292_306_258_1797_810.jpg			
fong-spivak-invitation_D14_292	292	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-292_306_258_1797_1287.jpg			
fong-spivak-invitation_D14_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_206_138_386_386.jpg			
fong-spivak-invitation_D14_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_176_304_404_804.jpg			
fong-spivak-invitation_D14_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_124_418_884_752.jpg			

fong-spivak-invitation_D12_294	294	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-294_188_448_2013_736.jpg	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f)\downarrow & & G(f)\downarrow & & H(f)\downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f)\downarrow & & G(f)\downarrow & & H(f)\downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$	$\begin{array}{ccccc} F(c) & \xrightarrow{\alpha_c} & G(c) & \xrightarrow{\beta_c} & H(c) \\ F(f)\downarrow & & G(f)\downarrow & & H(f)\downarrow \\ F(c') & \xrightarrow{\alpha_c} & G(c') & \xrightarrow{\beta_c} & H(c') \end{array}$
fong-spivak-invitation_D12_295	295	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-295_192_292_398_812.jpg	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f)\downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f)\downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$	$\begin{array}{ccc} F(c) & \xrightarrow{\text{id}_{F(c)}} & F(c) \\ F(f)\downarrow & & \downarrow F(f) \\ F(c') & \xrightarrow{\text{id}_{F(c')}} & F(c') \end{array}$
fong-spivak-invitation_D12_295	295	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-295_252_734_1569_570.jpg			
fong-spivak-invitation_D12_296	296	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_328_550_788_686.jpg			
fong-spivak-invitation_D12_296	296	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_214_198_1515_552.jpg			
fong-spivak-invitation_D12_296	296	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-296_300_548_1909_996.jpg			
fong-spivak-invitation_D12_297	297	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-297_180_700_1741_590.jpg			

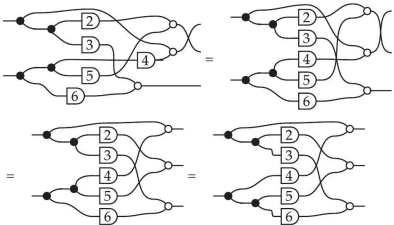
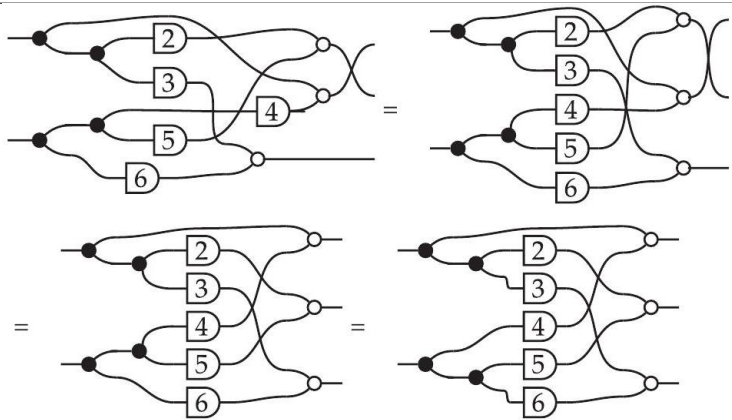
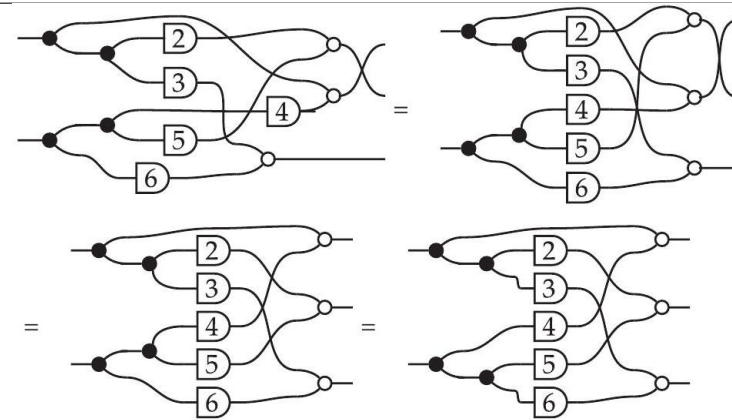
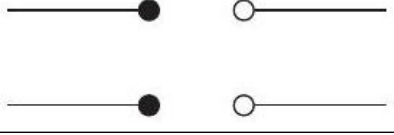
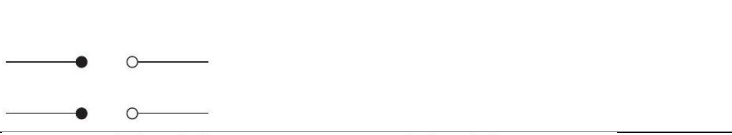
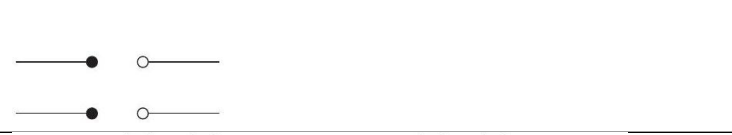
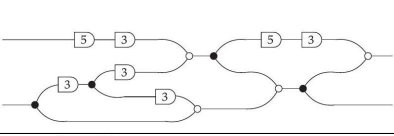
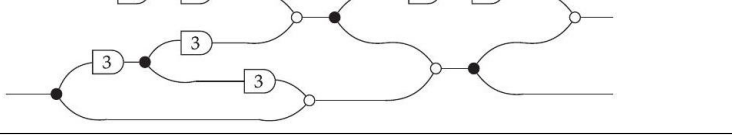
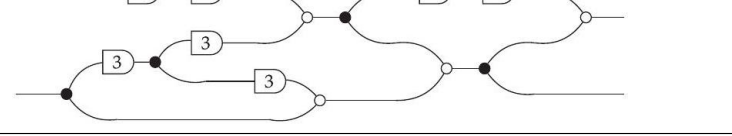
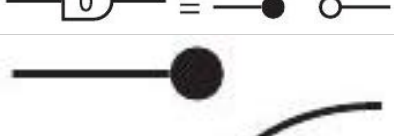
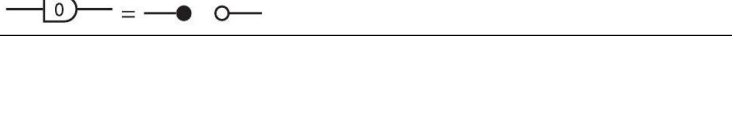
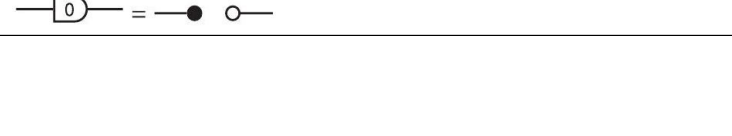
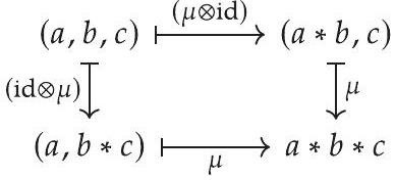
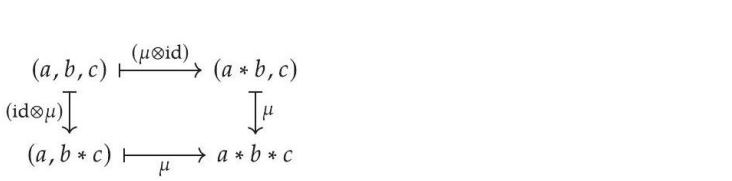
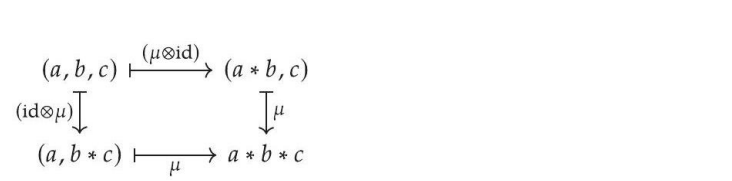
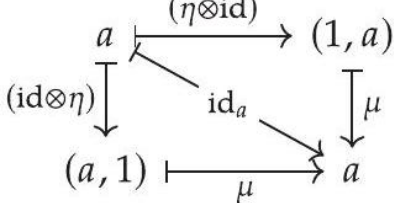
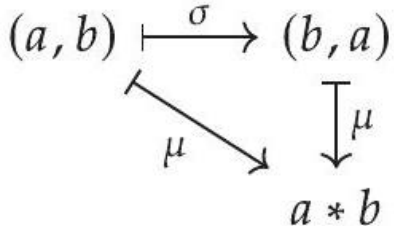
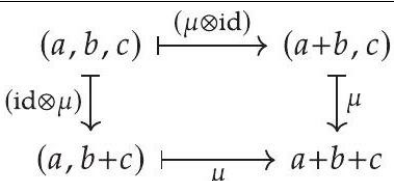
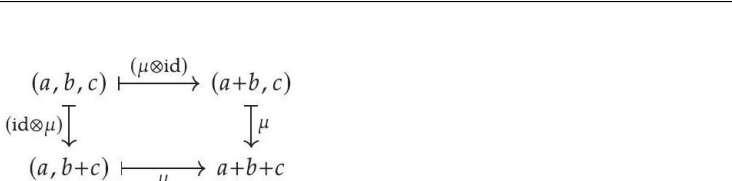
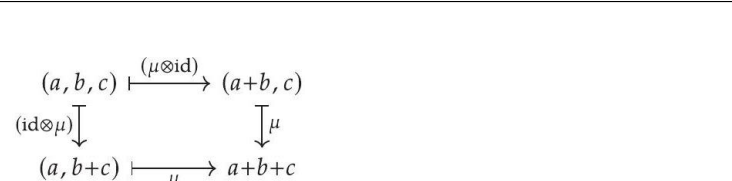
fong-spivak-invitation_D12_299.jpg	299	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-299_372_672_1325_622.jpg			
fong-spivak-invitation_D12_302.jpg	302	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-302_326_852_414_514.jpg			
fong-spivak-invitation_D12_303.jpg	303	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-303_280_662_1024_608.jpg			
fong-spivak-invitation_D12_304.jpg	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_310_124_1523_692.jpg			

fong-spivak-invitation_D12_044	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_296_222_1533_1004.jpg	 $\underline{3} \sqcup \underline{3}$	 $\underline{3} \sqcup \underline{3}$	 $\underline{3} \sqcup \underline{3}$
fong-spivak-invitation_D12_044	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_302_226_1937_694.jpg	$\underline{3} \sqcup \underline{3}$ 	 $\underline{3} \sqcup \underline{3}$	 $\underline{3} \sqcup \underline{3}$

fong-spivak-invitation_D12_044	304	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-304_308_122_1937_1106.jpg			
fong-spivak-invitation_D12_045	305	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-305_446_1074_338_422.jpg			
fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_196_784_1345_568.jpg			
fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_192_460_1637_732.jpg			

fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_192_242_2051_410.jpg			
fong-spivak-invitation_D12_046	306	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-306_192_532_2051_976.jpg			
fong-spivak-invitation_D12_047	307	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-307_194_544_346_688.jpg			
fong-spivak-invitation_D12_048	308	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-308_166_560_1719_388.jpg			
fong-spivak-invitation_D12_048	308	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-308_146_494_1737_1000.jpg			
fong-spivak-invitation_D12_048	308	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-308_170_928_1997_476.jpg			
fong-spivak-invitation_D12_049	309	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-309_136_262_518_546.jpg			
fong-spivak-invitation_D12_049	309	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-309_234_278_420_1052.jpg			
fong-spivak-invitation_D12_049	311	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-311_186_648_898_614.jpg			
fong-spivak-invitation_D12_049	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_202_630_376_626.jpg			

fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_82_184_1162_582.jpg			
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_84_182_1321_584.jpg			
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_144_424_1617_752.jpg	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_182_446_1815_742.jpg	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_86_144_2075_544.jpg	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
fong-spivak-invitation_D10_142	312	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-312_268_532_1977_848.jpg			
fong-spivak-invitation_D10_143	313	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-313_140_514_1649_730.jpg	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$	$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \sim$
fong-spivak-invitation_D10_143	313	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-313_106_528_1847_724.jpg	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \sim$
fong-spivak-invitation_D10_143	313	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-313_154_912_2091_528.jpg	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \sim$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \sim$	$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \sim$
fong-spivak-invitation_D10_144	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_156_1006_420_484.jpg			
fong-spivak-invitation_D10_144	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_70_876_650_548.jpg			

fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_544_954_800_510.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_92_274_1573_374.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_192_800_1521_750.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_42_342_1887_790.jpg			
fong-spivak-invitation_D19_144b	314	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-314_98_148_2147_888.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_184_388_740_400.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_182_320_742_864.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_152_248_760_1269.jpg			
fong-spivak-invitation_D19_144b	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_184_380_1056_404.jpg			

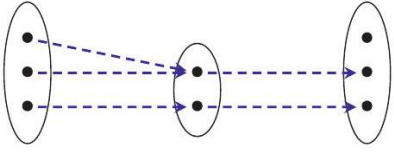
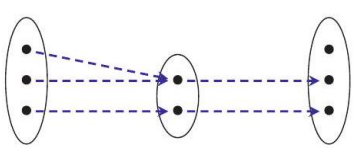
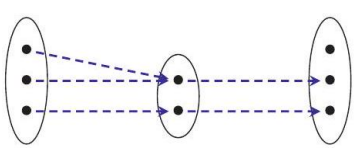
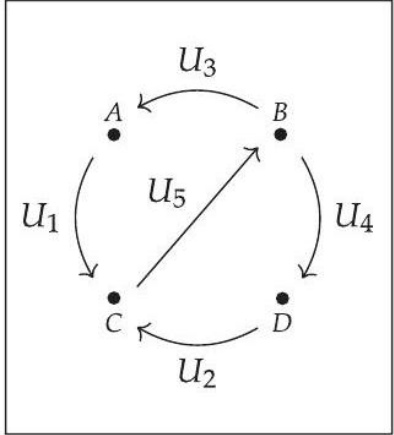
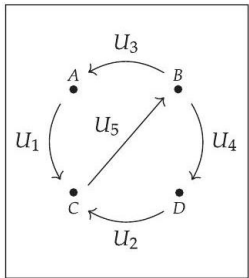
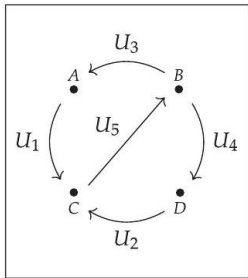
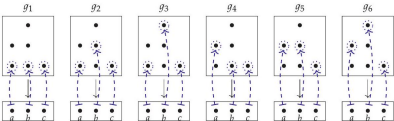
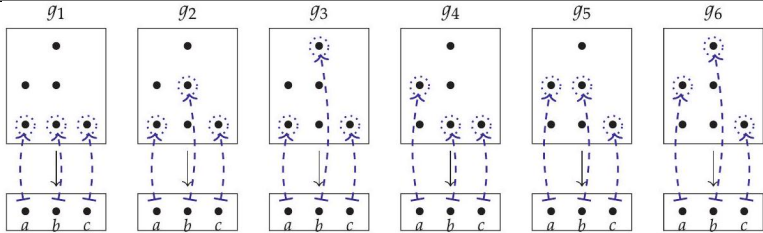
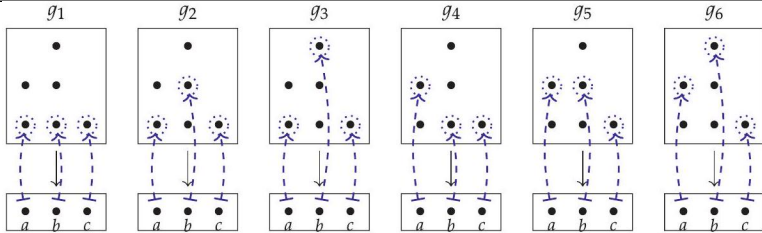
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fong-spivak-invitation_D12_145	315	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-315_152_248_1076_1265.jpg			
fong-spivak-invitation_D12_145	317	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-317_74_94_1443_1028.jpg			
fong-spivak-invitation_D12_145	317	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-317_72_114_1445_1166.jpg			
fong-spivak-invitation_D12_145	317	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-317_76_110_1445_1327.jpg			
fong-spivak-invitation_D12_145	319	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-319_306_394_978_760.jpg			
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fong-spivak-invitation_010_240	320	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-320_284_598_724_662.jpg			
fong-spivak-invitation_010_241	321	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-321_524_616_436_630.jpg			
fong-spivak-invitation_010_241	321	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-321_196_244_1164_816.jpg			
fong-spivak-invitation_010_242	321	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-321_172_180_1909_870.jpg			
fong-spivak-invitation_010_242	322	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-322_192_230_410_588.jpg			

fong-spivak-invitation_D19_2466	322	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-322_198_234_406_1060.jpg			
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fong-spivak-invitation_D19_2466	322	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-322_480_368_1703_756.jpg			
fong-spivak-invitation_D19_2466	323	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-323_300_1221_1144_328.jpg			
fong-spivak-invitation_D19_2466	323	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-323_172_274_1739_802.jpg			
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fong-spivak-invitation_D19_2476	324	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-324_242_310_470_786.jpg			
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fong-spivak-invitation_D19_2476	325	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-325_94_420_766_728.jpg			
fong-spivak-invitation_D19_2476	325	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-325_68_618_1084_630.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_410_932_492_476.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_412_970_1016_456.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_176_828_1641_548.jpg			
fong-spivak-invitation_D19_2476	326	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-326_192_836_2057_544.jpg			
fong-spivak-invitation_D19_2476	327	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-327_210_922_358_498.jpg			

fong-spivak-invitation_D19_2488	328	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-328_234_264_2011_808.jpg			
fong-spivak-invitation_D19_2489	329	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-329_180_184_764_868.jpg			
fong-spivak-invitation_D19_2490	329	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-329_180_156_1249_880.jpg			
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fong-spivak-invitation_D10_040	330	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-330_174_458_558_712.jpg			
fong-spivak-invitation_D10_040	330	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-330_344_560_1333_680.jpg	$\begin{array}{ccccccc} (0,0) & \leftarrow & (0,1) & \leftarrow & (0,2) & \leftarrow & (0,3) \cdots \\ \uparrow & & \uparrow & & \uparrow & & \\ (1,0) & \leftarrow & (1,1) & \leftarrow & (1,2) & \cdots & \\ \uparrow & & \uparrow & & & & \\ (2,0) & \leftarrow & (1,1) & & & & \\ \uparrow & & & & & & \\ (3,0) & \cdots & & & & & \\ & \vdots & & & & & \end{array}$	$\begin{array}{ccccccc} (0,0) & \leftarrow & (0,1) & \leftarrow & (0,2) & \leftarrow & (0,3) \cdots \\ \uparrow & & \uparrow & & \uparrow & & \\ (1,0) & \leftarrow & (1,1) & \leftarrow & (1,2) & \cdots & \\ \uparrow & & \uparrow & & & & \\ (2,0) & \leftarrow & (1,1) & & & & \\ \uparrow & & & & & & \\ (3,0) & \cdots & & & & & \\ & \vdots & & & & & \end{array}$	$\begin{array}{ccccccc} (0,0) & \leftarrow & (0,1) & \leftarrow & (0,2) & \leftarrow & (0,3) \cdots \\ \uparrow & & \uparrow & & \uparrow & & \\ (1,0) & \leftarrow & (1,1) & \leftarrow & (1,2) & \cdots & \\ \uparrow & & \uparrow & & & & \\ (2,0) & \leftarrow & (1,1) & & & & \\ \uparrow & & & & & & \\ (3,0) & \cdots & & & & & \\ & \vdots & & & & & \end{array}$
fong-spivak-invitation_D10_040	333	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-333_360_322_300_776.jpg			
fong-spivak-invitation_D10_040	333	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-333_398_690_1176_592.jpg	$X := \begin{array}{ccccccc} & & b_3 & & & & \\ a_2 & & b_2 & & & & e_2 \\ a_1 & & b_1 & c_1 & & & e_1 \end{array}$ $f \downarrow$ $Y := \begin{array}{ccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}$	$X := \begin{array}{ccccccc} & & b_3 & & & & \\ a_2 & & b_2 & & & & e_2 \\ a_1 & & b_1 & c_1 & & & e_1 \end{array}$ $f \downarrow$ $Y := \begin{array}{ccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}$	$X := \begin{array}{ccccccc} & & b_3 & & & & \\ a_2 & & b_2 & & & & e_2 \\ a_1 & & b_1 & c_1 & & & e_1 \end{array}$ $f \downarrow$ $Y := \begin{array}{ccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}$
fong-spivak-invitation_D10_040	333	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-333_398_690_1817_592.jpg	$X := \begin{array}{ccccccc} & & c_1 & & & & e_2 \\ a_2 & & b_1 & b_2 & b_3 & & a_1 \\ e_1 & & & & & & \end{array}$ $f \downarrow$ $Y := \begin{array}{ccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}$	$X := \begin{array}{ccccccc} & & c_1 & & & & e_2 \\ a_2 & & b_1 & b_2 & b_3 & & a_1 \\ e_1 & & & & & & \end{array}$ $f \downarrow$ $Y := \begin{array}{ccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}$	$X := \begin{array}{ccccccc} & & c_1 & & & & e_2 \\ a_2 & & b_1 & b_2 & b_3 & & a_1 \\ e_1 & & & & & & \end{array}$ $f \downarrow$ $Y := \begin{array}{ccccc} a & b & c & d & e \\ \bullet & \bullet & \bullet & \bullet & \bullet \end{array}$
fong-spivak-invitation_D10_040	334	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-334_364_1196_444_372.jpg			
fong-spivak-invitation_D10_040	334	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-334_152_1128_1100_376.jpg	$\begin{array}{ccccccc} (a_1, b_1, c_1) & & (a_1, b_2, c_1) & & (a_1, b_3, c_1) & & (a_2, b_1, c_1) \\ & \searrow & \downarrow & \swarrow & & \downarrow & \swarrow \\ & & (a_1, c_1) & & & & (a_2, c_1) \end{array}$	$\begin{array}{ccccccc} (a_1, b_1, c_1) & & (a_1, b_2, c_1) & & (a_1, b_3, c_1) & & (a_2, b_1, c_1) \\ & \searrow & \downarrow & \swarrow & & \downarrow & \swarrow \\ & & (a_1, c_1) & & & & (a_2, c_1) \end{array}$	$\begin{array}{ccccccc} (a_1, b_1, c_1) & & (a_1, b_2, c_1) & & (a_1, b_3, c_1) & & (a_2, b_1, c_1) \\ & \searrow & \downarrow & \swarrow & & \downarrow & \swarrow \\ & & (a_1, c_1) & & & & (a_2, c_1) \end{array}$

fong-spivak-invitation_D12_046	334	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-334_466_452_1499_464.jpg			
fong-spivak-invitation_D12_046	334	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-334_424_228_1541_1241.jpg			
fong-spivak-invitation_D12_046	335	— (dup)	05b3e583-fc72-4ff5-9914-6f882b9df5b5-335_230_1118_1721_378.jpg			